Premier University



Outcome Based Curriculum

Bachelor of Science (B.Sc. Engg.) in Computer Science & Engineering

Department of Computer Science & Engineering

Premier University

Academic Building#4, 44, HazariLane, Kotwali, Chattogram, Bangladesh

info@puc.ac.bd

Web: www.puc.ac.bd

1. Title of the Academic Program: B.Sc.(Engg.) in Computer Science & Engineering

2. Name of the University: Premier University

3. Vision of the university:
The vision of Premier University is to become a leading academia in the country and beyond for quality education, research and innovations in higher education.

4. Mission of the university:
The mission of Premier University is to develop human resources compatible to the 21st Century challenges and changes locally and globally through a state-of-the-art education, research and innovation. We focus on nurturing the hidden potentials of the younger generations of students through a holistic approach to quality teaching, learning and sharing the up-to-date knowledge among students, teachers and the employers both at home and abroad. We will ensure that our students develop both the skills that a sound education provides and the competencies essential for success and leadership in the emerging creative economy. We will also lead in generating practical and theoretical knowledge that enables people to better understand our world and improve conditions for local and global communities. Our students will possess a holistic, ethical and liberal psychology along with a keen and ready mind to contribute to the diverse needs of the society and country to enhance its overall progress. We will sustain the Quality Culture at campus, and will be enhancing the practice accordingly.

5. Name of the Program Offering Entity (Department/Faculty/Institute): Department of Computer Science and Engineering/ Faculty of Engineering

6. Vision of the program:

To be among the nation's leading universities renowned for world class is teaching and research aiming to produce 21st century competent professionals for information industry.

7. Mission of the program:

The mission of the CSE department at Premier University is to:
Mission 1: Emphasize analytical and creative assessment in the teaching learning to enable students to identify and analyze real world problems;

Mission 2: Equip graduates with contemporary knowledge and machine intelligence to contribute significantly to the local and global information industry;

Mission 3: Provide professional exposure to acquire experience on real-time problem-solving design, implementation and evaluation methods to address the actualities of 21st century;

Mission 4: Collaborate with local, national, and international entities in education and research;

Mission 5: Support access to contemporary research resources and facilities to engage teachers and students in research activities for quality innovation;

Mission 6: Nurture our graduates' generous and communication skills to make them visionary and frontier leaders in multicultural environment.

Mission 7: Engage students in co-curricular or extra-curricular activities, community services or service to humanity.

8. Name of the Degree: Bachelor of Science (B.Sc. Engg.) in Computer Science & Engineering

9. Description of the Program:

The B.Sc (Engg.) in Computer Science & Engineering (CSE) program is intended to produce skilled graduates in the field in order to meet the growing demand for computer engineering graduates both at home and abroad. It allows students to gain a broad understanding of computer science and computer engineering while also allowing them to tailor the program to their specific needs.

10. Program Educational Objectives (PEOs):

PEO1: Think Critically:

Use problem-solving, decision-making and research skills to identify and solve complex problems needed to pursue a diverse range of professions

PEO2: Implementation Efficiency:

Develop and implement efficient, sustainable, scalable, manageable, and future-proof solutions to problems through continuous learning.

PE03: Society, Ethics and Team Player:

Ethically manage independent or team work considering the societal, health and safety, and environmental impact.

PEO4: Communication:

Graduates will be able to disseminate information clearly and precisely to a broad range of audiences.

11. Program Learning Outcomes (POs):

The learning outcome of the degree program supports all the outcomes recommended by BAETE criteria. The students who complete the BSc in CSE program will be able to do the following:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization as specified in K1 to K4 respectively to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences. (K1 to K4)

PO3: Design/Development of Solutions Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. (K5)

PO4: Conduct investigations of complex problems: Conduct investigations of complex problems using research-based knowledge (K8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5: Modern Tool Usage: Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations. (K6)
PO6: The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. (K7)
PO7: Environment and Sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts. (K7)
PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice. (K7)
PO9: Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11: Project Management and Finance: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12: Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

12. Attributes of the Knowledge Profile, the ranges of Complex Problem Solving and Complex Engineering Activities:

In addition to incorporating the above-listed POs (graduate attributes), the educational institution may include additional outcomes in its learning programs. An engineering program that aims to attain the abovementioned POs must ensure that its curriculum encompasses all the attributes of the Knowledge Profile (K1 – K8) as presented in Table 1 and as included in the PO statements. The ranges of Complex Problem Solving (P1 – P7) and Complex Engineering Activities (A1 – A5) are given in Tables 2 and 3, respectively.

Table 1: Knowledge Profile

Knowledge Profile	Attribute
K1	A systematic, theory-based understanding of the natural sciences applicable to the discipline
K2	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline

K3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
K5	Knowledge that supports engineering design in a practice area
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
K7	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability
K8	Engagement with selected knowledge in the research literature of the discipline

Table 2: Range of Complex Engineering Problem Solving

Attribute	**Complex Engineering Problems have characteristic P1 and some or all of P2 to P7:**
Depth of knowledge required	P1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach
Range of conflicting requirements	P2: Involve wide-ranging or conflicting technical, engineering and other issues
Depth of analysis required	P3: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models
Familiarity of issues	P4: Involve infrequently encountered issues
Extent of applicable codes	P5: Are outside problems encompassed by standards and codes of practice for professional engineering
Extent of stakeholder involvement and conflicting requirements	P6: Involve diverse groups of stakeholders with widely varying needs
Interdependence	P7: Are high-level problems including many component parts or sub-problems

Table 3: Range of Complex Engineering Activities

Attribute	**Complex activities** mean (engineering) activities or projects that have some or all of the following characteristics:
Range of resources	A1: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies)
Level of interaction	A2: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues

Innovation	A3: Involve creative use of engineering principles and research-based knowledge in novel ways
Consequences for society and the environment	A4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
Familiarity	A5: Can extend beyond previous experiences by applying principles-based approaches

13. Mapping between Mission and PEOs

Missions	PEO-1	PEO-2	PEO-3	PEO-4
Mission 1:	√	√		
Mission 2:	√	√		
Mission 3:	√	√	√	
Mission 4:			√	√
Mission 5:	√	√	√	√
Mission 6:	√	√	√	√
Mission 7:	√	√	√	√

14. Program outcomes (PO) statements and the mapping with the corresponding PEOs

PO No.	PO Statement	PEO-1	PEO-2	PEO-3	PEO-4
		Thinking Critically	Implement Efficiently	Society, Ethics, and Team Player	Communication
PO1	Engineering Knowledge	√			
PO2	Problem Analysis	√			
PO3	Design/ Development of Solutions		√	√	√
PO4	Conduct investigations of complex problems	√			
PO5	Modern Tool Usage		√	√	
PO6	The Engineer and Society			√	
PO7	Environment and Sustainability		√	√	
PO8	Ethics			√	
PO9	Individual and Team Work			√	√
PO10	Communication			√	√
PO11	Project Management and Finance		√		
PO12	Life-long Learning		√		

15. Blooms Taxonomy:

Table 4: Cognitive Domain (C)

Level	Category or 'level'	Behavior descriptions
C1	Knowledge	Recall or recognize information

C2	Comprehension	understand meaning, re-state data in one's own words, interpret, extrapolate, translate
C3	Application	se or apply knowledge, put theory into practice, use knowledge in response to real circumstances
C4	Analysis	Interpret elements, organizational principles, structure, construction, internal relationships; quality, reliability of individual components
C5	Evaluation	Assess effectiveness of whole concepts, in relation to values, outputs, efficacy, viability; critical thinking, strategic comparison and review; judgement relating to external criteria
C6	Create	Develop new unique structures, systems, models, approaches, ideas; creative thinking, operations

Table 5: Affective Domain (A)

Level	Category or 'level'	Behavior descriptions
A1	Receive	Open to experience, willing to hear
A2	Respond	React and participate actively
A3	Value	Attach values and express personal opinions
A4	Organize or Conceptualize values	Reconcile internal conflicts; develop value system
A5	Internalize or characterize values	Adopt belief system and philosophy

Table 6: Psychomotor Domain (P)

Level	Category or 'level'	Behavior descriptions
P1	Imitation	Copy action of another; observe and replicate
P2	Manipulation	Reproduce activity from instruction or memory
P3	Precision	Execute skill reliably, independent of help
P4	Articulation	Adapt and integrate expertise to satisfy a non-standard objective
P5	Naturalization	Automated, unconscious mastery of activity and related skills at strategic level

16. Structure of the Curriculum

a) Duration of the program: Years: 4 Semesters: 8

b) Admission Requirements:

Admission Requirements: Minimum GPA of 2.5 (or second division) in SSC and HSC examinations (or their equivalent), OR at least one GPA 2.00 but aggregate GPA of 6.00 in SSC and HSC OR GCE “O” Level in 5 subjects and “A” Level in 2 subjects with at least 4 B Grade or GPA 4.00 and 3 C Grade or GPA 3.50 (using a scale of A=5, B=4, C=3.5, D=2, E=1) However:

- Sons/daughters of freedom fighters will be eligible for admission if they have an aggregate GPA of 5.0 in SSC and HSC.
- Equivalent performance under other educational systems (e.g. American High School Diploma, IB, etc.) will also be accepted.

• A combined SAT score of 1100 is also accepted in lieu of the Admission Test for High School Graduates in any system.
• Transfer of credits from comparable educational institutions may be considered after admission. • Students will have to pass an admission test to get admission at PU.

c) Total minimum credit requirement to complete the program: 160

Passing of all courses individually and maintaining a CGPA of 2.25.

Completion of Project/Thesis with a minimum grade of "C".

Letter grade and corresponding grade points will be awarded as follows:

Numerical Grade	**Letter Grade**	**Grade Points**
80% or above	A+ (A plus)	4.00
75% to less than 80%	A (A regular)	3.75
70% to less than 75%	A- (A minus)	3.50
65% to less than 70%	B+ (B plus)	3.25
60% to less than 65%	B (B regular)	3.00
55% to less than 60%	B- (B minus)	2.75
50% to less than 55%	C+ (C plus)	2.50
45% to less than 50%	C (C regular)	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

GPA of student in of a semester shall be calculated as follows:

$$\text{GPA} = \frac{\sum_{i=1}^{n} C_i G_i}{\sum_{i=1}^{n} C_i}$$

Where,

n is the number of courses enrolled by student in a particular semester;

C_i is the number of credits assigned to a particular course i; and

G_i is the grade point corresponding to the grade earned for i-th course.

Cumulative Grade Point Average (CGPA) will be calculated by dividing the total grade points accumulated up to the date by the total credits. Both GPA and CGPA will be rounded off to the second place of decimal for reporting.

d) Total class weeks in a Year/semester:

All programs at the PU follow two terms per academic year (bi-semester) system. The two terms per year system will be as follows:

• Term I (Spring semester) - between March and August,
• Term II (Fall Semester) - between September and February

The duration of each term in two terms per year system will be a minimum of 23 weeks, which will be used as follows:

1. Term-I

Classes	13 weeks
Mid Term (Generally after 50% of the Term Duration)	1 week

Term Final Examination Period Duration	*3 weeks 1 day (Minimum 2 days interval/course)
Total	18 weeks 1 day
Publication of result	2 weeks
Inter Term Break	1 week
Course enrollment and result correction	2 weeks
Total	23 weeks 1 day

2. Term-2

Classes	13 weeks
Mid Term (Generally after 50% of the Term Duration)	1 week
Term Final Examination Period Duration	*3 weeks 1 day (Minimum 2 days interval/course)
Total	18 weeks 1 day
Publication of result	2 weeks
Inter Term Break	1 week
Course enrollment and result correction	2 weeks
Total	23 weeks 1 day

3. Ramadan, Puja, and other Vacations throughout the Level 5 weeks 1 day 51 weeks 3 days

* Design for 05 theory courses ** There will be strictly no class during this period
** The remaining 3 days are for any special vacation/holidays decided by the university authority

The particular dates of the terms may vary from year to year due to official holiday variables or special circumstances, but the minimum duration of each term will always be maintained.

e) Minimum CGPA requirements for graduation: 2.0
f) Maximum academic years of completion: 7 academic years
g) Program- specific criteria:
The curriculum of the program offers 144 credits of **Required** courses and 16 credits of **Elective** courses including basic science, language, mathematics, core, and non-engineering category. The following table shows the breakdown of the curriculum of the program.

Category	**Required**	**Elective**
Basic Science	06 Credits	
Language	07 Credits	
Mathematics	12 Credits	
Core	107 Credits	16 Credits

Non-Engineering	12 redits	

i. Distribution of Courses

Term-wise course outline for the entire program:

Level one Term one			
Course No.	**Course Title**	**Credit Hours**	**Contact Hours**
ACC 101	Basic Accounting	3	3
CSE 110	Introduction to Computer Systems (Laboratory)	2	4
EEE 101	Electrical Circuits I	3	3
EEE 102	Electrical Circuits I Laboratory	1.5	3
ENG 101	General English	3	3
MAT 105	Engineering Mathematics I	3	3
ME 102	Mechanical Engineering Drawing & CAD (Laboratory)	1	2
PHY 101	Engineering Physics I	3	3
Total	**Theory: 5** **Sessional: 3**	19.5	24

Level one Term two			
Course No.	**Course Title**	**Credit Hours**	**Contact Hours**
CSE 103	Discrete Mathematics	3	3
CSE 111	Structured Programming	2	2
CSE 112	Structured Programming Laboratory	2	4
EEE 211	Electronics I	3	3
EEE 212	Electronics I Laboratory	1.5	3
ENG 103	Developing English Skills	2	2
MAT 107	Engineering Mathematics II	3	3
PHY 103	Engineering Physics II	3	3
Total	**Theory: 6** **Sessional: 2**	19.5	23

Level two Term one			
Course No.	**Course Title**	**Credit Hours**	**Contact Hours**
CSE 211	Object Oriented Programming	3	3
CSE 212	Object Oriented Programming Laboratory	1.5	3
CSE 221	Data Structures	3	3
CSE 222	Data Structures Laboratory	1.5	3
ECO 201	Basic Economics	3	3
EEE 311	Digital Electronics	3	3
EEE 312	Digital Electronics Laboratory	1.5	3

MAT 201	Engineering Mathematics III	3	3
Total	**Theory: 5** **Sessional: 3**	**19.5**	**24**

Level two Term two			
Course No.	**Course Title**	**Credit Hours**	**Contact Hours**
CSE 225	Algorithm Design and Analysis	3	3
CSE 226	Algorithm Design and Analysis Laboratory	1	2
CSE 237	Database Management Systems	3	3
CSE 238	Database Management Systems Laboratory	1.5	3
EEE 201	Signals & Systems	3	3
EEE 202	Signals & Systems Laboratory	1	2
MAT 203	Engineering Mathematics IV	3	3
MGT 203	Industrial and Business Management	3	3
Total	**Theory: 5** **Sessional: 3**	**18.5**	**22**

Level three Term one			
Course No.	**Course Title**	**Credit Hours**	**Contact Hours**
CSE 301	Computational Methods for Engineering Problems	3	3
CSE 302	Computational Methods for Engineering Problems Laboratory	1	2
CSE 305	Software Engineering & Information System Design	4	4
CSE 306	Software Engineering & Information System Design Laboratory	1.5	3
EEE 309	Communication Engineering	3	3
EEE 310	Communication Engineering Laboratory	1.5	3
EEE 371	Microprocessors & Microcontrollers	3	3
EEE 372	Microprocessors & Microcontrollers Laboratory	1.5	3
MGT 251	Organizational Behavior	3	3
Total	**Theory: 5** **Sessional: 4**	**21.5**	**27**

Level three Term two			
Course No.	**Course Title**	**Credit Hours**	**Contact Hours**
CSE 317	Artificial Intelligence	3	3
CSE 318	Artificial Intelligence Laboratory	1.5	3
CSE 333	Operating Systems	3	3
CSE 334	Operating Systems Laboratory	1.5	3

CSE 337	Computer Organization & Architecture	3	3
CSE 338	Software Development Project (Laboratory)	2	4
CSE 364	Data Communication	3	3
CSE 367	Computer Networks	3	3
CSE 368	Computer Networks Laboratory	1.5	3
Total	**Theory: 5** **Sessional: 4**	**21.5**	**26**

Level four Term one			
Course No.	**Course Title**	**Credit Hours**	**Contact Hours**
EEE 313	Control Systems	3	3
EEE 314	Control Systems Laboratory	1.5	3
ENG 401	Technical Writing & Presentation	2	2
CSE 309	Theory of Computation	2	2
CSE 451	Neural Network & Fuzzy Logic	3	3
CSE 452	Neural Network & Fuzzy Logic Laboratory	1	2
CSE 455	Computer Graphics & Image Processing (Elective)	3	3
CSE 456	Computer Graphics & Image Processing Laboratory (Elective)	1	2
CSE 437	Network and Computer Security	3	3
Total	**Theory: 6** **Sessional: 3**	**19.5**	**23**

Level four Term two			
Course No.	**Course Title**	**Credit Hours**	**Contact Hours**
CSE 453	Compiler Construction	3	3
CSE 454	Compiler Construction Laboratory	1.5	3
CSE 457	Machine Learning (Elective)	3	3
CSE 458	Machine Learning Laboratory (Elective)	1	2
CSE 481	Contemporary Courses of Computer Course (Elective)	3	3
CSE 482	Contemporary Courses of Computer Course Laboratory (Elective)	1	2
CSE 459	Pattern Recognition (Elective)	3	3
CSE 460	Pattern Recognition Laboratory (Elective)	1	2
CSE 400	Project/Thesis	4	4
Total	**Theory: 4** **Sessional: 5**	**20.5**	**25**

Four Optional courses have to be selected from the list given below:

CSE 411	Project Management for Information Systems	3
CSE 412	Project Management for Information Systems Laboratory	1

CSE 417	Advanced Software Engineering	3
CSE 418	Advanced Software Engineering Laboratory	1
CSE 451	Neural Network & Fuzzy Logic	3
CSE 452	Neural Network & Fuzzy Logic Laboratory	1
EEE 319	Telecommunication & Switching	3
EEE 320	Telecommunication & Switching Laboratory	1
EEE 403	Microwave & Antenna Engineering	3
EEE 404	Microwave & Antenna Engineering Laboratory	1
EEE 435	Cellular Mobile Communication	3
EEE 469	Optical Fiber Communication	3
EEE 470	Optical Fiber Communication Laboratory	1
CSE 431	Wireless Technologies	3
CSE 432	Wireless Technologies Laboratory	1
CSE 433	Advanced Switching & Routing Concepts	3
CSE 434	Advanced Switching & Routing Concepts Laboratory	1
CSE 435	WAN Technologies	3
CSE 436	WAN Technologies Laboratory	1
CSE 415	Human Computer Interaction	3
CSE 416	Human Computer Interaction Laboratory	1
CSE 419	Basic Multimedia Systems	3
CSE 420	Basic Multimedia Systems Laboratory	1
CSE 455	Computer Graphics & Image Processing	3
CSE 456	Computer Graphics & Image Processing Laboratory	1
CSE 457	Machine Learning	3
CSE 458	Machine Learning Laboratory	1
CSE 459	Pattern Recognition	3
CSE 460	Pattern Recognition Laboratory	1
CSE 461	Management Information System	3
CSE 462	Management Information System Laboratory	1
CSE 463	Graph Theory	3
CSE 464	Graph Theory Laboratory	1
CSE 465	Distributed AI and Intelligent Agent	3
CSE 466	Distributed AI and Intelligent Agent Laboratory	1
CSE 467	Parallel & Distributed Computing	3
CSE 468	Parallel & Distributed Computing Laboratory	1
CSE 471	VoIP	3
CSE 472	VoIP Laboratory	1
CSE 477	Computer Interfacing	3
CSE 478	Computer Interfacing Laboratory	1
CSE 483	Digital System Design	3
CSE 484	Digital System Design Laboratory	1
EEE 443	VLSI Design	3
EEE 444	VLSI Design Laboratory	1
EEE 455	Satellite Communication	3
EEE 456	Satellite Communication Laboratory	1
EEE 477	Digital Signal Processing	3
EEE 478	Digital Signal Processing Laboratory	1
CSE 481	Contemporary Courses of Computer Science	3
CSE 482	Contemporary Courses of Computer Science Laboratory	1

ii. Marks Distribution

Marks distribution of the courses are divided into four categories: theory, seminar type, laboratory, and project/thesis. Further, laboratory courses are divided into five sub-categories. The marks distributions are as follows:

Theory Course: All courses excluding ENG 103

Category	Marks %
Class Attendance	10
Class Test	20
Home Works/Assignments	10
Midterm	20
Final Exam	40
Total	**100**

Seminar type : Only the course: ENG 103

Category	Marks %
Class Assessment	10
Class participation/group work	30
Quiz/Class Test/Assignment	30
Presentation	20
Viva	10
Total	**100**

Laboratory Courses:

Marks distribution of Hardware/Networking lab

Category	Marks %
Class Assessment	10
Lab performance	40
Report	20
Quiz/written Exam	20
Viva	10
Total	**100**

Marks distribution of project-based lab (Hardware/programming)

Category	Marks %
Class Assessment	10
Lab Performance	20
Project	30
Presentation / Viva	20
Report	20
Total	**100**

Marks distribution of programming-based lab

Category	**Marks %**
Class Assessment	10
Lab Performance	20
Report	10
Lab Exam	30
Quiz/written Exam	20
Viva	10
Total	**100**

Marks distribution of project

Category	**Marks %**
Performance	40
Report	30
Presentation	30
Total	**100**

Annexure: 6.1
Course Profile

Course Title: Basic Accounting
Course Code: ACC 101

Credits: 3 **Class Hours/Week: 3**
Course Type: Non-engineering **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course helps students to get basic knowledge of accounts and other business studies.

Course Objectives:
The objectives of this course are:
1. To enable the students to study, understand and use the accounting process and accounting information.
2. To introduce the students to accounting as a valuable tool of business, record keeping, generation of accounting information, analysis and interpretation and communication of the same to different interest groups and thereby achieving the predetermined goals of the organization.
3. To teach the causes of non-equal balance of bank account of a company account and bank statement and also understand the bank transaction and function of bank.
4. To measure cost per unit and total cost of a product.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:
- **CO1** **Describe (C2)** various concepts related to accounting system.
- **CO2** **Calculate (C3)** the recording process of accounting system.
- **CO3** **Demonstrate (C3)** the process of preparing financial statement for a problem.
- **CO4** **Illustrate (C3)** various concepts and process of depreciation.
- **CO5** **Solve (C3)** various problems of cost accounting and standard coding.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√										
CO3	√	√										
CO4	√	√										
CO5	√	√										

Course Description:

SL No.	Course Contents	COs
1.	**Introduction:** Definition of Accounting and Financial Accounting, Users of Accounting Information, Branches of Accounting, Types of accounts, Determination of debit and credit, Accounting concepts and Conventions, Definition of business transaction, Nature of business transaction, Accounting Principles, GAAP, Accounting Cycle.	CO1

2.	**Recording Process:** Journal, Ledger, Trial Balance, Cash Book, adjusting entries, closing entries, classifying capital and revenue expenditure	CO2
3.	**Preparation of Financial Statement:** Definition, Features of financial statement, Component/parts of financial statement, Multi-step Income Statement, Owners Equity Statement, Classified Balance Sheet, Cash Flow Statement, Bank Reconciliation Statement.	CO3
4.	**Depreciation:** Definition of Depreciation, Objectives and methods for providing depreciation.	CO4
5.	**Cost Accounting:** Introduction, Objectives and advantages of cost accounting, Elements of cost, Preparation of cost sheet, Inventory valuation (Store ledger), Overhead allocation.	CO5
6.	**Standard Costing:** Calculation of material, labor and overhead variance, Break-even point, Margin of safety, Use of accounting information in project evaluation, Budget (cash budget and master budget).	CO5

Text Books, Reference Books and Other Resources:

1. Accounting Principles. by Weygandt, Kieso and Kimmel- Latest edition
2. Cost Accounting by Bhabatosh Banarjee

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2	C2	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO2	PO1, PO2	C3	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO3	PO1, PO2	C3	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO4	PO1, PO2	C3	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO5	PO1, PO2	C3	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam

Course Title: Introduction to Computer Systems Laboratory
Course Code: CSE 110

Credits: 2 **Class Hours/Week: 4**
Course Type: Core **Pre-requisite:**
CIE Marks: 40 **SEE Marks: 60**

Course Rationale:
This course is to introduce the basic computer system and computer programming principles.

Course Objectives:
The objectives of this course are:
1. To introduce the fundamentals of computer systems.
2. To help students hands-on use of: Word processing, Spreadsheet, and Presentation applications.
3. To familiarize with number systems.
4. To introduce basic programming language.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Define (C1)** the fundamental concepts of a computer system and number systems.
CO2 **Use (C3)** word processing, spreadsheet, and ppt presentation for solving engineering tasks.
CO3 **Execute (C3)** basic computer programming principles like data types, programming constructs, input and output operations.
CO4 **Report** (**A3**) lab results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√	√										
CO4										√		

Course Description:

SL No.	Course Contents	COs
1.	**Introduction:** Generation of Computer, Functions of the Different Computer Units, Input & Output Devices, Computer Memories and Software.	CO1, CO4
2.	**Number Systems:** Binary, Octal, Decimal and Hexadecimal.	CO1, CO4
3.	Office applications: MS Word, MS Excel, MS Power Point	CO2, CO4

4.	**Introduction to C Language:** Flow Charts, Identifiers, Data Types, Variables, Constants, Input/output Statements, Operators and Expressions, Conditional Statements (If, else, switch, etc.), Looping statements (for, while, do-while), arrays.	CO3, CO4

Text Books, Reference Books and Other Resources:

1. Introduction to computers ---- (Peter Norton)
2. C- The Complete Reference ---- (Herbert Schildt)
3. Schaum's Outline of Programming with C ---- (Byron S. Gottfried)
4. Programming in ANCI C ---- (E. Balagurusami)

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C1	Lecture notes, PPT slides, problem solving, web content	Lab Performance
CO2	PO1	C3	Lecture notes, PPT slides, problem solving, web content	Lab Performance
CO3	PO1, PO2	C3	Lecture notes, PPT slides, problem solving, web content	Lab Performance, Lab Exam
CO4	PO10	A3	Demonstration	Report

Course Title: Electrical Circuit I
Course Code: EEE 101

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course is intended to enable the students to learn the fundamentals of electrical circuits. Use the acquired knowledge to understand the working and operations of electrical circuits to solve the problems for industrial opportunities.

Course Objectives:
The objectives of this course are:
1. To provide the knowledge of Direct Current (DC) and Alternating Current (AC) Circuits.
2. Helping the students to develop the ability to solve the unknown electrical parameters which are used to design electrical circuits.
3. To develop skills to analyze the behavior of electrical circuits.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Demonstrate** (**C3**) the electrical parameters which are used to design electrical and electronic devices.

CO2 **Illustrate** (**C3**) the circuit laws, network analysis and methodologies to solve DC as well as AC circuits/networks.

CO3 **Solve** (**C3**) different types of circuit theorems to find the unknown parameters of an electrical circuit/network.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√										
CO3	√	√										

Course Description:

SL No.	Course Contents	COs
1.	**Circuit Theorems:** Fundamental electrical concepts, Electrical parameters in AC and DC circuits, Measuring devices in the electrical system, Different types of materials, Temperature effect on materials, Ohm's Law, KVL, KCL, VDR, CDR.	CO1, CO2
2.	**DC Circuits (DC):** Series DC circuits, Parallel DC Circuits, Series-parallel circuits.	CO1, CO2
3.	**Method of Analysis and Theorems:** Different types of circuit conversions, Branch Current analysis, Mesh analysis, Nodal analysis, Network theorems.	CO2, CO3
4.	**AC Circuits (AC):** The Basic Elements and Phasors: R branch, L branch, C branch, and RLC branch Series AC circuits, Parallel AC circuits, Series-parallel AC circuits. Reasoning Circuit: Series and Parallel circuits.	CO1, CO2

Text Books, Reference Books and Other Resources:

1. Fundamentals of Electric Circuits, Charles K. Alexander & Matthew N. O. Sadiku
2. Introductory Circuit Analysis, Robert L. Boylestad

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1, PO2	C3	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam
CO2	PO1, PO2	C3	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam
CO3	PO1, PO2	C3	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam

Course Title: Electrical Circuit I Laboratory
Course Code: EEE 102

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite:**
CIE Marks: 70 **SEE Marks: 30**

Course Rationale:
This course is intended to enable the learners to analyze the behavior of electrical circuits, use the acquired knowledge to implement or design efficient electrical circuits practically to solve real world problems.

Course Objectives:
The objective of the course is to enable the students to
1. To provide practical knowledge of the electrical circuits as well as the technical skills to implement such electrical circuits.
2. To help the students to pursue further studies in electrical or telecommunications engineering as well as some other related engineering disciplines including computer engineering.
3. This course covers practical works that will be helpful for students to build and analyze devices in real world.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Execute** (**C3**) the circuit network and theorems related to DC and AC circuits/networks.

CO2 **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2										√		

Course Description:

SL No.	Course Contents	COs
1	Laboratory work based on theory course **EEE 1101:** Introduction to Electrical Engineering	CO1, CO2

Text and Reference books:

1. Electrical Circuit Lab Manual Book

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2	C3	Demonstration, discussion, experiment	Quiz/Written Exam, Performance
CO2	PO10	A3	Demonstration	Report

Course Title: General English
Course Code: ENG 101

Credits: 3 **Class Hours/Week: 3**
Course Type: Language **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course aims to introduce the basic grammar of English language.

Course Objectives:
The objectives of this course are:
1. To focus on the basics of English grammar for developing integrated language skills.
2. To develop the students' habit of thinking logically and clearly.
3. To provide basic grammar rules and involve students into extensive practice.
4. To make learners competent in English by the accurate use of grammar.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Discuss (C2, A2)** the properties of spoken English (speaking and listening skills) and written English (reading and writing skills).

CO2 **Discuss (C2, A2)** the factors how grammar and vocabulary choices together create a range of different meanings in speech and writing.

CO3 **Outline (C3, A2)** the different ways in which grammar has been described.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√									√		
CO2	√									√		
CO3	√									√		

Course Description:

SL No.	Course Contents	COs
1.	**Sounds and Pronunciation:** English sounds and pronunciation development	CO1
2.	**Grammar and Vocabulary:** Content related words and expressions and how to apply them in written texts	CO2
3.	**Tense and types, Parts of Speech:** Noun, pronoun, adjective, adverb, verb, **Preposition and article, Conditionals**	CO3
4.	**Phrasal Verb**	CO2
5.	**Wh/yes-no questions**	CO3

Text Books: (for grammar purpose)

Author	***Book Name***
1. Raymond Murphy	Intermediate English Grammar
2. John Eastwood	Oxford English Grammar
3. Cliff's TOEFL - a handbook	
4. P.K. Dey Sarkar	A textbook Higher English Grammar
5. R. A. Close	The English We Use
6. W. Stannard Allen	Living English Structure

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1, PO10	C2, A2	Lecture notes, PPT slides, problem solving, web content	Class Test, Assignment, Midterm, Final Exam
CO2	PO1, PO10	C2, A2	Lecture notes, PPT slides, problem solving, web content	Class Test, Assignment, Midterm, Final Exam
CO3	PO1, PO10	C3, A2	Lecture notes, PPT slides, problem solving, web content	Class Test, Assignment, Midterm, Final Exam

Course Title: Engineering Mathematics I
Course Code: MAT 105

Credits: 3 **Class Hours/Week: 3**
Course Type: Mathematics **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
It is essential for the engineering/science students to develop strong analytic skills which cannot be achieved without learning calculus of one and multiple variables. This course is intended to build up strong base in calculus as well as to include the experience of applications of its contents in the relevant courses of science/engineering.

Course Objectives:
The objectives of this course are:
1. To enhance the understanding and problem-solving skills on related/necessary basic mathematical topics required for learning ordinary and partial differentiation.
2. To develop the basic concepts of differentiation in its wide scope.
3. To build up problem solving experience in each relevant calculus topic along with addition of application familiarity in the related areas of science/engineering.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:
- **CO1** **Explain (C2)** different ways of representing functions: graphical, numerical, analytical or verbal and understand the connection among these representations.
- **CO2** **Use (C3)** derivatives to solve a variety of problems.
- **CO3** **Illustrate (C3)** series expansion of functions and apply series concept in engineering problems.
- **CO4** **Calculate (C3)** geometric properties and relationships to solve multistep problems in two dimensions.

Mapping of Course Outcomes to Program Outcomes:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√										
CO3	√	√										
CO4	√	√										

Course Description:

SL No.	**Course Contents**	**COs**
1.	Function, Various types of functions, Domain and range of the function, Graphical representation of function, Limit of function, Continuity of function	CO1
2.	Differentiability of function	CO1, CO2

3.	Differential co-efficient of various types of functions	CO2
4.	Successive Differentiation	CO2, CO3
5.	Leibnitz Theorem, Exercises on Leibnitz Theorem	CO1, CO2, CO3
7.	Roll's Theorem and exercises, Mean value theorem and exercises	CO1
8.	Expansion of functions	CO3
9.	Euler's Theorem	CO2
10.	Partial Differentiation	CO3
11.	Transformation of co-ordinate axes, Pair of straight lines, equations of conics and reduction to standard forms, Tangent, Normal, Asymptote, Second degree equation	CO4

Text Books, Reference Books and Other Resources:

1. Calculus—A New Horizon: By Howard Anton
2. The Elements of Coordinate Geometry: by S. L. Loney
3. Calculus & Analytic Geometry: by George B Thomas, Jr.
4. The Analytical Geometry of the Conic Sections: by The Rev. E. H. Askwith

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment tools**
CO1	PO1, PO2	C2	Lecture, Notes, Problem solution	Class Test, Assignment, Midterm, Final Exam
CO2	PO1, PO2	C3	Lectures, Notes, Practice Problems	Class Test, Assignment, Midterm, Final Exam
CO3	PO1, PO2	C3	Lectures, Notes, Practice Problems	Class Test, Assignment, Midterm, Final Exam
CO4	PO1, PO2	C3	Lectures, Notes, Practice Problems	Class Test, Assignment, Midterm, Final Exam

Course Title: Mechanical Engineering Drawing & CAD Laboratory
Course Code: ME 102

Credits: 1 **Class Hours/Week: 2**
Course Type: Core **Pre-requisite:**
CIE Marks: 40 **SEE Marks: 60**

Course Rationale:
This course leads students to an understanding of engineering drawing, an essential means of communication in engineering.

Course Objectives:
The main objectives of this course are:
1. Familiarization and use with drawing instruments and conventions in making engineering drawings.
2. Freehand single view and multi-view sketching.
3. Isometric sketching techniques and multi-view projections.
4. Tolerance and dimensioning and applying them in drawing exercises.
5. Use of AutoCAD (Or Solid Works) in making engineering drawings with dimensions and taking sectional views.
6. Drawing, manipulation and modification of solids using AutoCAD (Or Solid Works) Extraction of multiple views from a solid model in AutoCAD (Or Solid Works).

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Use (C3)** drawing instruments in making an engineering drawing.
CO2 **Describe (C2)** different types of lines, drawing paper sizes, and grades of pencils.
CO3 **Illustrate (C3)** freehand single view, multi-view sketches, and isometric sketches.
CO4 **Use (C3)** AutoCAD (Or Solid Works) to draw plane drawing.
CO5 **Demonstrate (C3)** multi view projections using AutoCAD (Or Solid Works).
CO6 **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√										
CO3	√	√										
CO4	√	√			√							
CO5	√	√			√							
CO6										√		

Course Description:

SL No.	Course Contents	COs
1.	This course contributes towards the engineering topics component of the mechanical engineering curriculum by familiarizing students with the state-	CO1, CO2,

	of-the-art CAD and FEA software for modeling, analyzing and designing mechanical components. This course brings into focus the Introduction, Orthographic drawings, First and third angle projections, scale drawing, sectional view, isometric views, missing line, auxiliary view, detail and assembly drawing, project on engineering drawing and CAD using AutoCAD or contemporary packages instructed by the teachers. The students also understand the requirements for good engineering drawings, and are able to apply these to their work/projects.	CO3, CO4, CO5, CO6

Text Books, Reference Books and Other Resources:

1. AutoCAD – 2002 instant reference, by Goerge Omura & B. Robert Colloria.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2	C3	Demonstration, PPT slides	Lab performance, Quiz, Lab Exam
CO2	PO1, PO2	C2	Demonstration, PPT slides	Lab performance, Quiz, Lab Exam
CO3	PO1, PO2	C3	Demonstration, PPT slides	Lab performance, Quiz, Lab Exam,
CO4	PO1, PO2, PO5	C3	Demonstration, PPT slides	Lab performance, Quiz, Lab Exam,
CO5	PO1, PO2, PO5	C3	Demonstration, PPT slides	Lab performance, Quiz, Lab Exam,
CO6	PO10	A3	Demonstration	Report

Course Title: Engineering Physics I
Course Code: PHY 101

Credits: 3 **Class Hours/Week: 3**
Course Type: Basic Science **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course is intended to build up strong base in mechanics, waves and oscillations, and thermal science as well as to include the experience of applications of its contents in the relevant courses of science/engineering.

Course Objectives:
The objectives of this course are:
1. To enhance the understanding and problem-solving skills on related/necessary basic topics in mechanics, waves and oscillations, and thermal science required for learning the related topics in the courses of undergraduate program in computer science and engineering.
2. To develop the basic concepts of mechanics, waves and oscillations, and thermal science in their relevant scope.
3. To builds up problem solving experience in each relevant topic along with addition of application familiarity in the related areas of science/engineering. Etc.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Identify (C1)** typical simple and moderately complex problems of selected topics and of the linked topics of science/engineering courses of undergraduate programs;

CO2 **Develop (C2)** mathematical expressions of different physical laws of knowledge and skills acquired in the related more advanced courses.

CO3 **Apply (C3)** different physical laws for solving related unusual/advanced problems.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											

Course Description:

SL No.	Course Contents	COs
1.	Mechanics: Kinematics, Graphical representations of displacement-time, velocity-time and acceleration-time.	CO1
2.	Motion in two and three dimensions-projectile motion, Application of Newton's laws of motion; Equilibrium forces. Work-kinetic energy theorem. Power; Conservation of energy, Conservation of linear momentum for a system of particles. Center-of-mass motion. Elastic and inelastic collision in one dimension, Gravitation: Gravitational field, Kepler's laws	CO1, CO2, CO3

3.	Waves & Oscillations: Simple Harmonic motion, damped simple harmonic oscillation, forced oscillations, Combination and Composition of simple harmonic motions, Lissajous figures, Transverse and Longitudinal nature of waves, traveling and standing waves, intensity of waves, Energy calculation of progressive & stationary waves, Phase velocity, group velocity, Velocity of Longitudinal wave in a gaseous medium, Doppler effect.	CO1, CO2, CO3
4.	Thermodynamics: Heat and temperature. Zeroth and 1st law of thermodynamics, Isothermal and adiabatic relations, Second law of thermodynamics. Reversible and irreversible processes, Carnot cycle, Auto cycle, Diesel cycle and their efficiency, Clausius theorem. Entropy. Absolute scale of temperature. Clausius Clapeyron equation. Thermodynamic functions, Maxwell's thermodynamic relations. Problem involving thermodynamic relations and functions. Gibb's phase rule	CO1, CO2, CO3

Text Books, Reference Books and Other Resources:

1. An Introduction to Mechanics by Daniel Kleppner, Robert Kolenkow
2. A text book of Sound by Brij Lal and N. Subrahmanyam
3. A Textbook of Heat by M. N. Saha and B. N. Srivastava
4. Heat and Thermodynamics by Brij Lal and N. Subrahmanyam
5. Physics for Engineers Part 1 by Dr. Giasuddin Ahmed
6. Physics Vol. 1 by Robert Resnick, David Halliday, Kenneth S. Krane
7. Heat and Thermodynamics: An Intermediate Textbook by Richard H.Dittman and Mark W. Zemansky
8. Acoustics & Vibrational Physics by Stephenson

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1	C1	Lecture notes, PPT slides, problem solving, web content	Assignment, Class Test, Midterm, Final exam
CO2	PO1	C2	Lecture notes, PPT slides, problem solving, web content	Assignment, Class Test, Midterm, Final exam
CO3	PO1	C3	Lecture notes, PPT slides, problem solving, web content	Assignment, Class Test, Midterm, Final exam

Course Title: Discrete Mathematics
Course Code: CSE 103

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course will introduce the basic elements of discrete mathematics for an understanding of algorithms and data structures used in computing.

Course Objectives:
The main objectives of this course are:
1. To acquaint students with a good understanding of the concepts and methods of discrete mathematics
2. Helping the students to develop ability in evaluating elementary mathematical arguments and analyzing reasons.
3. To help the students to apply appropriate mathematical concepts and operations to interpret data in problem solving.
4. To Foster the students to enhance logical thinking for demonstrating the basic algorithms.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Demonstrate (C3)** a solid understanding of fundamental concepts in discrete mathematics.
CO2 **Prove (C2)** elementary properties of modular arithmetic and explain their applications in computer science.
CO3 **Apply (C3)** graph theory models of data structures and state machines to solve problems of connectivity and constraint satisfaction.
CO4 **Comprehend (C2**) basic concepts in formal language and computability.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2		√										
CO3		√	√									
CO4	√											

Course Description:

SL No.	Course Contents	COs
1.	Graph theory problem and applications	CO3
2.	Basic number theory and modular arithmetic	CO2

3.	Propositional and Predicate Logic	CO4
4.	Method of Proofs	CO1

Text Books, Reference Books and Other Resources:

1. Discrete Mathematics and Its Applications, Seventh Edition by Kenneith H. Rosen
2. Discrete Mathematics, Seventh Edition by Richard Johnson

Mapping Course Outcomes with the Teaching – Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C3	Lecture, Problem solution, Web link, PPT slides	Assignment, Class Test, Midterm Exam, Final Exam
CO2	PO2	C2	Lecture, Problem solution, Web link, PPT slides	Assignment, Class Test, Midterm Exam, Final Exam
CO3	PO2, PO3	C3	Lecture, Problem solution, PPT slides, Web link	Assignment, Class Test, Midterm Exam, Final Exam
CO4	PO1	C2	Lecture, Problem solution, PPT slides Web link	Assignment, Class Test, Midterm Exam, Final Exam

Course Title: Structured Programming
Course Code: CSE 111

Credits: 2 **Class Hours/Week: 2**
Course Type: Core **Pre-requisite: ICS**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course introduces computer programming and problem solving in a structured program logic environment.

Course Objectives:
The objectives of this course are:

1. To introduce the student with fundamental principles, typical characteristics and mechanism of structured programming.
2. To introduce students with the algorithmic way of thinking and problem solving by computers.
3. To help the student to enhance their analysing skills and use the same for writing program in C.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Comprehend (C2)** the fundamental concepts of structured programming such as variables, data types, operators, and control statements.

CO2 **Apply (C3)** fundamental programming constructs such as control structures, functions, dynamic memory allocation, file management to solve programming problems.

CO3 **Develop (C3)** modular and reusable code by implementing user-defined functions and data types.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√	√										
CO3	√											

Course Description:

SL No.	Course Contents	COs
1.	Basic overview of C tokens, Variable, Variable declaration, Operators, input / output, opening and closing a file, creating a file, processing a file.	CO1
2.	**Control Statement:** Decision Making and Branching (if-else, nested if-else, switch-case, goto)	CO2

3.	**Control Statement:** Looping (for, while, do-while, nested looping), *break* and *continue statement*	CO2
4.	One-dimensional Array, Multidimensional array, String, Array of Strings, user defined function, recursion function, pointers, structure & union.	CO2, CO3

Text Books, Reference Books and Other Resources:

1. Programming in ANSI C. (Latest Edition) ---- (*E. Balagurusamy)*
2. Teach Yourself C –-- (*Herbert Schild*)
3. Programming with C ---- (*Schaum's Outlines*)
4. The C Programming Language ---- (*C Kernighan & D.M. Ritchie*)
5. Theory and Problems of Programming with C ---- (*Byron S. Gottfried*)

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture notes, PPT slides, problem solving	Assignment, Class Test, Midterm Exam, Final Exam
CO2	PO1, PO2	C3	Lecture notes, PPT slides, problem solving	Assignment, Class Test, Midterm Exam, Final Exam
CO3	PO1	C3	Lecture notes, PPT slides, problem solving	Assignment, Class Test, Midterm Exam, Final Exam

Course Title: Structured Programming Laboratory
Course Code: CSE 112

Credits: 2 **Class Hours/Week: 4**
Course Type: Core **Pre-requisite: ICS**
CIE Marks: 40 **SEE Marks: 60**

Course Rationale:
Intended to familiarize with various techniques of programming aimed at developing the skills of analyzing and solving problems by using functions/methods, block structures, control statements, file etc.

Course Objectives:
The objectives of this course are:
1. To introduce the student's various concepts and structures for writing programs.
2. To evaluate the opportunities to develop basic programming skills with respect to program design and development.
3. To develop logical and analytical ability through problem solving.
4. To make the students interested in Competitive Programming.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

- **CO1** **Demonstrate (C2)** a sound understanding of fundamental programming concepts, including variables, data types, control structures, functions, and arrays.
- **CO2** **Apply (C3)** different types of structured programming tools to solve real world problems.
- **CO3** **Debug (C3)** structured programs to identify and correct errors.
- **CO4** **Analyze (C4)** codes to improve readability, maintainability, and efficiency.
- **CO5** **Describe (A3)** the importance of Competitive Programming like IUPC, NCPC, ICPC etc.
- **CO6** **Report (A3)** lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√				√							
CO3					√							
CO4		√										
CO5	√											√
CO6										√		

Course Description:

SL No.	Course Contents	COs
1.	Compiler/IDE setup, Basic input/output, file-handling, C program without using conditional and loop statements	CO1, CO6
2.	Problem solving using Control Structures and loops (if-else, nested if-else, switch-case, for loop, nested for loops, while loops, do-while loops), one dimensional array and multi-dimensional array, string processing, user defined functions, structure and unions.	CO1, CO2, CO6
3.	Where Do Errors Come from in Coding? Common programming errors and how to handle them. Error handling, Debugging and Testing	CO3, CO6
4.	Reusability, Maintainability, Scalability and performance	CO4, CO6
5.	Introduction to competitive programming (NCPC, ICPC, Online Judge, Contest criteria etc.), problem solving in different online platforms.	CO5, CO6

Text Books, Reference Books and Other Resources:

1. Online references
2. Programming in ANSI C. (Latest edition) ---- (E. Balagurusamy)
3. Schaum's Outline of Programming with C (Schaum's Outline Series) ---- (Byron Gottfried, Latest Edition)

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Demonstration, Discussion, Experiment	Lab Performance, Lab Exam, Quiz
CO2	PO1, PO5	C3	Demonstration, Discussion, Experiment	Lab Performance, Lab Exam, Quiz
CO3	PO5	C3	Demonstration, Discussion, Experiment	Lab Performance, Lab Exam, Quiz
CO4	PO2	C4	Demonstration, Discussion, Experiment	Lab Performance, Lab Exam, Quiz
CO5	PO1, PO12	A3	Demonstration	Programming Contest Problem Solving
CO6	PO10	A3	Demonstration	Report

Course Title: Electronics I
Course Code: EEE 211

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: ECKT I**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course is focused on beginners to provide foundation knowledge of electronics devices that are used in modern day electronic system.

Course Objectives:
The objectives of this course are -
1. To understand the basic knowledge of electronics and devices.
2. To provide foundation knowledge to solve unknown parameters of electronic circuits
3. and devices.
4. To develop skills to design electronic circuits/networks.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Explain (C2)** the knowledge about electronic devices and parameters.
CO2 **Interpret (C2)** of electronics parameters that are used to design electronics devices.
CO3 **Apply (C3)** the problems related to electronic systems and designs.
CO4 **Illustrates (C3)** electronics-related circuitry and its interfacing.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											
CO4	√											

Course Description:

SL No.	Course Contents	COs
1.	**P-N junction as a circuit element:** Intrinsic and extrinsic semiconductors, operational principle of p-n junction, contact potential, current-voltage characteristics of a diode, simplified dc and ac diode models, dynamic resistance and capacitance.	CO1, CO2, CO3
2.	**Diode circuits:** Half wave and full wave rectifiers, rectifiers with filter capacitor, characteristics of a zener diode, zener shunt regulator, clamping and clipping circuits.	CO1-CO4

3.	**Bipolar junction transistor:** Current components, BJT characteristics and regions of operation, BJT as an amplifier, biasing the BJT for discrete circuits, small signal equivalent circuit models, BJT as a switch. Single stage mid-band frequency BJT amplifier circuits and different type of biasing techniques. Voltage and current gain, input and output impedance of common base, common emitter and common collector amplifier circuits.	CO1, CO2, CO4
4.	**Metal-oxide-semiconductor field-effect-transistor (MOSFET):** Structure and physical operation of an enhancement MOSFET, threshold voltage, Body effect, current- voltage characteristics of an enhancement MOSFET biasing discrete and integrated MOS amplifier circuits, Current Mirror: Simple CMOS current mirror, Source degenerated current mirror, high output impedance current mirror, Bipolar current mirror; Single-stage MOS amplifiers, MOSFET as a switch, CMOS inverter.	CO1-CO4
5.	**Junction field-effect-transistor (JFET):** Structure and physical operation of JFET, transistor characteristics, pinch-off voltage. Differential and multistage amplifiers, Description of differential amplifiers, small-signal operation, differential and common mode gains, RC coupled mid-band frequency amplifier.	CO1-CO4
6.	**Active filters:** Basics of Op-Amp, its characteristics, Different types of Op-Amp	CO1, CO3, CO4

Text Books, Reference Books and Other Resources:

1. Electronics Devices and Circuit Theory by Robert L. Boylestad & Louis Nashelsky.
2. Electronics Devices by Thomas L. Floyd.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture, Problem solving, PPT slide	Assignment, Class Test, Midterm Exam, Final Exam
CO2	PO1	C2	Lecture, Problem solving, PPT slide	Assignment, Class Test, Midterm Exam, Final Exam
CO3	PO1	C3	Lecture, Problem solving, PPT slide	Assignment, Class Test, Midterm Exam, Final Exam
CO4	PO1	C3	Lecture, Problem solving, PPT slide	Assignment, Class Test, Midterm Exam, Final Exam

Course Title: Electronics I Laboratory
Course Code: EEE 212

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: ECKTL**
CIE Marks: 70 **SEE Marks: 30**

Course Rationale:
This laboratory course is focused on the practical aspects of the course Electronics I course.

Course Objectives:
The objectives of this course are:
1. Familiarize with the different component of electronics system such as diode, transistor
2. To provide adequate knowledge on electronic devices.
3. To make students learn how to use the measurement equipment for electronics system.
4. Test the physical characteristics of different component in lab.
5. Develop some practical circuits that describe the basic phenomenon of electronics.
6. Provide design concept of complex electronics system.
7. Introduce with the design issue of the electronic system.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Demonstrate (C2, P2)** different electronic circuits and power supply as well as the BJT amplifier/JFET amplifier, and their operation.
CO2 **Conclude (C5)** and **show (P3)** the result from experimental data.
CO3 **Report (A3)** lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2				√								
CO3										√		

Course Description:

SL No.	Course Contents	COs
1.	Familiarization with the components & devices used in Electronics Lab	CO1
2.	Find out the v-i characteristics of a semiconductor diode under forward & reverse biased condition	CO1, CO2, CO3
3.	Design a Half wave rectifier	CO1, CO2, CO3
4.	Design a Full wave rectifier using center taped transformer.	CO1, CO2, CO3

5.	Design a Full wave bridge rectifier.	CO1, CO2, CO3
6.	Find out the transistor characteristics curve for common emitter configuration.	CO1, CO2, CO3
7.	Find out the transistor characteristics curve for IB vs VBE.	CO1, CO2, CO3
8.	Find out the transistor characteristics curve for common base configuration.	CO1, CO2, CO3
9.	Single stage Common emitter configured NPN transistor amplifier.	CO1, CO2, CO3
10.	Observation of Common collector configured NPN transistor amplifier	CO1, CO2, CO3
11.	Observation of FET performance	CO1, CO2, CO3

Text Books, Reference Books and Other Resources:

Lab Manual

CO Delivery and Assessment:

COs	POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2, P2	Lecture & Laboratory Experiments	Quiz, Performance Test, Report
CO2	PO4	C5, P3	Lecture & Laboratory Experiments	Performance Test, Report
CO3	PO10	A3	Demonstration	Report

Course Title: Developing English Skills
Course Code: ENG 103

Credits: 2 **Class Hours/Week: 2**
Course Type: Language **Pre-requisite: GE**
CIE Marks: 100 **SEE Marks:**

Course Rationale:
This is an advanced course for students to learn reading, writing, listening and speaking skills.

Course Objectives:
The objectives of this course are to:
1. Develop a student's writing skill, listening skill.
2. Improve the ability to note taking.
3. Grow students speaking English through group discussion.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Demonstrate (C2) and practice (A2)** English listening and speaking effectively.
CO2 **Demonstrate (C2) and practice (A2)** English writing and reading skills.
CO3 **Communicate (C3) and work (A2**) collaboratively with others.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3									√	√		

Course Description:

SL No	Course Contents	COs
1.	**Developing Writing Skill:** Letter Writing: formal and informal, Report writing; Business communication and tenders, business letters, letters of opinion, application and CV writing, e-mail, memo, etc.	CO1
2.	**Listening Skill and Note Taking:** Listening to recorded texts and class lectures and learning to take useful notes based on listening.	CO2
3.	**Developing Speaking Skill:** Oral skills including communicative expressions for personal identification, life at home, giving advice and opinion, instruction and directions, requests, complaints, apologies, describing people and places, narrating events.	CO3

4.	**Discussion:** A group of students to be brought on the dais at a time. Other students of the class will be interrogating and likewise every student should be brought in turn and questions should be asked from the fields of literature, science, current politics, international affairs, games and sports, etc. The Instructor will act as a conductor.	CO3

Text Books, Reference Books and Other Resources:

1. English Grammar in Use by Raymond Murphy.
2. Cambridge English advanced vocabulary
3. Cambridge English for IELTS

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1	C2, A2	Lecture notes, PPT slides, web content	Class Test, Assignment, Midterm, Final Exam
CO2	PO1	C2, A2	Lecture notes, PPT slides, web content	Class Test, Midterm, Assignment, Final Exam
CO3	PO9, PO10	C3, A2	Lecture notes, PPT slides, web content	Class Test, Assignment, Midterm, Assignment, Final Exam

Course Title: Engineering Mathematics II
Course Code: MAT 107

Credits: 3 **Class Hours/Week: 3**
Course Type: Mathematics **Pre-requisite: EM I**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course aims to build up basics in integral calculus & linear differential equations in the context of engineering applications.

Course Objectives:
The objectives of this course are:
1. To help the students to determine length, area & volumes using integration.
2. To develop the ability to solve the differential equations in the context of engineering problems.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Determine (C2)** indefinite and definite integral of algebraic, exponential, logarithmic, inverse functions.

CO2 **Apply (C3)** integration to calculate Arc, Areas of regions in a plane, Volumes of solids, surface area of solid revolution.

CO3 **Identify (C2)** order and degree of the differential equation. Distinguish between linear, non- linear, homogenous and Exact differential equation also solve 1st order and 1st degree differential equation.

Mapping of Course Outcomes to Program Outcomes:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											

Course Description:

SL No.	Course Contents	COs
1.	Integration of various types of functions, Substitution method, Integration by Parts, Properties of definite integral, Beta and Gamma function	CO1
2.	Integration of various types of functions, Substitution method Integration by Parts, Properties of definite integral, Beta and Gamma function, Rectification, Area of a surface, Volume of solids of revolution	CO2

3.	Integration of various types of functions, Substitution method, Integration by Parts, Properties of definite integral, Beta and Gamma function, Rectification Area of a surface, Volume of solids of revolution, Order and Degree of differential equation, Variable separable, Homogenous differential equation, Linear differential equation Exact differential equation solution of linear differential equations with constant coefficients series solution of differential equations	CO3

Text Books, Reference Books and Other Resources:

1. Calculus & Analytic Geometry: by George B Thomas, Jr.
2. Differential Equations: by Erinu Kresgige
3. Advanced Engineering Mathematics: by Peter V. O'Neil
4. An Introduction to Differential Equation and Tensor Analysis: by Syed Md. Farid Ordinary and Partial Differential Equations: by M D. Raisinghania

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1	C2	Lecture, Notes, Problem solution	Assignment, Class Test, Midterm Exam, Final Exam
CO2	PO1	C3	Lectures, Notes, Practice Problems	Assignment, Class Test, Midterm Exam, Final Exam
CO3	PO1	C2	Lectures, Notes, Practice Problems	Assignment, Class Test, Midterm Exam, Final Exam

Course Title: Engineering Physics II
Course Code: PHY 103

Credits: 3 **Class Hours/Week: 3**
Course Type: Basic Science **Pre-requisite: EP I**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course is intended to build up strong base in electromagnetism, optics, and modern physics, as well as to include the experience of applications of its contents in the relevant courses of science/engineering.

Course Objectives:
1. This course enhances the understanding and problem-solving skills on related/necessary basic topics in electromagnetism, optics and modern physics required for learning the related topics in the courses of undergraduate program in computer science and engineering.
2. Toward the above goal, it develops the basic concepts of electromagnetism, optics, and modern physics in their relevant scope.
3. While doing the above it builds up problem solving experience in each relevant topic along with addition of application familiarity in the related areas of science/engineering. Etc.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Identify (C1)** typical simple and moderately complex problems of selected topics and of the linked topics of science/engineering courses of undergraduate programs;
CO2 **Develop (C2)** mathematical expressions of different physical laws of knowledge and skills acquired in the related more advanced courses.
CO3 **Apply (C3)** different physical laws for solving related unusual/advanced problems.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											

Course Description:

SL No.	Course Contents	COs
1.	**Electricity and Magnetism:** Electromagnetism: Magnetic fields, Maxwell's equations, Ampere's law, Faraday's law, Lenz's law. Inductance: Self mutual inductance.	CO1, CO2, CO3

2.	Magnetic properties of matter: Magnetomotive force, magnetic field intensity, permeability and susceptibility	CO1, CO2, CO3
3.	classification of magnetic materials, magnetization curve of ferromagnetic materials, magnetic circuits, magnetostriction;	CO1, CO2, CO3
4.	**Optics:** Theories of light; Huygen's principles and constructions;	CO1, CO2, CO3
5.	**Interference of light:** Young's double slit experiment, Fresnel bi-prism, Newton's ring, interferometers	CO1, CO2, CO3
6.	**Diffraction of light:** Diffraction, Fresnel Fraunhoffer diffraction, Diffraction by single slit, Diffraction by double slit,	CO1, CO2, CO3
7.	**Polarization of light:** Polarization of electromagnetic waves, production and analysis of polarized light, optical activity, Optics of crystals;	CO1, CO2, CO3
8.	**Lasers and their Applications:** Laser introduction, Stimulated and spontaneous radiation's coherence, Resonators Ruby, and other laser. Material processing with shaping	CO1, CO2, CO3
9.	**Modern physics:** Relativity, Michelson-Morley experiment, Lorenz-Einstein transformation, Mass energy relation, Quantum effect, Photoelectric effect, Compton Effect;	CO1, CO2, CO3
10.	**Atomic Physics:** De-Broglie wave, correspondence principles, uncertainty principle, The Rutherford-Bohr model of the atom, Energy levels and spectra,	CO1, CO2, CO3
11.	The Zeeman effect, Electron spin, Many electron atoms and the exclusion principle, vector atom model;	CO1, CO2, CO3
12.	**Nuclear Physics:** Introduction, Nuclear constituents, Nuclear binding and nuclear structure, Radioactivity, Radioactive decay, Half-life,	CO1, CO2, CO3
13.	Law of successive disintegration, Radioactive equilibrium, The nucleus, Properties of nucleus, Binding energy, Nuclear reactions, Nuclear fission and fusion, nuclear reactors.	CO1, CO2, CO3

Text Books, Reference Books and Other Resources:

Text books:

1. M. S. Huq, A. K. Rafiqullah, A. K. Roy — Concept of Electricity & Magnetism
2. Dr. Giasuddin Ahmed — Physics for Engineers part 2
3. Kailash K. Sharma — Optics: Principles and Applications
4. S. N. Ghoshal — Nuclear Physics

References:

1. K. K. Tewari — Electricity and Magnetism with Electronics
2. D. N. Vasudeva — Fundamentals of Magnetism and Electricity
3. Francis A. Jenkins and Harvey E. White — Fundamentals of Optics
4. Floyd Karker Richtmyer, Earle Hesse Kennard, John N. Cooper — Introduction to Modern Physics
5. Irving Kaplan — Nuclear Physics

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C1	Lecture notes, PPT slides, problem solving, web content	Class Test, Assignment, Midterm, Final Exam
CO2	PO1	C2	Lecture notes, PPT slides, problem solving, web content	Class Test, Assignment, Midterm, Final Exam
CO3	PO1	C3	Lecture notes, PPT slides, problem solving, web content	Class Test, Assignment, Midterm, Final Exam

Course Title: Object Oriented Programming
Course Code: CSE 211

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: SP**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
To empower the learner to perceive the fundamental knowledge of object-oriented programming paradigm and aimed at developing the skills of analyzing and solving real-world problems.

Course Objectives:
The objectives of this course are to:
1. Build an understanding of the object-oriented approach in programming.
2. Familiarize the student with the writing of computer programs to solve real-world problems in Java.
3. To make students learn and appreciate the importance and merits of proper comments in source code and API documentation.
4. Be able to write GUI interfaces for a computer program to interact with users, and to understand the event-based GUI handling principles.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Explain (C2)** the fundamental OOP concepts (Classes, Operators, Variables, Keywords, Objects, Methods, Constructors, and Packages).

CO2 **Illustrate (C3)** different object-oriented principles (Encapsulation, Polymorphism, Abstraction and Inheritance) to solve engineering problems.

CO3 **Demonstrate (C3)** object-oriented features: exception handling, multi-threading, generics, collection framework, file handling, and GUI.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓											
CO2	✓	✓										
CO3	✓	✓										

Course Description:

SL No.	Course Contents	COs
1.	**Introduction:** Basic concepts on object-oriented programming (OOP), benefits and application areas of Object-Oriented Programming, procedural vs. OOP programming, Comparison with OOP and other languages paradigm, and important features of OOP	CO1
2.	**Class, objects, and constructor:** Introducing classes, objects and constructors (class fundamentals, declaring objects, assigning object reference variables)	CO1

3.	**Method basics and method overloading:** Introducing Methods, this Keyword, Garbage Collection, the finalize () Method, Overloading Methods, using objects as parameters, returning objects, constructor, and constructor overloading.	CO1, CO2,
4.	**Inheritance:** Method overriding, Inheritance, different types of inheritance	CO2
5.	**Miscellaneous class:** Understanding Static, Introducing Super, Final, Nested, abstract, wrapper, and Inner class	CO1, CO2
6.	**Interface:** Basics of interface, class vs. interface, multiple inheritances, defining an interface, dynamic initialization	CO2
7.	**Multithreaded Programming:** Basics of thread, differences between multithreading and multitasking, the concept of multithreaded programming, and different ways to create a new thread.	CO3
8.	**Multithreaded Programming:** Implementing the Runnable interface, the life cycle of a thread, deadlock, synchronization, and set thread's priority.	CO3
9.	**Managing Errors and Exceptions:** Explain different types of errors, Errors vs. Exceptions, and common Java exceptions with the cause of occurrence. Exception handling mechanism, finally blocks, illustrating the usage of throw and throws, user-defined exception.	CO3
10.	**Generics:** Basic concepts of generics in Java, advantages and use cases.	CO3
11.	**Java Graphics:** Enumerate the basic concepts of the Swing package.	CO3
12.	**Event Handling and Layout Managers:** AWT (Abstract Window Toolkit),	CO3
13.	**File Handling (Input/Output):** Stream in Java, the hierarchy of java stream classes, and purposes of the various stream functions. Stream Tokenizer, random and sequential access file, basic file-related programs to check the I/O. Basic file-related programs to check the I/O.	CO3
14.	**Java Collections Framework:** Java Map methods e.g., tree map, hash map. relevant programs. Java collections framework and iterator. relevant programs. List, set, and relevant programs.	CO3

Textbooks, Reference Books and Other Resources:

1. Programming with Java A Primer--- (E. Balagurusamy)
2. Java the Complete Reference --- (Herbert Schildt)
3. Java How to Program--- (Deitel and Deitel)

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam

CO2	PO1, PO2	C3	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam
CO3	PO1, PO2	C3	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam

Course Title: Object Oriented Programming Laboratory
Course Code: CSE 212

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: SPL**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
To empower the learner to perceive the fundamental knowledge of object-oriented programming paradigm and aimed at developing the skills of analyzing and solving real-world problems.

Course Objectives:
The objectives of this course are to:
1. Build an understanding of the object-oriented approach in programming.
2. Familiarize the student with the writing of computer programs to solve real-world problems in Java.
3. To make students learn and appreciate the importance and merits of proper comments in source code and API documentation.
4. Be able to write GUI interfaces for a computer program to interact with users, and to understand the event-based GUI handling principles.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Demonstrate (C3)** the fundamental OOP concepts (Classes, Objects and Packages) using Object Oriented Language.
CO2 **Implement (C3)** different object-oriented principles (Encapsulation, Polymorphism, Abstraction and Inheritance) to solve engineering problems.
CO3 **Implement (C3)** object-oriented features: exception handling, multi-threading, generics.
CO4 **Implement (C3)** GUI features to develop lightweight application based on constraints.
CO5 **Report (A3)** lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓											
CO2	✓	✓										
CO3	✓											
CO4	✓	✓										
CO5										✓		

Course Description:

SL No.	Course Contents	COs
1.	**Introduction:** Basics of Java programming language, equation solving, conditional statement-related problem solving, and loop	CO1, CO5

2.	**Class, objects, and constructor:** Programs related to Class, Objects and Constructor	CO1, CO5
3.	Method overloading, Method Overriding, Inheritance, Abstraction, Polymorphism, Encapsulation	CO2, CO5
4.	**Multithreaded Programming:** Programs related to multithreading, creating a new thread, implementing the Runnable interface, synchronization, and setting the thread's priority.	CO3, CO5
5.	**Managing Errors and Exceptions:** Programs related to errors and exceptions, common Java exceptions, exception handling mechanism, and finally block, illustrating the usage of throw and throws, user-defined exception.	CO3, CO5
6.	**Applet:** HTML and applet program integration, graphics programming with the applet, repaint () method, Swing package, Event Handling and Layout Managers	CO4, CO5

Text Books, Reference Books and Other Resources:

1. Programming with Java A Primer--- (E. Balagurusamy)
2. Java the Complete Reference --- (Herbert Schildt)
3. Java How to Program--- (Deitel and Deitel)

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C3	Demonstration, discussion, experiment	Quiz/Written Exam, Performance, Report
CO2	PO1, PO2	C3	Demonstration, discussion, experiment	Quiz/Written Exam, Performance, Report
CO3	PO1	C3	Demonstration, discussion, experiment	Quiz/Written Exam, Performance, Report
CO4	PO1, PO2	C3	Demonstration, discussion, experiment	Quiz/Written Exam, Performance, Report
CO5	PO10	A3	Demonstration	Report

Course Title: Data Structures
Course Code: CSE 221

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: SP**
CIE Marks: 60 **SEE Marks: 40**

Rationale:
Data structure provides a good understanding for organizing and storing data in a computer such that it can be stored, retrieved, and updated frequently.

Course Objectives:
The main objectives of this course are to:
1. Make the students understand the data structures and their implementations.
2. Facilitate necessary knowledge about the importance of data structures in the context of writing efficient programs.
3. Develop the ability to implement appropriate data structures in problem solving.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Interpret (C2**) the basic concepts of data structures, their types and basic operations.
CO2 **Summarize (C2)** the strength and weaknesses of different data structures.
CO3 **Use (C3)** the appropriate data structures in the context of the solution to any given problem.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓											
CO2		✓										
CO3	✓	✓										

Course Description:

SL No.	Course Content	COs
1.	Introduction to Data Structures, Basic Operations, and Performance Measurement.	CO1
2.	Introduction to Array, Search, and Sorting Strings: Pattern Matching Algorithm (Naive and KMP)	CO1, CO2, CO3
3.	Types of Linked List and Basic Operations on Linked List	CO1, CO2, CO3
4.	Basic Operations on Stack and Use of Recursion Types of Queue and Basic Operations on Queue	CO1, CO2, CO3

5.	Types of Trees, Representation of Tree, Binary Search Tree, Balanced Search Tree, Efficient Heap	CO1, CO2, CO3
6.	Graph Representation, Traversing Graph, Topological Sorting, Path Matrix, Warshall Algorithm	CO1, CO2, CO3
7.	Hash Function and its Application.	CO1, CO2

Text Books, Reference Books and Other Resources:

1. Seymour Lipschutz, Data Structures, McGraw-Hill, 2014.
2. Debasis Samanta, Classic Data Structures, Prentice-Hall of India Private Limited, 2009.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's taxonomy domain/level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery methods and activities**	**Assessment tools**
CO1	PO1	C2	Lecture, Web link, PPT slide	Assignment, Class Test, Midterm examination, Final
CO2	PO2	C2	Lecture, Web link, PPT slide	Assignment, Class Test, Midterm examination, Final Examination
CO3	PO1, PO2	C3	Lecture, Web link, PPT slide	Assignment, Class Test, Midterm examination, Final Examination

Course Title: Data Structures Laboratory
Course Code: CSE 222

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: SPL**
CIE Marks: 40 **SEE Marks: 60**

Rationale:
To practically implement as well analyze the various data structures and basic algorithm analysis.

Course Objectives:
The main objectives of this course are to:
1. Help the students to understand the analysis and design of linear and non-linear data structures in the context of writing efficient programs.
2. Develop the ability in identifying and applying the suitable data structure for the given real- world problem.

Course Outcomes (COs):

CO1 **Describe (C2)** the operations of basic data structures.
CO2 **Classify (C2)** the appropriate data structure to solve specific problems.
CO3 **Solve (C3)** problems or improve existing code using learned data structures.
CO4 **Report (A3)** lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓											
CO2		✓										
CO3		✓										
CO4										✓		

Course Description:

SL No.	**Course Contents**	**COs**
1.	Identify and apply the basic operations of data structures: Array, Linked List, Stack, Queue, Tree, Graph.	CO1, CO4
2.	Identify the basic data structures and apply solutions: Bubble Sort, Selection Sort, Insertion Sort, Linear Search, Binary Search, Pattern matching: Naïve, KMP, Binary Search Tree, Heap Tree (Using Priority Queue), Breadth-first Search, Depth-first Search	CO2, CO3, CO4

Text Books, Reference Books and Other Resources:

1. Data Structures (Third Edition), by Seymour Lipschutz, Publishers: McGraw-Hill.
2. Classic Data Structures, by D. Samanta, Publishers: Prentic-Hall of India Private Limited.
3. Data Structures and Algorithms in Python, by Michael T. Goodrich, publisher: VP & PUBLISHER

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture, Practical Implementation	Lab Performance, Quiz, Report, Lab exam
CO2	PO2	C2	Lecture, Practical Implementation	Lab Performance, Quiz, Report, Lab exam
CO3	PO2	C3	Lecture, Practical Implementation	Lab Performance, Quiz, Report, Lab exam
CO4	PO10	A3	Demonstration	Report

Course Title: Basic Economics
Course Code: ECO 201

Credits: 3 **Class Hours/Week: 3**
Course Type: Non-engineering **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course is a foundation course that will prepare students to understand fundamental microeconomic and macroeconomic concepts. The goal of this course is to provide some indispensable lessons about individual decision making, the role of the market as well as government in an economy.

Course Objectives:
The main objectives of this course are:
1. Edify the concept of fundamental demand and supply mechanism in relation to individual decision making.
2. Figure out the measurement of individual satisfaction and find out basic economic problems of society as well as make clear the basic ideas on firm and production.
3. Understand and verify the objectives, goals, tools, and problems of macroeconomic concepts and public policies.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:
- **CO1** **Explain (C2)** Demand and supply concept as an apparatus of individual decision making and express individual satisfaction level numerically, graphically.
- **CO2** **Describe (C2)** the acquired concepts to understand the nature of firm and production.
- **CO3** **Summarize (C2)** the advantages and disadvantages of different macroeconomic situations and identify required policies.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											

Course Description:

SL No.	Course Contents	COs
1.	The Economic Problem, The Language of economics, Economic System, How the Economy works, Demand and Supply Theory, Elasticity, Consumer behavior and market demand, Production and Costs Theory, The perfectly competitive market and its characteristics, Imperfectly competitive markets, Basic principles of factor pricing, Resource allocation and the market, International trade, Non – Renewable resources, Key macroeconomics phenomena, Income determination, Balance of payments and the exchange rates	CO1, CO2, CO3

Text Books, Reference Books and Other Resources:

1. N. Gregory Mankiw: Principles of Economics; 6th edition
2. W.J. Baumol and A.S. Blinder: Economics: Principles and Policies
3. Robin Bade and Michael Parkin: Foundations of Microeconomics, 8th Edition
4. S. Fisher, R. Dornbusch and Schmalensee: Economics, McGraw Hill Company, New York
5. Modern Microeconomics: Theory and Application. H.L. Ahuja and S. Chand. Revised Edition.

CO Delivery and Assessment:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO2	PO1	C2	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO3	PO1	C2	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam

Course Title: Digital Electronics
Course Code: EEE 311

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course aims to empower the learner to inspect and perceive the fundamental knowledge of digital electronics and develop sufficient background for advanced digital electronic courses.

Course Objectives:
The objectives of this course are to:
1. Provide a modern introduction to logic design and the basic building blocks used in technical areas in particular digital computers.
2. Familiarize the student with fundamental principles of 21st century's digital design.
3. Provide coverage of classical hardware design for both combinational and sequential logic circuits.
4. Develop research skills and aspects of professional practice through group-bases assignments and team works that leads them to become successful self-employed, motivator, entrepreneur and manager.

Course Outcomes (COs):

Upon successful completion of this course, students will be able to:

CO1 **Describe** (**C2**) the fundamental concepts, theories and techniques used in digital electronics.
CO2 **Interpret** (**C2**) various combinational and sequential circuits.
CO3 **Implement (C3)** descriptions of logical problems to efficient digital logic circuits.
CO4 **Demonstrate** (**C3**) memory, interfacing and converter.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√	√										
CO4	√	√										

Course Description:

SL No.	Course Contents	COs
1.	**Number systems:** Representation of numbers in different bases, Addition and subtraction in different bases, complement: Subtraction using complements, Binary multiplication & division.	CO1

2.	**Binary codes:** Different coding system, Boolean algebra, various gates, Sum of products and product of sums, Standard and canonical forms and other logical operations.	CO1
3.	**Simplification of Boolean functions:** Karnaugh map method, Tabular method of simplification; Implementation of logic circuit using various gates, Universal gates.	CO2
4.	**Combinational logic circuit:** Design procedure: Adder, Subtractor, Code converters, Parity bit checker and magnitude comparator, Analysis of different combinational circuits, Encoder, decoder, Multiplexer, Demultiplexer, ROM, PLA and their applications.	CO3
5.	**Flip-flops:** SR, JK, Master slave, T and D type flip-flops and their characteristic tables & equations; Triggering of flip-flops, Flipflop, Excitation table.	CO3
6.	**Sequential circuits:** Introduction to sequential circuits, Analysis and synthesis of synchronous and asynchronous sequential circuits.	CO3
7.	**Counters:** Classifications, Synchronous and asynchronous counter design and analysis, Ring counter, Johnson counters, Ripple counter and counter with parallel load.	CO3
8.	**Registers:** Classification, Shift registers, Circular registers and their applications and registers with parallel load. Basic Concept of Application Specific IC (ASIC) design.	CO3
9.	**Digital IC logic families:** Brief description of TTL, DTL, RTL, ECL, I2L, MOS and CMOS logic and their characteristics, principles of operation and application.	CO2
10.	**Digital IC logic families:** Brief description of TTL, DTL, RTL, ECL, I2L, MOS and CMOS logic and their characteristics, principles of operation and application.	CO2
11.	**Memory Units:** Various memory devices and their interfacing.	CO4
12.	**Converters:** Digital to Analog (D/A), Analog to Digital (A/D) converters, and their applications.	CO4

Text Books, Reference Books and Other Resources:

1. Digital Logic and Computer Design by M. Morris Mano
2. Modern Digital Electronics, by R.P. Jain.
3. Digital Electronics, by Tocci.
4. Computer Engineering by M. M. Mano,
5. Digital Computer Electronics by A.P. Malvino.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lectures, Notes, Practice Problems	Assignment, Class Test, Mid Term, Final Examination
CO2	PO1	C2	Lectures, Notes, Practice Problems	Assignment, Class Test, Mid Term, Final Examination
CO3	PO1, PO2	C3	Lectures, Notes, Practice Problems	Assignment, Class Test, Mid Term, Final Examination
CO4	PO1, PO2	C3	Lectures, Notes, Practice Problems	Assignment, Class Test, Mid Term, Final Examination

Course Title: Digital Electronics Laboratory
Course Code: EEE 312

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite:**
CIE Marks: 70 **SEE Marks: 30**

Course Rationale:
This subject aims to get acquainted with integrated circuits, to design and implement digital circuits, and to be able to troubleshoot implemented circuits.

Course Objectives:
The objective of the course is to enable the students to:

1. Make students understand the operations and applications of various digital electronic circuits.
2. Help students develop the analyzing skills of digital electronic circuits.
3. Help students develop the ability to present the results of investigations orally and in writing.

Course Outcomes (COs):

Upon successful completion of this course, students will be able to

CO1 **Demonstrate (C3, P2)** different logic gate circuits as well as the multiplexer/demultiplexer and counter operation.

CO2 **Conclude** (**C4**) and **show (P3)** the result from experimental data.

CO3 **Report (A3)** lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2		√										
CO3										√		

Course Description:

SL No.	Course Contents	COs
1.	Familiarization of Digital Electronics Laboratory	CO1
2.	Familiarization of the Digital Logic gates ICs, truth tables & verification of Fundamental logic gates.	CO1, CO2, CO3
3.	Boolean function implementation.	CO1, CO2, CO3
4.	Verification of universality of NAND & NOR gates	CO1, CO2, CO3
5.	Verification of Half adder & Full adder circuit.	CO1, CO2, CO3
6.	Verification of Half subtractor & Full subtractor circuit.	CO1, CO2, CO3
7.	Implementation of 4-Bit Parallel Adder Using 7483	CO1, CO2, CO3
8.	Verification of basic operation of a Multiplexer and Demultiplexer.	CO1, CO2, CO3

9.	Design and verify the 4-bit Synchronous counter.	CO1, CO2, CO3
10.	Design and verify the 4-bit Asynchronous counter.	CO1, CO2, CO3
11.	Familiarization of different types of flip-flops.	CO1, CO2, CO3

Text and Reference books:

1. Digital Logic and Computer Design by M. Morris Mano
2. Lab Manual

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's taxonomy domain/level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery methods and activities**	**Assessment tools**
CO1	PO1	C3, P2	Demonstration, Discussion, Experiment	Quiz/Written Exam, Performance, Report
CO2	PO2	C4, P3	Demonstration, Discussion, Experiment	Quiz/Written Exam, Performance, Report
CO3	PO10	A3	Demonstration,	Report

Course Title: Engineering Mathematics III
Course Code: MAT 201

Credits: 3 **Class Hours/Week: 3**
Course Type: Mathematics **Pre-requisite: EM II**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course is intended to build up strong bases in the complex number system and its relevant calculus, in vector calculus, and in statistics and probability, as well as to include the experience of applications of its contents in the relevant courses of science/engineering.

Course Objectives:
The objectives of this course are to:
1. Enhance and broaden the contents covered in pre-requisite math courses, and builds up a moderately strong base in statistics and probability.
2. This course also highlights the basic concept of vector calculus and Statistics (Probability, Distribution, Sampling, etc.).
3. While doing the above it builds up problem solving experience in each relevant topic along with the addition of application familiarity in the related areas of science/engineering.

Course Outcomes (COs):

Upon successful completion of this course, students will be able to
- **CO1** **Demonstrate (C2)** complex analysis techniques for geometric interpretation.
- **CO2** **Calculate (C3)** vector field parameters using vector integration.
- **CO3** **Comprehend (C2)** the formulation of statistical data from various distributions.
- **CO4** **Analyze (C4)** real-world problems using various probability models.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											
CO4	√											

Course Description:

SL No.	Course Contents	COs
1.	Complex Number System, Analytic Function, Harmonic Function Cauchy-Riemann theorem, function, exercise Construction of analytic functions, Cauchy's theorem, Cauchy's Integral formula, exercises Cauchy's Residue theorem, Exercises on Series Theorem.	CO1
2.	Vector Differentiation, Del operator, gradient, divergence, curl, Laplacian operator, Line, Surface and Volume integral, Exercises, Gauss's Theorem, exercises, Green's Theorem, exercises, Stoke's theorem, exercises	CO2

3.	Basic concepts of frequency distribution, Measures of location and variation Permutation, Combination	CO3
4.	Probability Distribution, Binomial, Poisson's Distribution	CO4

Text Books, Reference Books and Other Resources:

1. A Text Book on Co-ordinate Geometry with Vector Analysis: By A. F. M. Abdur Rahman and P. K. Bhattacharjee
2. Advanced Engineering Mathematics: by Peter V. O'Neil
3. Basic Statistics & Probability: by S. K. Gupta
4. An Introduction to Statistics and Probability: by M. Nurul Islam
5. Theory and Problems of Complex Variables: by Murray R. Spiegel

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1	C2	Lecture, Problem solution	Assignment, Class Test, Mid Term, Final
CO2	PO1	C3	Lecture notes, Problem solution	Assignment, Class Test, Mid Term, Final
CO3	PO1	C2	Lectures, Notes, Practice Problems	Assignment, Class Test, Mid Term, Final
CO4	PO1	C4	Lectures, Notes, Practice Problems	Assignment, Class Test, Mid Term, Final

Course Title: Algorithm Design and Analysis
Course Code: CSE 225

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: DS**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
To practically implement basic algorithms, as well as to analyze the runtime and memory use.

Course Objectives:
The main objectives of this course are:
1. General Understanding of Basic Algorithms.
2. Developing Programming Skills for Solving Real Time Problem using Algorithm
3. Conduct experiments to get a sense of the time and storage (memory) efficiency of the program codes.
4. Demonstrate advantages and disadvantages of specific algorithms.

Course Outcomes:

Upon successful completion of the course the students will be able to:

CO1 **Interpret (C2)** the fundamental concepts and design paradigms of algorithms.

CO2 **Calculate (C2)** the time and space complexities of various algorithms.

CO3 **Implement (C3)** efficient algorithms in simple engineering design situations.

Mapping of Course Outcomes to Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓											
CO2	✓	✓										
CO3	✓	✓										

Course Description:

SL No.	Course Contents	COs
1.	**Fundamental Algorithm Paradigms:** Deterministic and nondeterministic algorithm, Divide and conquer algorithms, Dynamic Programming, Greedy algorithms, Minimum Spanning Tree, Single Source Shortest path algorithms, Incremental Improvement, Backtracking, Assignment Problem, Branch & Bound and Approximation Problem.	CO1
2.	**Complexity Analysis and Evaluation:** Space and time complexity, Order of growth, Recurrences relations, Amortization, Complexity classes.	CO2
3.	Real time applications and complexity comparison for engineering problems.	CO3

Recommended and Supplementary Books:

1. Introduction to Algorithms **(Third Edition),** by Coreman *et* al., Publishers: McGraw-Hill
2. Fundamentals of Computer Algorithms, Sartaj Sahni and Sanguthevar Rajasekaran Ellis Horowitz

Mapping of Course Outcomes with Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1	C2	Lecture, Web Content, PPT slide	Class Test, Assignment, Midterm Exam, Final Examination
CO2	PO1, PO2	C2	Lecture, Web Content, PPT slide	Class Test, Assignment, Midterm Exam, Final Examination
CO3	PO1, PO2	C3	Lecture, Web Content, PPT slide	Class Test, Assignment, Midterm Exam, Final Examination

Course Title: Algorithm Design and Analysis Laboratory
Course Code: CSE 226

Credits: 1 **Class Hours/Week: 2**
Course Type: Core **Pre-requisite: DSL**
CIE Marks: 40 **SEE Marks: 60**

Rationale:
This course introduces students to the general tools and techniques for analyzing and designing computer algorithms.

Course Objectives:
Upon completion of this course, students will be able to do the following:
1. Acquaint students with the major algorithms and data structures.
2. Foster the analytical and critical analysis of algorithmic design paradigms and methods.
3. Apply the knowledge of asymptotic performance of algorithms.
4. To facilitate necessary knowledge about algorithms in common engineering design situations.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to
- **CO1** **Use (C3)** the appropriate algorithms to solve problems belonging to randomized, iterative, recursive, dynamic, greedy, graph, backtracking, branch and bound paradigms.
- **CO2** **Argue (C5)** the efficiency and correctness of the algorithms in CO1.
- **CO3** **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		√										
CO2		√										
CO3										√		

Course Description:

SL No.	Course Contents	COs
1.	Analysis, Implementation and Evaluation of Divide & conquer algorithms: Merge sort, Quick sort, Dynamic Programming: 0/1 knapsack, Subset Problem; BFS, DFS, Topological Sorting, Minimum Spanning Tree :Prim's Algorithm, Krushkal's Algorithm, Single Source Shortest paths: Bellman-Ford, Dijkstra's algorithm; Backtracking: Hamiltonian Cycle, N Queen's problem, Branch & Bound : Travelling Salesman Problem	CO1, CO2, CO3

Text Books, Reference Books and Other Resources:

1) Introduction to Algorithms **(Third Edition),** by Coreman *et* al., Publishers: McGraw-Hill
2) Fundamentals of Computer Algorithms, **Sartaj Sahni and Sanguthevar Rajasekaran Ellis Horowitz**
3) Data Structures and Algorithms in Python, by Michael T. Goodrich, publisher: VP & PUBLISHER

Mapping of Course Outcomes with Teaching -Learning and Assessment Delivery and Assessment:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO2	C3	Lecture, Practical Implementation	Lab Performance, Quiz, Lab Exam
CO2	PO2	C5	Lecture, Practical Implementation	Lab Performance, Quiz, Lab Exam
CO3	PO10	A3	Demonstration	Report

Course Title: Database Management System
Course Code: CSE 237

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: ICS**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course will cover the principles of relational database management systems, as well as recent advances in database theory and practice.

Course Objectives:
1. To make the students understand the different issues involved in the design and implementation of a database system in a modern information system.
2. To acquaint students with database modeling techniques/tools of the modern era.
3. To learn various techniques for managing a database-oriented system.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to
CO1 **Demonstrate (C2)** key concepts of database management system and its security.
CO2 **Write (C3)** database query to serve user requirements.
CO3 **Plan (C6)** to design various data models minimizing database anomalies.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2		√										
CO3		√	√	√								

Course Description:

SL No.	Course Contents	COs
1.	**Introduction:** database systems, purposes, applications, database architecture, database administrator **Relational Model:** structure, database schema, keys, schema diagrams, relational query, relational algebra **Distributed Database (DDB):** centralized vs distributed database, homogenous and heterogenous DDB, major objective, storage technique, deadlock, ACID properties, data structure, data mining and warehousing, indexing and hashing	CO1
2.	**Query Language:** SQL, basic query structure, set operations, NULL values, aggregate functions, nested subqueries, database modification **Intermediate SQL:** join, views, transactions, integrity constraints, SQL data types, index in SQL, authorization, **Advanced SQL:** SQL from application program, function, procedure, trigger	CO2

3.	**ER model:** different attribute, mapping cardinalities, primary key for entity and relationship, strong and weak entity, redundant attribute, reducing ER diagram to relational schema **Enhanced ER diagram:** supertype and subtype, specialization, generalization, aggregation, various constraints, different ER notations, entity clustering **ER diagram to relation:** cardinality constraints to relational mapping, converting different types of attributes in relation, practice problems, **Database Normalization:** various anomalies, importance of normalization, functional dependency, partial dependency, transitive dependency, 1NF, 2NF, practice problems, 3NF, BCNF, practice problems.	CO3

Text Books, Reference Books and Other Resources:

1. Database System Concepts (7th Edition) ***by*** Silbarchatz
2. Fundamentals of Database Systems (7th Edition) ***by*** Elmasri
3. Database Management System (3rd Edition) ***by*** Ramakrishnan
4. Modern Database Management (10th Edition) ***by*** Jeffrey A. Hoffer

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1, PO2	C2	Lecture notes, PPT slides	Class Test, Midterm, Assignment, Final
CO2	PO2	C3	Lecture notes, PPT slides	Class Test, Midterm, Assignment, Final
CO3	PO2, PO3, PO4	C6	Lecture notes, PPT slides	Class Test, Midterm, Assignment, Final

Course Title: Database Management System Laboratory
Course Code: CSE 238

Credits: 1.5 | **Class Hours/Week: 3**
Course Type: Core | **Pre-requisite: ICS**
CIE Marks: 30 | **SEE Marks: 70**

Course Rationale:
This course is designed to introduce the concept of real-world database design through the implementation of a database-driven real-time project.

Course Objectives:
1. To facilitate necessary knowledge about the design of a database based on various scenarios.
2. To develop a real-world database application.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Execute (C3)** query language using database panel or application programming.
CO2 **Model (C3)** a database for a given problem considering different anomalies.
CO3 **Develop (C6)** and **Report (A3)** a database solution to an engineering problem as an individual or group project.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		√			√							
CO2		√	√	√	√							
CO3		√	√	√	√			√	√	√	√	√

Course Description:

SL No.	Course Contents	COs
1.	**Familiarization** with database management system, Basic, Intermediate and Advanced SQL queries, executing queries from application programming	CO1
2.	**Modeling:** ER and EER, ER to relation, relation to ER, normalization	CO2
3.	**Project**: Activities, report, presentation slides.	CO3

Text Books, Reference Books and Other Resources:
1. Database System Concepts (7th Edition) ***by*** Silbarchatz
2. Fundamentals of Database Systems (7th Edition) ***by*** Elmasri
3. Database Management System (3rd Edition) ***by*** Ramakrishnan
4. Modern Database Management (10th Edition) ***by*** Jeffrey A. Hoffer
5. Web references: www.w3schools.com, www.mysqltutorial.org

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO2, PO5	C3	Demonstration, PPT slides	Lab Performance, Assignment
CO2	PO2, PO3, PO4, PO5	C3	Demonstration, PPT slides	Lab Performance, Assignment
CO3	PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO11, PO12	C6, A3	Demonstration, PPT slides	Project Demo and Report

Course Title: Signals & Systems
Course Code: EEE 201

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: EM II**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course aims to provide basic knowledge about various types of signals and systems along with different analysis techniques.

Course Objectives:
The objectives of this course are:
1. To help students conceptualize the relation between signals & systems.
2. To make the students understand classification of signals & properties of systems.
3. Accumulate basic ideas about analogous system and differential equations.
4. Facilitate students with necessary knowledge to conceptualize the properties of Fourier Series
5. Acquaint students with the basic theory of Fourier and Laplace transformation.
6. Develop skills to apply Fourier and Laplace transformation to solve real-time problems.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

- **CO1** **Apply** (**C3**) Linearity, Causality, Time invariance, Memory, Stability and Inevitability of any system.
- **CO2** **Analyze** (**C4**) how signals transform from time domain to frequency domain & transform from frequency domain to time domain.
- **CO3** **Apply**(**C3**) convolution on signals.
- **CO4** **Analyze**(**C4**) how signals convert between mechanical system and electrical system.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2		√										
CO3	√											
CO4		√										

Course Description:

SL No.	Course Content	COs
1.	**Introduction to Signals:** Introduction to the Signals - classification, Basic operation on signals, Elementary signals. Representation of signals using impulse function.	CO1

2.	**Introduction to Systems:** Introduction to the Systems – classification. Properties of System. Identification of systems by analyzing properties.	CO1
3.	**Analogues system:** Analogues system and their solution, Linear mechanical elements, Element representation, D'Alembert's Principle, F-V analogy, F-I analogy, Mechanical	CO4
4.	**Linear differential equation-I:** Overview of linear differential equation-I, Solution of higher order linear differential equation with constant coefficient.	CO1
5.	**Linear differential equation-II:** Overview of linear differential equation-II, Solution of higher order linear differential equation with undetermined coefficient.	CO1
6.	**Lumped element electrical systems:** Lumped element electrical systems using linear differential equation-I Lumped element electrical systems using linear differential equation-II	CO4
7.	**Fourier Series:** Overview of Fourier Series. Fourier series- properties, Harmonic representation, System response, Frequency response of LTI systems.	CO2
8.	**Fourier transformation:** Fourier transformation- properties, System transfer function, System response and distortion-less systems. Fourier transformation-	CO2
9.	**Laplace transformation:** Fourier transformation to Laplace transformation (LT), LT of important functions. LT: gate function, LT of periodic functions.	CO2
10	**Application of Laplace Transformation:** Application of Laplace Transformation, inverse transform, solution of system equations, system transfer function. Inverse Laplace Transform. LT: Impulse	CO2
11	**Convolution of Signals:** Convolution of Signals. Introduction to Random signal, Stationery, Ergodicity, Noise models. Correlation and power spectrum, Distribution	CO3

Text Books, Reference Books and Other Resources:

1. Signals and Systems by Haykin
2. Linear System Analysis by D.K. Cheng.
3. Signal & Systems Analysis using Transform Methods and MATLAB by M.J. Roberts

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**POs**	**Bloom's Taxonomy domain/level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1	C3	Lecture notes, PPT slides, problem solving, web content	Class Test, Assignment, Midterm, Final
CO2	PO2	C4	Lecture notes, PPT slides, problem solving, web content	Class Test, Assignment, Midterm, Final
CO3	PO1	C3	Lecture notes, PPT slides, problem solving, web content	Class Test, Assignment, Midterm, Final
CO4	PO2	C4	Lecture notes, PPT slides, problem solving, web content	Class Test, Assignment, Midterm, Final

Course Title: Signals and Systems Laboratory
Course Code: EEE 202

Credits: 1 **Class Hours/Week: 2**
Course Type: Core **Pre-requisite:**
CIE Marks: 40 **SEE Marks: 60**

Course Rationale:
This course aims to develop students' knowledge to implement different transform methods in MATLAB environment.

Course Objectives:
The objective of the course is to enable the students to

1. Familiarize students with the MATLAB environment
2. Introduce with plotting & sub-plotting concept.
3. Learn simulation of Fourier, Laplace & Z-transformation.

Course Outcomes (COs):

Upon successful completion of this course, students will be able to

CO1 **Evaluate** (**A3, P4, C6**) different types of Signals in Matlab or CAD tools.

CO2 **Respond** (**A2, P2**) in writing comprehensive reports on the work done in laboratory in a group and orally present the findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1					√							√
CO2									√	√		√

Text and Reference books:
1. Signals and Systems by Haykin.
2. Linear System Analysis by D.K. Cheng.
3. Signal & Systems Analysis using Transform Methods and MATLAB by M.J. Roberts
4. Lab Manual

Course Description:

SL No	Course Contents	CLOs
1.	Laboratory work based on theory course **EEE 201:** Signals and Systems	**CO1 – CO2**

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO5, PO12	A3, P4, C6	Lectures, Web	Lab Performance, Quiz/Viva
CO2	PO9, PO10, PO12	A2, P2	Discussion and Presentation	Report and Presentation

Course Title: Engineering Mathematics IV
Course Code: MAT 203

Credits: 3 **Class Hours/Week: 3**
Course Type: Mathematics **Pre-requisite: EM III**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
It is essential for the engineering/science students to develop strong analytic skills which cannot be achieved without extending knowledge and skills obtained in the mathematics courses (MAT 105, MAT 107, MAT 201) and without including matrix methods in particular and Linear Algebra in general. This course is intended to build up strong bases of Laplace transforms, Fourier analysis, partial differential equations (P. D. E.), matrix, and Linear Algebra, as well as to include the experience of applications of its contents in the relevant courses of science/engineering.

Course Objectives:

The objectives of this course are to:
1. This course enhances and broadens the contents covered in MAT 105, MAT 107, & MAT 201.
2. Toward the above goal, it develops the basic concepts of Laplace transforms, Fourier analysis, matrix, and Linear Algebra in their wide scopes, and of P. D. E in its relevant scopes.
3. While doing the above it builds up problem solving experience in each relevant topic along with the addition of application familiarity in the related areas of science/engineering. Etc.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Apply (C3)** different types of matrices and determinants in solving real time engineering problems.
CO2 **Solve (C3)** different types of matrix and linear transformation.
CO3 **Apply (C3)** Fourier analysis to convert time domain signal to frequency domain signal and vice-versa.
CO4 **Solve (C3)** basic properties, inverse properties, differential and integral equations using Laplace Transformation.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											
CO4	√											

Course Description:

SL No.	Course Content	COs
1.	Basic definition of matrix, Inverse and Rank of a matrix Test of solution system, State and prove Cayley-Hamilton theorem & verify some matrix	CO1
2.	Eigen values and Eigen vectors, Applications of Eigenvalues and Eigenvectors, Geometrical meaning of Linear Independence and discussion on CT answer script, Application, Vector Space and Subspace, Linear combination, Application	CO2
3.	Formulation of Fourier series, Formulation of Fourier series, Some example of Fourier series with graph, Review	CO3
4.	Some important properties of Laplace transformation and calculate problems, Established basic formula of Laplace transformation in different function, Some important properties of Laplace transformation and calculate problems, Laplace transform of first derivative to n^{th} derivatives, Definition of inverse	CO4

Text Books, Reference Books and Other Resources:

1. Linear Algebra By Howard Anton
2. Mathematical Physics By B.D.Gupta
3. Linear Algebra by Abdur Rahman
4. Linear Algebra & Its Applications by David C. Lay
5. Advanced Engineering Mathematics by Peter V. O'Neil
6. Fourier Series and Boundary Value Problems by Charchil
7. Laplace Transform by M. R. Spigel

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C3	Lecture, Notes, Problem solution	Class Test, Midterm, Assignment, Final
CO2	PO1	C3	Lectures, Notes, Practice Problems	Class Test, Midterm, Assignment, Final
CO3	PO1	C3	Lectures, Notes, Practice Problems	Class Test, Midterm, Assignment, Final
CO4	PO1	C3	Lectures, Notes, Practice Problems	Class Test, Midterm, Assignment, Final

Course Title: Industrial & Business Management
Course Code: MGT 203

Credits: 3 **Class Hours/Week: 3**
Course Type: Non-engineering **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course is designed to familiarize students with the current management practices and the business environment in the modern world.

Course Objectives:
The main objectives of this course are:
1. Introduce students to the industrial management concepts and functional areas of business.
2. Acquaint students with the management skills needed for different organizational activities.
3. Provide the knowledge of organizational structure & design
4. Impart knowledge on basic Marketing & HRM concepts and functions
5. Provide an understanding of financial and operational management tools and techniques.
6. Apply the knowledge of organizational innovations to gain competitive advantages
7. Facilitate students with necessary knowledge regarding leadership and ethical issues in organizations.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Demonstrate (C3)** an understanding of industrial management concepts and practices.

CO2 **Identify (C1)** the basic managerial role, skills, functions & the nature of managerial work for upcoming challenges in an organization.

CO3 **Analyze (C4)** the implication of numerous elements of organizational structure & design.

CO4 **Explain (C2)** the basic principles and functions of marketing & HRM.

CO5 **Understand (C2)** and **apply (C3)** a variety of technical and analytical tools used by the financial and operations managers for decision making.

CO6 **Understand (C2)** how to integrate organizational innovations to gain competitive advantages.

CO7 **Apply (C3)** appropriate leadership practices and ethics in complex business environment.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											
CO4	√											
CO5	√											
CO6	√											
CO7	√											

Course Description:

SL No.	Course Contents	COs
1.	**Introduction:** Basic concepts-Industry, Business, management, industrial management, Types of Industry, why study industrial business management, objectives, Importance, Applications, Problems of industrial management.	CO1
2.	**Management:** Nature of management, levels of management, managerial roles, managerial skills, organizational hierarchy, managerial functions, the science and art of management, productivity, effectiveness and efficiency. Management theories-Classical & behavioral management, Principles of management, challenges of a manager.	CO1
3.	**Organization:** Meaning, organizational structure, departmentalization, work specialization, chain of command, span of control, centralization vs decentralization, formalization, common organizational design, simple structure, bureaucracy, matrix structure, team structure, virtual organization, boundary less organization. learning organization.	CO2
4.	**Marketing:** Definition, basic concepts (needs, wants and demands), marketing strategy: segmentation; targeting; positioning, marketing mix, holistic marketing, green marketing.	CO4
5.	**Financial Management:** Functions & scope, key activities of financial managers, principles of financial management, financial statements and reports, financial ratio analysis, types of financial ratios: liquidity ratios; asset management ratios; profitability ratios and market ratios, net present value (NPV), internal rate of return (IRR), break-even analysis, depreciation, time value of money, future value vs present value.	CO5
6.	**Human Resource Management:** Meaning, HRM planning, job analysis, recruitment, selection, selection process, selection techniques, employee training and development, performance appraisal, conflict management, health and safety measures.	CO4
7.	**Operations Management:** Definition, facility location, plant layout, inventory management, productivity, quality theories and characteristics, project planning (PERT, CPM).	CO5
8.	**Innovation Management:** Innovation and discovery, sources of innovative, types of innovation idea, role of creative thinking, creative climate, innovation process, principles of innovation, product life cycle, service innovation, intellectual property, patent, copyrights.	CO6
9.	**Leadership & Ethics in Business:** Definition and components, trait versus process leadership, assigned versus emergent leadership, leadership and power, leadership and management, transformational leadership, servant leadership, team leadership. Concept, domains of ethical theories, principles of personal & professional ethics, concepts of business ethics, importance of business ethics, values & ethics, characteristics of an ethical organization, legal vs ethical.	CO7

Text Books, Reference Books and Other Resources:

1. Griffin, R. W. (2005). Management (8th ed.). Houghton Mifflin Company.
2. Robins, S. P., & Coulter, M. (2004). Management (8th ed.). Prentice Hall.
3. Lee, J. K., Larry, P. R., & Manoj, K. M. (2006). *Operations Managemen: Processes & Value Chains* (8th ed.). New Jersey, USA: Prentice Hall.
4. Kotler, P., Armstrong, G., Agnihotri, P. Y., & Haque, E. (2010). *Principles of Marketing: A South Asian Perspective* (13th ed.). Delhi: Pearson Prentice Hall.
5. Brigham, E. F., & Houston, J. F. (2009). Fundamentals of Financial Management (10th/Latest ed.). Cengage Learning.

CO Delivery and Assessment:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C3	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO2	PO1	C1	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO3	PO1	C4	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO4	PO1	C2	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO5	PO1	C2, C3	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO6	PO1	C2	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam
CO7	PO1	C3	Lecture, PPT slides, problem solving	Class Test, Assignment, Midterm, Final Exam

Course Title: Computational Methods for Engineering Problems
Course Code: CSE 301

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: EM III**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course is designed to make the understanding about computational concepts.

Course Objectives:
The objectives of this course are to:
1. Provide knowledge for solving various scientific and engineering problems.
2. Analyze and evaluate the accuracy of common numerical methods in its applications.
3. Develop mathematical modeling skills for experiments and research.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Implement (C3)** techniques to minimize errors of numerical methods.
CO2 **Use (C3)** numerical techniques for solving linear and nonlinear complex optimization problems.
CO3 **Implement (C3)** complex systems of interpolation, curve fitting, and numerical differentiation and integration.
CO4 **Implement (C3)** numerical algorithms for solving ordinary and partial differential equations.
CO5 **Apply (C3)** numerical methods to solve problems in computational linear algebra, such as matrix operations and eigenvalue calculations.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√										
CO3	√	√										
CO4	√	√										
CO5	√	√										

Course Description:

SL No.	Course Contents	COs
1.	Discuss Geometrical method to find real root of the equation, Bisection method, False position method, Fixed point iteration, Newton Raphson method, Solution of linear system, Difference table, Forward Difference table and Backward Difference table; Fundamental theorem of the difference Calculus, Interpolation with equal interval, Interpolation with unequal interval, Numerical Differentiation, Numerical Integration, Solution of Ordinary Differential Equations	CO1 – CO5

Text Books, Reference Books and Other Resources:

1. Numerical Analysis, by Vasistha
2. Numerical Analysis, by Richard L. Burden,
3. Numerical methods for scientific and engineering computation, by Mahinder Kumar Jain, S. R. K. Iyengar, Rajendra K. Jain
4. Advanced Engineering Mathematics by H.K. Das
5. Numerical Analysis, by Vasistha
6. Numerical Analysis by Richard L. Burden

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1, PO2	C3	Lecture, Notes, Problem solution	Class Test, Assignment, Midterm, Final Exam
CO2	PO1, PO2	C3	Lectures, Notes, Practice Problems	Class Test, Assignment, Midterm, Final Exam
CO3	PO1, PO2	C3	Lectures, Notes, Practice Problems	Class Test, Assignment, Midterm, Final Exam
CO4	PO1, PO2	C3	Lectures, Notes, Practice Problems	Class Test, Assignment, Midterm, Final Exam
CO5	PO1, PO2	C3	Lectures, Notes, Practice Problems	Class Test, Assignment, Midterm, Final Exam

Course Title: Computational Methods for Engineering Problems Laboratory
Course Code: CSE 302

Credits: 0.75 **Class Hours/Week: 1.5**
Course Type: Core **Pre-requisite: EM I, EM II, EM III**
CIE Marks: 70 **SEE Marks: 30**

Course Rationale:
This course equips students with the knowledge and skills to solve complex mathematical problems encountered in computer science using efficient and accurate numerical techniques.

Course Objectives:
The objectives of the course are:

1. To brings light the numerous uses of numerical methods in engineering.
2. To obtain error estimates for numerical derivative formulae.
3. To demonstrate implementation of numerical problems for engineering problems

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Apply (C3)** numerical methods for solving engineering problems.

CO2 **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√			√							
CO2										√		

Course Description:

SL No.	Course Contents	COs
1.	Lab based on theory course of Computational Methods for Engineering Problems	CO1, CO2

Text Books, Reference Books and Other Resources:

1. Computer Networking: A Top-Down Approach---- (James F. Kurose, Keith W. Ross)-- Instructor contents and PPT slides: http://gaia.cs.umass.edu/kurose_ross/instructor.php
2. Cisco Packet Tracer Simulation
3. Additional contents: Cisco PowerPoint slides, pdf tutorials will be provided.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2, PO5	C3	Lecture, Laboratory Experiments	Quiz, Performance, Assignment
CO2	PO10	A3	Lecture, Laboratory Experiments	Quiz, Performance, Assignment

Course Title: Software Engineering & Information System Design
Course Code: CSE 305

Credits: 4 **Class Hours/Week: 4**
Course Type: Core **Pre-requisite: DMS**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
Software Engineering and Information Systems is a profile where the software development is studied in a systematic, controllable and efficient way.

Course Objectives:
The objectives of this course are:
1. To introduces the concept and method required for the construction of large software intensive system. It aims to develop a broad understanding of the discipline of software engineering.
2. To complement this with detail knowledge of technique for the analysis and design of complex software system. It aims to set these techniques in an appropriate engineering and management context.
3. To provide a brief account of associated professional and legal issues.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Explain (C2)** different concepts of software engineering and information system design.
CO2 **Describe (C2)** conventional and agile software engineering process models.
CO3 **Illustrate (C3)** different modeling techniques to represent the working process of a software system.
CO4 **Demonstrate (C3)** different types of software testing strategies.
CO5 **Illustrate (C3)** various types of software project variables to estimate software project and maintaining project risks for quality improvement.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√	√										
CO3		√	√									
CO4	√											
CO5	√											

Course Description:

SL No.	Course Contents	COs
1.	**Introduction Software Engineering** Introduction to Software Engineering, Purpose, Applications, failure curves, software applications, legacy software, Umbrella Activities.	CO1

2.	**Software Process Model:** Process Flow, Process Models (RAD, Waterfall, Incremental, Spiral, Agile, Prototype, Concurrent), Advantages, Disadvantages, Framework Activities**,** Task set, Agile Development, Agility, Human Factors, Extreme Programming	CO1, CO2
3.	**Unified Modeling Language:** Notations and basic concepts, Use Case Diagram, Sequence Diagram, Activity Diagram	CO3
4.	**Software Testing:** Exhaustive Testing**,** Selective Testing, Good Test, Testing Principles**,** Black box Testing, White box Testing, Equivalence Partitioning, Boundary Value Analysis, Cause-Effect, Unit Testing, Integration Testing, System Testing Graphing**,** Testing, Error, Fault, Failure, Acceptance Testing	CO4
5.	**Estimation and Software Risk:** Introduction to software project variables, project estimation, risk management, software quality management	CO5
6.	**Information System Design:** Information System Design Life Cycle**,** Requirements engineering**,** Role and attributes of a system analyst**,** Feasibility Analysis**,** Data Flow Diagrams, Document Flow Diagrams**,** Process specification, Data Input Methods	CO1, CO3, CO5

Text Books, Reference Books and Other Resources:

1. Analysis and Design of Information Systems (2nd Edition) V. Rajaraman.
2. Software Engineering: A Practitioner's Approach (8th Edition) R. S. Pressman.
3. Software Engineering (9th Edition) Martin L.Shooman.
4. Software Engineering (9th Edition) by Ian Sommerville

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1	C2	Lecture, Textbook, slides	Class test, Midterm, Assignment, Final
CO2	PO1, PO2	C2	Lecture, Textbook, slides	Class test, Midterm, Assignment, Final
CO3	PO2, PO3	C3	Lecture, Textbook, slides	Class test, Midterm, Assignment, Final
CO4	PO1	C3	Lecture, Textbook, slides	Class test, Midterm, Assignment, Final
CO5	PO1	C3	Lecture, Textbook, slides	Class test, Midterm, Assignment, Final

Course Title: Software Engineering & Information System Design Laboratory
Course Code: CSE 306

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: DMSL**
CIE Marks: 30 **SEE Marks: 70**

Course Rationale:
Software Engineering and Information Systems is a profile where the software development is studied in a systematic, controllable and efficient way.

Course Objectives:
The objectives of the course are:
1. To impart state-of-the-art knowledge on Software Engineering and UML in an interactive manner through the Web.
2. To present case studies to demonstrate practical applications of different concepts.
3. To provide a scope to students where they can solve small, real life problems.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Use (C3)** appropriate tools for managing a software project.

CO2 **Implement (C3)** various client-side and server-side concepts utilizing design pattern to develop a software.

CO3 **Investigate (C4)** a system and **Report (A2)** the overall working process using modern tools.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1					√							
CO2	√	√	√		√							
CO3		√		√	√				√	√		

Course Description:

SL No.	Course Contents	COs
1.	**Project Management:** Project collaboration, Task Management	CO1
2.	**Implementation:** Client side and server-side feature implementation, design pattern and modular approach, software testing	CO2
3.	**System Modeling:** Project Proposal, Process model selection, UML diagram design, SRS document	CO3

Text and Reference Books:

1. Head First Design Patterns by Eric Freeman, Elisabeth Robson, Bert Bates, Kathy Sierra
2. Downloaded Software Engineering with UML by Bhuvan Unhelkar
3. Practical UI Patterns for Design Systems. Fast-Track Interaction Design for a Seamless User Experience by Diana MacDonald

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy domain/level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO5	C3	Lecture, slides	Class performance, Assignment, Report
CO2	PO1, PO2, PO3, PO5	C3	Lecture, slides	Class performance, Assignment, Report
CO3	PO2, PO4, PO5, PO9, PO10	C4, A2	Lecture, slides	Class performance, Assignment, Report

Course Title: Communication Engineering
Course Code: EEE 309

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This Course is offered to introduce the basic principles and applications of analog and digital communication in our daily life.

Course Objectives:
The objectives of this course are:
1. To introduce students with the fundamental concepts of communication systems.
2. To provide the knowledge of different analog modulation schemes and digital modulation schemes.
3. To make the students understand the multiple access techniques and key aspects of noise.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Explain (C2)** different types of analog modulation and digital modulation
CO2 **Solve (C3)** mathematical problems related to Communication engineering.
CO3 **Analyze (C4)** various channels and noises of communication system.
CO4 **Analyze (C4)** graphical and mathematical system of multiplexing & De-multiplexing.
CO5 **Design (C6)** Communication system

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√	√										
CO3		√										
CO4		√										
CO5		√										

Course Description:

SL No.	Course Contents	COs
1.	**Introduction of communication systems**: Basic principles, fundamental elements, system limitations.	CO2
2.	**Information Theory:** Information and system capacity, Information transmission, Entropy, Continuous channel capacity, Transmission through electrical network.	CO2

3.	**Analog communication:** AM, FM, PM, DSB, SSB, VSB, ISB.	CO1
4.	**Digital communication:** Introduction, Nyquist sampling theorem, Quantization of analog system, Quantization noise, PAM, PWM, PPM, PCM, LOGPCM, and systems, Digital modulations, ASK, FSK, PSK, DPSK, MSK, M-array digital modulation, QAM, QPSK, Delta modulation, Multi carrier	CO1, CO5
5.	**Multiplexing:** Space division multiplexing, frequency division multiplexing, time division multiplexing, and code division multiplexing.	CO4, CO5
6.	**Noise:** Physical sources of noise, types of noise, calculation of noise, SNR & noise figure, and calculation of noise figure, noise temperature, equivalent noise resistance.	CO3

Text Books, Reference Books and Other Resources:

1. Electronic Communication System by George Kennedy.
2. Modern Digital & Analog Communication Systems by B.P. Lathi.
3. Communication System by Simon Haykin.
4. Principles of Communication Systems by Herbert Taub and Donald L. Schilling

CO Delivery and Assessment:

COs	Corresponding POs	Bloom's taxonomy domain/level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture, PPT slide, Mathematical Problem solving	Class Test, Assignment, Mid Term, Final Exam
CO2	PO1, PO2	C3	Lecture, PPT slide, Mathematical Problem solving	Class Test, Assignment, Mid Term, Final Exam
CO3	PO2	C4	Lecture, PPT slide, Mathematical Problem solving	Class Test, Assignment, Mid Term, Final Exam
CO4	PO2	C4	Lecture, PPT slide, Mathematical Problem solving	Class Test, Assignment, Mid Term, Final Exam
CO5	PO2	C6	Lecture, PPT slide, Mathematical Problem solving	Class Test, Assignment, Mid Term, Final Exam

Course Title: Communication Engineering Laboratory
Course Code: CSE 310

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite:**
CIE Marks: 40 **SEE Marks: 60**

Course Rationale:
This course aims to introduce students with basic techniques and tools for electrical circuit simulation.

Course Objectives:
The objective of the course is to enable the students to

1. To develop knowledge of using simulation tools for simulating electric circuit.
2. To distinguish the difference between the simulation behavior and behavior of practical circuit.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Demonstrate** (**C5, A4, P3**) different types of analog and digital modulation and demodulation techniques using experimental setup as an individual or as a member of a team.

CO2 **Conclude** (**C6, A3, P1**) the result from experimental data.

CO3 **Interpret** (**A4, P2**) the laboratory work done in a group to present the findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1					√				√	√		
CO2				√								√
CO3									√	√		√

Course Description:

SL No.	Course Contents	COs
1.	Laboratory work using hardware based on theory course: EEE 309: **Communication Engineering**. It will cover AM FM, ASK, FSK, PSK modulation and demodulation practical laboratory work.	CO1, CO2, CO3
2.	Report writing based on laboratory work.	CO3
3.	Oral presentation on mini project work (design project/analytical project/ experimental project/industrial tour).	CO1, CO2

Text and Reference books:

1. Lab Manuals
2. Additional contents: Communication Engineering PowerPoint slides, pdf tutorials will be provided.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO5, PO9, PO10	C5, A4, P3	Lecture, Demonstration, discussion, experiment	Quiz/Written Exam, Performance, Report
CO2	PO4, PO12	C6, A3, P1	Lecture, Calculation, Graph Drawing, Report	Class Performance, Report
CO3	PO9, PO10, PO12	A4, P2	Lecture, Mini Project, Presentation, Report Writing	Class Performance, Quiz/Written Exam, Presentation

Course Title: Microprocessors and Microcontrollers
Course Code: EEE 371

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: DE**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
Intended to enable the learners to familiar with different types of microprocessors and microcontrollers, use the acquired knowledge to understand computer systems and embedded systems circuitry or architectures.

Course Objectives:
The objectives of this course are to:

1. Provide a theoretical &empirical introduction to microcontrollers and microprocessors, assembly language programming techniques, design of hardware interfacing circuit, microcontroller, and microprocessor system design considerations.
2. Provide the rudimentary concepts to identify the basic elements and functions of contemporary microprocessors and microcontrollers and implicating programming in the instruction sets of microprocessors and microcontrollers (Basically Intel family).
3. Designing and maintaining different types of digital projects to fulfill the quotidian life problem in embedded system.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

- **CO1** **Describe** (**C2**) the basics of microprocessor and microcontroller architecture, interfacing, and operations.
- **CO2** **Execute** (**C3**) assembly language instructions or programs to control microprocessor and microcontroller operations.
- **CO3** **Solve** (**C3**) the real-world problems related with embedded systems and designs.
- **CO4** **Illustrate** (**C2**) microprocessor and microcontroller related circuitry and sensor interfacing.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3			√									
CO4	√											

Course Description:

SL No.	Course Contents	COs
1.	**Introduction to different type of microprocessors:** 8 bit, 16 bit, 32 bit and their architectures; pin diagram & junction; Intel series microprocessor and Co-	CO1

	processor; RISK & CISK processor; Assembly Language: Basic Instruction Sets and Assembly language Programming based on 8086 microprocessor.	
2.	**Microprocessor peripherals:** Introduction to some available microprocessor peripherals IC's and their application such as 8251, 8253, 8254, 8255, 8257, 8259, 8279, A/D and D/A converter interfacing, Timing Diagram, Interrupts, I/O systems, DMA-based data transfer, memory interfacing. MMX and SIMD technologies. The above peripheral is based on 8085 and 8086 processor.	CO2
3.	**Interfacing:** Introduction, interfacing to microprocessor to keyboards, alphanumeric displays. Introduction to microcomputers and interfacing to microcomputer ports to high power devices.	CO3
4.	**Basic Micro controller:** Introduction of microcontroller; embedded system design; microcontroller programming environment; Architecture of different microcontroller such as PIC, MSP, ARM etc. Real time application design based on microcontroller.	CO3, CO4
5.	**PIC Microcontroller:** Introduction to PIC Microcontroller, Internal Architecture, Components & Programming. Some PIC Based Project work.	CO3, CO4
6.	**Arduino & Raspberry Pi:** Introduction of Development boards.	CO3, CO4

Text Books, Reference Books and Other Resources:

1. The Intel Microprocessor Architecture, Programming and Interfacing, Barry B. Brey.
2. Microprocessors and Interfacing: Programming and Hardware,Douglas V. Hall.
3. Advanced Microprocessor and Interfacing, Badri Ram.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment tools
CO1	PO1	C2	Lecture, Slide, Problem Solving, and examples	Class Test, Midterm, Assignment, Final Exam
CO2	PO1	C3	Lecture, Slide, Problem Solving, and examples	Class Test, Midterm, Assignment, Final Exam
CO3	PO3	C3	Lecture, Slide, Problem Solving, and examples	Class Test, Midterm, Assignment, Final Exam
CO4	PO1	C2	Lecture, Slide, Problem Solving, and examples	Class Test, Midterm, Assignment, Final Exam

Course Title: Microprocessors and Microcontrollers Laboratory
Course Code: EEE 372

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: DEL**
CIE Marks: 30 **SEE Marks: 70**

Course Rationale:
The course is intended to introduce the basic concepts of microprocessor and to develop in students the assembly language programming skills and real time applications of Microprocessor as well as microcontroller.

Course Objectives:
The objectives of this course are to:
1. Learn about different types of microcontroller & Microprocessors
2. Gather knowledge about different types of sensors used in our day-to-day life.
3. Learn about microcontroller programming.
4. Design simple project based on microcontroller

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Describe** (**P1)** and write program in assembly language for 8086 microprocessors.
CO2 **Design** (**C3, P3**) micro-controller based Real Time Projects
CO3 **Design** (**C3, P3**) SBC based system.
CO4 **Perform** work (**C6, A3, P3**) in individual and or group project.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2			√		√							
CO3			√		√							
CO4			√		√				√	√		

Course Description:

SL No.	Course Contents	COs
1.	Study on micro controller and tinkercad simulator	CO1
2.	Familiarizing with Arduino, Arduino IDE & design a simple LED looping Circuit.	CO2
3.	Interfacing with stepper motor.	CO2
4.	Keyboard & LCD display interfacing	CO2
5.	PWM & its use in Arduino using Servo Motor.	CO2
6.	Interfacing with different types of sensors with Arduino module	CO2
7	Interfacing with Bluetooth module	CO2

8	Introduction to Single board computer (SBC)	CO3, CO4
9	Project Work	CO3, CO4

Text Books, Reference Books and Other Resources:

1. Lab manual

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	P1	Lecture, Slide, Problem Solving, and examples	Quiz, Performance Test
CO2	PO3, PO5	C3, P3	Lecture, Slide, Problem Solving, and examples	Quiz, Performance Test
CO3	PO3, PO5	C3, P3	Lecture, Slide, Problem Solving, and examples	Quiz, Performance Test
CO4	PO3, PO5, PO9, PO10	C6, A3, P3	Lecture, Slide, Problem Solving, and examples	Project submission, Presentation

Course Title: Organizational Behavior
Course Code: MGT 251

Credits: 3 **Class Hours/Week: 3**
Course Type: Non-engineering **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course helps students to identify the human behavior in organizational settings. It studies individual and group behavior, organizational communication and structure, conflicts, and stress have on behavior within organization and the proper use of such knowledge improve an organization's effectiveness.

Course Objectives:
The main objectives of this course are:
1. Identify the complex nature of human beings in organizations through causes and effects of that behavior.
2. Provide the knowledge of organizational communication, structure and culture.
3. Discuss conflict resolution and change management in order to indoctrinate students how to handle a new workforce and cope with the challenges of new environment.
4. Acquaint students with numerous approaches of stress management.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Describe (C2)** numerous concepts, factors, challenges related to human behavior in organization.
CO2 **Explain (C2)** the individual and team behavior in the organization.
CO3 **Discuss (C2)** the structure and process of organization.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											

Course Description:

SL No.	Course Contents	COs
1.	**Introduction:** Meaning of OB, importance, goals of OB, developing an OB model, challenges and opportunities, organizational culture and its impacts, socialization practice, globalization, cultural dimension, cross-cultural management	CO1

2.	**Behavior within Organization (The Individual):** Workplace diversity, workplace discrimination, biographical characteristics, characteristics factor, intellectual and physical abilities, attitudes, job satisfaction, emotions and mood, affective events theory, emotional intelligence, personality, Myers-Briggs Type Indicator (MBTI) personality framework and the Big Five mode, core self-evaluation (CSE), self-monitoring, proactive personality, Hofstede's five value dimensions and the GLOBE framework, perception, attribution theory, decision making, three-stage model of creativity, Contrast Effect, Halo Effect, Stereotyping. motivation, different motivational theories, justice, job engagement, changing work environment	CO2
3.	**Behavior within Organization (Group and Interpersonal Influence):** Types of groups, group development model, groups vs team, group effectiveness, group decision making, team arrangement, communication, types of communication, interpersonal communication, barriers to effective communication, cross-cultural communication, leadership, different theories of leadership, roles of leaders, leadership vs power, types of power, abuse of power, politics and political behavior, conflict, types and loci of conflict, conflict process, bargaining, negotiation, organizational structure, elements and characteristics of organizational structure, structural models, downsizing and its effects, behavioral implications	CO2
4.	**Organizational Structure and Process:** Organizational culture and its effects, factors for organizational culture, types of culture, effect of national culture on organizational culture, recruitment and selection methods, types of training, methods for performance evaluation, leadership role of HR, Forces for change, individual and organizational resistance to change, overcoming resistance to change. Stress management: Definition, Sources and Consequences, Managing Stress: Individual Approach and Organizational Approach.	CO3

Text Books, Reference Books and Other Resources:

1. Organizational Behavior. 15th ed. New Delhi (2014): Pearson/Prentice Hall by Robbins, S. P., Judge, T.A & Vohra, N.
2. Organizations: behavior, structure, processes. 12th ed. McGraw Hill (2006) by Gibson, J., Ivanchevich, J., Donnelly, J. &Konopaske R.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level (C: Cognitive, P: Psychomotor A: Affective)	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture, Notes, Case Studies	Class Test, Assignment, Mid Term, Final Exam
CO2	PO1	C2	Lecture, Notes, Case Studies	Class Test, Assignment, Mid Term, Final Exam
CO3	PO1	C2	Lecture, Notes, Case Studies	Class Test, Assignment, Mid Term, Final Exam

Course Title: Artificial Intelligence
Course Code: CSE 317

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: SP**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course introduces the basic concepts and techniques of Artificial Intelligence (AI) to creating software and hardware to get computers to do things that would be considered intelligent as if people did them.

Course Objectives:
1. Introducing the students to AI's fundamental problems and state-of-the-art algorithms.
2. Appreciating the role of knowledge representation, logic, and language in understanding human intelligence from a computational perspective.
3. Helping the students conceptualize the far-reaching goal of AI in shaping our future.
4. Exposing the students to the frontiers of AI-intensive computing while providing a sufficiently strong foundation to encourage further research.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Interpret** (**C2**) the key components and classical search algorithms of AI to solve various real-life problems.
CO2 **Investigate (C4)** statistical methods and machine learning techniques for solving complex AI-related problems
CO3 **Use (C3)** Natural Language Processing methods for real-life problem solving.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√										
CO3	√	√			√							

Course Description:

SL No.	Course Contents	COs
1.	**Introduction to Artificial Intelligence** AI Subfields, AI agent, Task Environment, Environment Type	CO1
2.	**Searching Techniques:** BFS, DFS, UCS, Best First Search, A*, Genetic Algorithm	CO1

3.	**Introduction to Natural Language Processing (NLP)** Introduction, Intuitions, Application Fields, Text Representation, Different Language Models (char/word n-grams), Text Similarity Measure, Machine Translation.	CO3
4.	**Introduction to Uncertainty** Quantifying Uncertainty, Basic Probability Notation, Inference using full joint distributions, Bayes Rule, Semantics of Bayesian network (BN), Inference in BN, Probabilistic Reasoning over time, Hidden Markov Models.	CO2
5.	**Introduction to Learning** Nearest Neighbors, Linear Regression, Logistic Regression, Decision Tree, K Means Clustering, Naïve Bayes Classifier, Reinforcement Learning	CO2

Text Books, Reference Books and Other Resources:

1. "*Artificial Intelligence – A Modern Approach*", by Stuart Russell and Peter Norvig.
2. "*Speech and Language Processing*", Jurafsky & Martin, Third Edition (Draft Link)
3. "*Artificial Intelligence*", by Elaine Rich, Kevin Knight.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2	C2	Lecture, Problem solution, Video, Web Link	Class Test, Assignment, Midterm, Final Exam
CO2	PO1, PO2	C4	Lecture, Problem solution, Video, Web Link	Class Test, Assignment, Midterm, Final Exam
CO3	PO1, PO2, PO5	C3	Lecture, Problem solution, Video, Web Link	Class Test, Assignment, Midterm, Final Exam

Course Title: Artificial Intelligence Laboratory
Course Code: CSE 318

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: SPL**
CIE Marks: 30 **SEE Marks: 70**

Course Rationale:
This subject aims to practically implement as well as analyze the various procedures, methods, and algorithms related to artificial intelligence for real-world applications.

Course Objectives:
The objective of the course is to enable the students to

1. Provide students with a practical aspect of various AI algorithms
2. Motivate them towards developing their careers in AI applications

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Demonstrate (C2)** the application of modern tools for data preprocessing and analysis.

CO2 **Apply (C3)** search strategies for solutions to problems with complete information.

CO3 **Develop (C3)** an intelligent system capable of learning from data.

CO4 **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√			√							
CO2	√	√			√							
CO3	√	√	√						√			
CO4										√		

Course Description:

SL No.	Course Contents	COs
1.	**Introduction to Python** Basic Syntax, Functions, Machine Learning Tools	CO1
2.	**Search Techniques Implementation:** Uninformed Search Techniques, Informed Search Techniques	CO2
3.	**Project Implementation** Text Data Reading and Preprocessing, Text Representation, Model Implementation	CO3

Text and Reference books:

1. "Artificial Intelligence – A Modern Approach", by Stuart Russell and Peter Norvig

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's taxonomy domain/level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2, PO5	C2	Lecture, Lab Task	Lab Performance, Assignment
CO2	PO1, PO2, PO5	C3	Lecture, Lab Task	Lab Performance, Assignment
CO3	PO1, PO2, PO3, PO9	C3	Lecture, Web links	Project Report, Performance, Presentation, Viva
CO4	PO10	A3	Demonstration	Report

Course Title: Operating Systems
Course Code: CSE 333

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course provides a comprehensive understanding of the modern Operating System and examines the ways that design goals can be achieved.

Course Objectives:
The objectives of this course are:
1. To enhance students' understanding of the function and structure of operating systems
2. To facilitate necessary knowledge about how the system interacts and provides services for the execution of application software.
3. To evaluate underlying hardware, virtual memory, concurrency, and synchronization.
4. To enable them to illustrate key operating system aspects by concrete examples, and prepare them for future courses.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Describe (C2)** the basic structure and characteristics of a modern operating system, the concept for virtualization, cloud, and multiple processor systems.
CO2 **Explain (C2)** the different process and thread synchronization methods and the tradeoffs between them.
CO3 **Examine (C4)** the algorithms on which the functions of the Operating Systems are built.
CO4 **Compare (C2)** different types of memory-management schemes for a system.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√	√										
CO3	√	√										
CO4	√	√										

Course Description:

SL No.	Course Contents	COs
1.	**Introduction:** Operating system, Computer-system organization, and architecture, Operations, Computing environments.	CO1

2.	**Operating-System Structures:** Operating system services, System calls, Operating system structure.	CO1
3.	**Processes:** Process concept, Process scheduling, Operations on processes, Communication	CO2
4.	**Multithreaded Programming:** Threading overview, Multithreading models	CO2
5.	**Process Scheduling:** Basic concept, Criteria, Algorithms	CO3
6.	**Deadlocks:** Characterization, Methods for handling deadlocks, Deadlock Prevention, Avoidance, Detection.	CO3
7.	**Main Memory:** Background, Swapping, Segmentation, Paging	CO4
8.	**Virtual Memory:** Background, Demand Paging, Page replacement	CO4
9.	**File System Interface:** File Concept, Access methods, Directory structure	CO1

3

Text Books, Reference Books, and Other Resources:

1. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne, "Operating System Concepts", (8th Edition) ISBN: 978-0-470-23399-3.
2. Andrew S. Tanenbaum (Author), Herbert Bos, "Modern Operating Systems"

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture, Problem solution	Assignment, Class Test, Midterm, Final
CO2	PO1, PO2	C2	Lecture, Problem solution	Assignment, Class Test, Midterm, Final
CO3	PO1, PO2	C4	Lecture, Problem solution	Assignment, Class Test, Midterm, Final
CO4	PO1, PO2	C2	Lecture, Problem solution	Assignment, Class Test, Midterm, Final

Course Title: Operating Systems Laboratory
Course Code: CSE 334

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite:**
CIE Marks: 40 **SEE Marks: 60**

Course Rationale:
This course provides a comprehensive understanding of the modern Operating System and examines the ways that design goals can be achieved.

Course Objectives:
The objectives of this course are to:
1. Learn and use the shell commands of Operating Systems.
2. Familiarize students with the architecture of an OS.
3. Provide necessary skills for developing and debugging programs in different environments.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Interpret (C2)** directory operation using GUI & Terminal.
CO2 **Use (C3)** different shell commands for basic system operation.
CO3 **Configure (C3)** different servers (such as Web, DNS, FTP, Email, etc.) through the knowledge gained in this course.
CO4 **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√											
CO3			√									
CO4										√		

Course Description:

SL No.	Course Contents	COs
1.	Introduction to various Operating Systems	CO1, CO4
2.	Basic Overview of Boot Process	CO1, CO4
3.	User management and Basic Shell commands	CO1, CO4
4.	Essential Commands	CO1, CO4
5.	Advanced commands & tools for System Administration	CO1, CO4

6.	Shell script and Scheduling	CO1, CO4
7.	Configuring a basic server	CO1, CO4

Text Books, Reference Books, and Other Resources:

1. Red Hat Enterprise Linux 6 Administration By *Sander Van Vugt*.
2. https://linuxconfig.org
3. https://www.linuxfoundation.org

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1, PO2	C2	Demonstration, discussion, experiment	Assignment, Performance
CO2	PO1	C3	Demonstration, discussion, experiment	Assignment, Performance, Report, Quiz
CO3	PO3	C3	Demonstration, discussion, experiment	Assignment, Performance, Lab Exam
CO4	PO10	A3	Demonstration	Report

Course Title: Computer Organization & Architecture
Course Code: CSE 337

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
Intended to enable the learners to gather proper knowledge in hardware design by understanding the fundamentals of core and modern computer architectures and hardware.

Course Objective:
The objectives of this course are:
1. To understand the basic structure and operation of a digital computer.
2. To familiarize students with hardware design and the operation of the arithmetic unit including the algorithms.
3. To learn different ways to communicate with I/O devices and standard I/O interfaces.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to-

CO1 **Describe (C2)** the basics of computer hardware and how software interacts with computer hardware.
CO2 **Demonstrate (C2)** computer performance using various assessing methods.
CO3 **Explain (C2)** how computers represent and manipulate data.
CO4 **Assemble (C3)** a simple computer with hardware design including data format, instruction format, instruction set, addressing modes, bus structure, input/output, memory, arithmetic/logic unit, control unit, and data, instruction and address flow.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											
CO4	√	√	√									

Course Description:

SL No.	Course Contents	COs
1.	**Instructions:** Language of the computer, introduction, Operations of the Computer Hardware, Operands of the Computer Hardware, Representing Instructions in Computer, Logical Operations.	CO1
2.	**Assessing and Understanding Performance:** CPU Performance and its Factors, Evaluating Performance, Fallacies and Pitfalls.	CO1

3.	**Basic Computer Organization & Design:** Instruction Codes, Direct/Indirect Address, Computer Registers and Bus systems	CO2
4.	**Design of a Basic Computer**: Computer Instruction, Timing and Control, Instruction Cycle, Interrupts	CO2
5.	**Enhancing Performance with Pipelining**: An overview of Pipelining, A Pipelined Datapath, Pipelined Control, Data Hazards and forwarding, Branch Hazards, Fallacies and Pitfalls	CO3
6.	**Central Processing Unit**: General Register Organization, Stack Organization, Instruction Formats, Reverse polish notation, Instruction Formats, Address Instructions, Addressing Modes, Data Transfer and Manipulations.	CO3
7.	**Computer Arithmetic**: Multiplication Algorithm, Division Algorithm, Floating-Point Arithmetic Operation.	CO3
8.	**I/O Organization**: I/O Interface, Asynchronous Data Transfer, Strobe Control Handshaking, Asynchronous Serial Transfer, Asynchronous Communication Interface, DMA, DMA Controller, DMA Transfer in a computer system	CO4
9.	**Memory Organization**: Memory Hierarchy, Main Memory, Auxiliary Memory, Cache Memory, Memory Mapping	CO4
10.	**Multiprocessors**: Characteristics of multiprocessors, Interconnection Structures, Cache Coherence	CO4

Text Books, Reference Books, and Other Resources:

1. M.Morris Mano, "Computer System Architecture", 3rd Edition, Prentice-Hall India, 2003, ISBN-81-203-0855-7
2. David A. Patterson, "Computer Organization and Design", 3rd Edition.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture, Slide Presentation	Class Test, Assignment, Midterm, Final Exam
CO2	PO1	C2	Lecture, Slide Presentation	Class Test, Assignment, Midterm, Final Exam
CO3	PO1	C2	Lecture, Slide Presentation	Class Test, Assignment, Midterm, Final Exam
CO4	PO1, PO2, PO3	C3	Lecture, Slide Presentation	Class Test, Assignment, Midterm, Final Exam

Course Title: Software Development
Course Code: CSE 338

Credits: 2 **Class Hours/Week: 4**
Course Type: Core **Pre-requisite: DMS**
CIE Marks: 30 **SEE Marks: 70**

Course Rationale:
In the Software Development course, the students will learn how to build software from scratch. The program focuses on developing responsive designs, validation and testing, analyzing different methods for solving a particular problem, working in a group for continuous delivery etc. Students will learn about the latest project management tools, how to model a problem from various domains.

Course Objectives:
The objectives of this course are:
1. To contribute effectively to the development process of a company.
2. To make students eligible to deal with changes during the middle of the development of software.
3. To prepare students to adapt to new technologies in the future.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Implement (C3)** various server-side and client-side concepts using latest framework.

CO2 **Use (C3)** 3rd party packages for developing software.

CO3 **Work (A4)** in an individual or group project maintaining engineering ethics.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√			√							
CO2	√	√			√							
CO3	√	√	√	√	√	√	√	√	√	√	√	√

Course Description:

SL No.	Course Content	COs
1.	**Introduction:** Familiarization to development framework, Client-side technologies, Server-side technologies, Report Writing with LATEX, Project Requirements	CO1, CO2, CO3
2.	**Server-side and Client-side concepts:** Design pattern, Routing, Interaction with database, CRUD operation, Authentication and Authorization, Validation, File Uploading, Client-Server	CO1, CO2

	Communication, API integration, Responsive Designs, 3rd Party package Integration	
3.	**Project:** Requirements analysis, identifying scopes, modeling the specifications, implementation, and representing all the steps via report	CO3

Text Books, Reference Books, and Other Resources:

1. World-wide web, YouTube Tutorials

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1, PO2, PO5	C3	Demonstration, experiment	Performance, Assignment
CO2	PO1, PO2, PO5	C3	Demonstration, experiment	Performance, Assignment
CO3	PO1, PO2, PO3, PO4, PO5, PO6, PO7 PO8, PO9, PO10, PO11, PO12	A4	Project Related Instruction	Project, Report, Presentation

Course Title: Data Communication
Course Code: CSE 364

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite:**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
Data communication, which is the transmission of digital data through a network or to a device external to the sending device, is the cornerstone of modern telecommunications. Given the importance of different communication systems, this course is designed for Computer Science and Engineering students.

Course Objectives:
1. Enable the students to understand the fundamental concepts and techniques of Data Communication.
2. Equip the skill required to apply encoding and communication protocols.
3. Facilitate students to analyze the error detection and correction techniques.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Discuss** (**C2**) the concepts of data communication systems and their components.
CO2 **Explain** (**C2**) the digital and analogue representations and channels.
CO3 **Demonstrate** (**C3**) the mechanism and techniques of encoding.
CO4 **Describe** (**C2**) the general principles of communication protocol switching techniques
CO5 **Implement** (**C3**) different error detection, correction and flow control techniques.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3	√											
CO4	√											
CO5	√											

Course Description:

SL No.	Course Content	COs
1.	**Introduction**: Data Communications, components, Data Representation, Data Flow	CO1
2.	**Data and Signals**: Analog and Digital, Periodic Analog signals, Digital Signals, Time and Frequency Domain, Composite Signal. Bandwidth, Digital Signal, Transmission of digital signal, Transmission Impairment, Noise, SNR. Data Rate Limits, capacity Theorem, Nyquist Bit rate, Shannon Bit rate. Performance, bandwidth Delay Product.	CO2

3.	**Modulation and demodulation**: amplitude modulation, frequency and phase Modulation Analog Transmission: Digital-to-Analog Conversion, Constellation Diagram	CO3
4.	**Modulation** and **demodulation**: amplitude modulation, frequency and phase Modulation Analog Transmission: Digital-to- Analog Conversion, Constellation Diagram	CO3
5.	**Bandwidth Utilization**: Multiplexing, Analog Hierarchy, Interleaving, Data Rate Management, Digital Hierarchy STDM, Spread Spectrum	CO2
6.	**Error Detection and Correction**: Introduction, Single Bit Error, Burst Error, Detection Vs Correction, Forward Error correction Vs retransmission, Block Coding Hamming Distance, Linear Block Codes, Flow control techniques.	CO5
7.	Circuit and Packet Switching techniques, different communication protocols	CO4

Text and Reference books:

1. Data Communication and Networking by Behrouz A Forouzan
2. Data and Computer Communications (9th Edition), by William Stallings, Publishers: Prentice Hall

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1	C2	Lectures/Instructions Reading Materials/PPT slides	Class Test, Midterm, Assignment, Final
CO2	PO1	C2	Lectures/Instructions Reading Materials/PPT slides	Class Test, Midterm, Assignment, Final
CO3	PO1	C3	Lectures/Instructions Reading Materials/PPT slides	Class Test, Midterm, Assignment, Final
CO4	PO1	C2	Lectures/Instructions Reading Materials/PPT slides	Class Test, Midterm, Assignment, Final
CO5	PO1	C3	Lectures/Instructions Reading Materials/PPT slides	Class Test, Midterm, Assignment, Final

Course Title: Computer Networks
Course Code: CSE 367

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: ICS**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course aims to introduce the basic concept and essential knowledge of computer networks.

Course Objectives:
The objectives of this course are:
1. Building an understanding of the fundamental concepts of computer networking.
2. Familiarizing the student with the basic taxonomy and terminology of the computer networking area.
3. Introducing the students to advanced networking concepts, preparing the student for getting entry into advanced courses in computer networking.
4. Allowing the students to gain expertise in some specific areas of networking such as the design and maintenance of individual networks.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Interpret** (**C2**) the OSI and TCP/IP layered architecture, different TCP/IP layer protocols, network organization, the operations of the key networking components, advance network architecture: software defined networking, 4G/5G, etc.

CO2 **Describe** (**C2**) the functionalities and performance of different routing and transport layer protocols, flow and congestion control mechanisms, link layer protocols, error detection and correction mechanisms, OpenFlow protocols.

CO3 **Practice** (**C3**) the acquired knowledge to configure/design wired/wireless networks and sub-networks, routing protocols, security principles to networking, solve different network issues, etc.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√										
CO3	√	√										

Course Description:

SL No.	Course Content	COs
1.	**Computer Networks and the Internet:** Operations: protocol layers, service models, Performance Metrices: delay, loss, throughput	CO1

2.	**Application Layer:** Principles of network applications, Web and HTTP, electronic mail SMTP, POP3, IMAP, DNS, socket programming with UDP and TCP	CO1, CO2
3.	**Transport Layer:** Transport-layer services, 3.2 multiplexing and demultiplexing, connectionless transport: UDP function, The TCP Connection, TCP Segment Structure, Round-Trip Time Estimation and	CO1, CO2, CO3
4.	**Network Layer: Data Plane** Overview of Network layer, data plane, control plane, What's inside a router? IP: Internet Protocol, datagram format, Fragmentation, IPv4 addressing,	CO2, CO3
5.	**Network Layer: Control Plane** Introduction, routing protocols, link state, distance vector, intra-AS routing in the Internet: OSPF, routing among the ISPs: BGP	CO2, CO3
6.	**The Link Layer and LANs** Link Layer introduction, services, The Services Provided by the Link Layer, Where Is the Link Layer Implemented? Error-Detection and -Correction	CO1, CO2, CO3
7.	**Wireless and Mobile Networks:** Elements of a wireless network, Wireless network taxonomy, Wireless link characteristics, Hidden Terminal Problem and signal attenuation, CDMA, 802.11, IEEE 802.11 MAC Protocol,	CO1, CO2

Text Books, Reference Books and Other Resources:

1. Computer Networking: A Top-Down Approach---- (James F. Kurose, Keith W. Ross)
Instructor contents and PPT slides: **http://gaia.cs.umass.edu/kurose_ross/instructor.php**
2. TCP/IP illustrated. --- (Richard Stevens)
3. Data Communications and Networking: The Basic Toolbox--- (Behrouz A. Forouzan)
4. Computer Networks--- (Andrew S. Tanenbaum and David J)
5. Additional contents: Cisco PowerPoint slides, pdf tutorials will be provided.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2	C2	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam
CO2	PO1, PO2	C2	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam
CO3	PO1, PO2	C3	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam

Course Title: Computer Networks Laboratory
Course Code: CSE 368

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: ICS**
CIE Marks: 70 **SEE Marks: 30**

Course Rationale:
This subject aims to teach an understanding of computer networks and systems design through hands-on lab works and analysis with real world applications.

Course Objectives:
The objective of the course is to enable the students to

1. Understand and implement the functionalities of major networking components, network topologies, and connections.
2. Become familiar with configuration of networking and routing protocols, network operations, state-of-the-art developments and applications.
3. Administer advanced topics such as WLAN, network security, threats mitigation, security best practices, network management, and/or other topics.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Identify** (**C1**) basic networking and end devices: real equipment and simulation, install a home or small business network, connection to the Internet.

CO2 **Illustrate** (**C2**) different application layer protocols, distance vector and link-state routing protocols, and transport layer protocols

CO3 **Enhance** (**C2, A3**) network security using access control lists and security best practices.

CO4 **Use** (**C3**) the acquired knowledge to establish, verify and troubleshoot network and Internet connectivity.

CO5` **Complete** (**P3**) different types of network cables and device connections by hands-on.

CO6 **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√										
CO3		√			√			√				
CO4		√			√							
CO5	√				√							
CO6										√		

Course Description:

SL No.	Course Content	COs
1.	Introduction to network simulation tools, Introduction to different types of network cables	CO1, CO5, CO6
2.	Configure Application layer services	CO2, CO6
3.	Basic Router and Switch Configuration, Routing protocol configuration, Network Address translation	CO2, CO4, CO6
4.	Subnetting, VLSM, and IPv6	CO2, CO4, CO6
5.	Access control lists	CO3, CO6
6.	Wireless router configuration	CO3, CO4, CO6
7.	Implement a custom topology	CO4, CO6

Text Books, Reference Books and Other Resources:

1. Computer Networking: A Top-Down Approach---- (James F. Kurose, Keith W. Ross)
   -- Instructor contents and PPT slides:
   http://gaia.cs.umass.edu/kurose_ross/instructor.php
2. Cisco Packet Tracer Simulation
3. Additional contents: Cisco PowerPoint slides, pdf tutorials will be provided.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2	C1	Demonstration, discussion, experiment	Quiz/Written Exam, Performance
CO2	PO1, PO2	C2	Demonstration, discussion, experiment	Quiz/Written Exam, Performance
CO3	PO2, PO5, PO8	C2, A3	Demonstration, discussion, experiment	Quiz/Written Exam, Performance
CO4	PO2, PO5	C3	Demonstration, discussion, experiment	Quiz/Written Exam, Performance
CO5	PO1, PO5	P3	Demonstration, discussion, experiment	Quiz/Written Exam, Performance
CO6	PO10	A3	Demonstration	Report

Course Title: Theory of Computation
Course Code: CSE 309

Credits: 2 **Class Hours/Week: 2**
Course Type: Core **Pre-requisite: DM**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
Theory of computation lays a strong foundation for abstract areas in computer science. Theory of computation teaches the learners about the elementary ways in which a computer can be made to think, how efficiently problems can be solved on a model of computation, using an algorithm, tends to set the foundation for understanding Compilers and interpreters.

Course Objectives:
The objectives of this course are:
1. To identify with the fundamental limits on what can be efficiently computed in our universe and other possible universes.
2. To analyze limits to reveal deep and mysterious properties about information, knowledge and processing, as well as practical issues about what can and cannot be computed.
3. To design & build models for complex computation.

Course Outcomes (COs):
After successful completion of this course, students will be able to:

CO1 **Comprehend (C2)** formal reasoning about languages.
CO2 **Categorize (C4)** types of different computing languages.
CO3 **Investigate (C4)** advanced knowledge of formal computation and its relationship to languages.
CO4 **Illustrate (C2)** a competent understanding of the concepts of complexity theory.

Mapping, of Course Learning Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√	√										
CO3		√		√								
CO4	√	√										

Course Description:

SL No.	Course Contents	COs
1	Language theory, Regular Language: Deterministic and nondeterministic finite automata and their equivalence, conversion of deterministic and nondeterministic finite automata	CO1
2	Equivalence with regular expressions, Closure properties, The pumping lemma and applications, pushdown automata	CO2

3	Context-free Grammars and Languages: Definitions, Parse trees, Ambiguous Grammar	CO2
4	Turing Machines: basic machines, configuration, computing with Turing machines, combining Turing machines	CO3
5	The pumping lemma for CFLs and applications, Chomsky Normal forms	CO3
6	General parsing, Sketch of equivalence with pushdown automata.	CO4
7	Decidability: The undecidability of the halting problem. Reductions to other problems	CO4

Text Books, Reference Books and Other Resources:

1. Michael Sipser: Introduction to the Theory of Computation
2. Jeffrey Ullman, John Hopcroft, Rajeev Motwani: Introduction to Automata Theory, Languages, and Computation
3. Anil Maheshwari, Michiel Smid: Introduction to Theory of Computation

Mapping CO with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1	C2	Lecture, Practical Implementation	Class Test, Assignment Midterm, Final Exam
CO2	PO1, PO2	C4	Lecture, Practical Implementation	Class Test, Assignment Midterm, Final Exam
CO3	PO2, PO4	C4	Lecture, Practical Implementation	Class Test, Assignment Midterm, Final Exam
CO4	PO1, PO2	C2	Lecture, Practical Implementation	Class Test, Assignment Midterm, Final Exam

Course Title: Network and Computer Security
Course Code: CSE 437

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: CN**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course is meant to offer Computer Science undergraduate students a broad overview of the field of computer security. Students will learn the basic concepts in computer security including software vulnerability analysis and defense, networking and wireless security, and applied cryptography. Students will also learn the fundamental methodology for how to design and analyze security critical systems.

Course Objectives:
1. The course covers principles of computer systems and network security.
2. We will discuss various attack techniques and how to defend against them.
3. Topics include network attacks and defenses, web security, e-mail, malware, mobile security and privacy.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Elaborate** (**C2**) the concepts, issues, principles and theories of computer, web, and Internet security

CO2 **Apply** (**C3**) and **explain** (**A3**) a range of computer network security technologies and tools and services in critical situations

CO3 **Analyze** (**C4**) solutions to security challenges using common network security tools and formal methods.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√			√			√				
CO3		√		√	√							

Course Description:

SL No.	Course Contents	COs
1.	Principle of Security, Security goals, Integer Algorithms, Modular Arithmetic and Linear Congruence	CO1
2.	Cryptography (Symmetric key cryptography): Data Encryption Standard (DES), Advanced Encryption Standard (AES)	CO1, CO2, CO3
3.	Cryptography (Asymmetric key cryptography): RSA, Diffie Helman	CO2, CO3

4.	Cryptosystems: ElGamal and Rabin's Cryptosystems	CO2, CO3
5.	Authentication, Digital Signatures, Digital Signature Standard (DSS), Randomized Encryption; Rabin and ElGamal signature schemes, Cryptographically Secure Hashing; Message Authentication Codes	CO2, CO3
6.	Web/Network security: IPSec, SSL/TLS, Kerberos, PGP, Packet-Filter Firewall, Proxy Firewall, System Security Issues: Malware Analysis.	CO2, CO3
7.	Different types of web vulnerabilities, Introduction to digital forensics concepts, Security in mobile platform, mobile threats and malware.	CO2, CO3

Text Books, Reference Books and Other Resources:

1. Behrouz A Forouzan: Cryptography and Network Security
2. Contemporary Reference books/materials.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2	C2	Lectures notes, PPT slides, Practice Problems, Video	Assignment, Class Test, Midterm, Final Exam
CO2	PO1, PO2, PO5, PO8	C3, A3	Lectures notes, PPT slides, Practice Problems, Video	Assignment, Class Test, Midterm, Final Exam
CO3	PO2, PO4, PO5	C4	Lectures notes, PPT slides, Practice Problems, Software, Video	Assignment, Class Test, Midterm, Final Exam

Course Title: Neural Network & Fuzzy Logic
Course Code: CSE 451

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: SP**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course tends to help the learners to gain knowledge about design and working principle of artificial neural network and its relation to biological neuron. It helps to develop and design different neuron models according to problem, introduces artificial intelligence and its application areas, application and evaluation of fuzzy logic and fuzzy set theories.

Course Objectives:
The objectives of this course are:
1. Facilitating necessary knowledge about the design and working principle of Artificial Neural Networks (ANN) and Deep Learning (DL).
2. Teaching the students how the mathematics behind Neural Networks (NN) Learning actually works, rather than presenting only a cursory or surface-level description.
3. Helping the students conceptualize and understand the key parameters in a neural network's architecture and apply them to their own applications.
4. Gaining the basics of different application areas and evaluation of fuzzy logic and fuzzy set theories.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Illustrate (C3)** how different components are combined to form a Single-layer (Perceptron) and Multi-layer NN and their basic working mechanism.
CO2 **Use (C3)** the knowledge about the principle and mathematics behind the Training of a simple NN (Loss/Cost Function, Gradient Descent, Computation Graph).
CO3 **Explain (C2)** fuzzy systems' fundamental concepts (fuzzy number, fuzzy relation, composition of fuzzy relation).
CO4 **Implement (C3)** a fuzzy system representing uncertain knowledge using fuzzy rules.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√										
CO3	√	√										
CO4	√	√		√								

Course Description:

SL No.	Course Contents	COs
1.	**A Gentle Introduction to NN:** Differences among AI/ML/DL, Subcategories of ML and their working principle, Decision Boundary, Classification Example: AND, OR, XOR, Biological Vs. Artificial NN	CO1
2.	**Applications of Neural Network:** An abstract example of NN, Supervised Learning with NN, Popular NN Architectures (ANN, CNN, RNN, Transformers), Reason behind Deep Learning's success	CO1
3.	**Structural Building Blocks of NN:** Input/Output Layers, Weights and Bias, Activation Function (non-linearity), Summation Vs. Matrix.	CO2
4.	**Single Layer Perceptron (A Simple NN):** Linear Regression, Training set, Input/Output, Forward Propagation, Pitfall of Single Layer Perceptron.	CO2
5.	**Multi-Layer NN:** Concepts of Hidden Layer, XOR Calculation Using Summation Formula, ReLU Activation Function, Vectorized Representation of Multi-Layer NN, Adding Bias as a weight.	CO2
6.	**How NN's Learns:** Weight Initialization (Random vs. Zero), Expected Vs Predicted Output, Loss and Cost Function, Back-propagation Basics.	CO3
7.	**Logistic Regression (LR):** The pitfalls of Linear Regression, Binary Classification, From Linear to Logistic Regression, Sigmoid Activation Function.	CO1
8.	**Loss and Cost Function:** Training/Dev/Test set, Loss function: M.S.E, Convex Vs. Non-convex, Local Vs. Global Optima, Log/Cross-Entropy Loss., L1 and L2 M.A.E, Cost Function, SoftMax Activation Function.	CO3
9.	**Gradient Descent (G.D):** Finding the Global Optima, Derivatives, Weights and Bias Update, Learning Rate, and Computation Graph.	CO3
10.	**Backward Propagation:** Back-prop using Computation Graph, G.D with Backpropagation for LR, G.D with 'M' training examples, G.D Algorithm using Summation, Vectorized G.D, Updating Weights and Bias, Iteration Vs. Batch Vs. Epochs.	CO3
11.	**Introduction to Fuzzy Logic:** Fuzzy Set, Fuzzy Set Operations - Union, Intersection, Complement, Properties of Fuzzy Set, Extension Principles, Alfa-cuts.	CO4
12.	**Fuzzy Relations:** Properties, Basic Operations, Compositions of Fuzzy Relations.	CO4
13.	**Fuzzy Number:** Representation, Properties, Addition, Subtraction of Discrete and Continuous Fuzzy Number, Addition and Subtraction of Discrete Fuzzy Number through Extension Principle, Multiplication and Division of Fuzzy Number.	CO4

14.	**Fuzzy Linguistic Description:** Linguistic Variables and Values, Implication Relations, Fuzzy Inference, and Composition.	CO4

Text Books, Reference Books and Other Resources:

1. Dive into Deep Learning – By Amazon Research Scientists.
2. "Make Your Own Neural Network" – By Tariq Rashid
3. "Fundamentals of Deep Learning", O'Reilly – By Nikhil Buduma.
4. "Deep learning" – By Ian Goodfellow and Yoshua Bengio and Aaron Courville.
5. Neural Networks and Deep Learning – By Michael Nielsen.
6. "Fuzzy and Neural Approach in Engineering" – Lefteri H. Tsoukalas, Robert E. Uhrig.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2	C3	Lecture, Problem solution, Video, Web Link	Class Test, Assignment, Midterm, Final Exam
CO2	PO1, PO2	C3	Lecture, Problem solution, Video, Web Link	Class Test, Assignment, Midterm, Final Exam
CO3	PO1, PO2	C2	Lecture, Problem solution, Video, Web Link	Class Test, Assignment, Midterm, Final Exam
CO4	PO1, PO2, PO4	C3	Lecture, Problem solution, Video, Web Link	Class Test, Assignment, Midterm, Final Exam

Course Title: Neural Network & Fuzzy Logic Laboratory
Course Code: CSE 452

Credits: 1 **Class Hours/Week: 2**
Course Type: Core **Pre-requisite: SPL**
CIE Marks: 30 **SEE Marks: 70**

Course Rationale:
This subject aims to gain knowledge about design and working principle of neural network (NN) as well as practically implement NN for real-world applications.

Course Objectives:
The objectives of the course are:

1. To teach students the foundations of implementing NN's using modern tools and libraries.
2. To disseminate the knowledge of how to appropriately report on the experiments they run, as done in academic publications.

Course Outcomes:
Upon successful completion of this course, students will be able to

CO1 **Use (C3)** proper environment and software packages necessary for preprocessing, developing, and analyzing the performance of neural network.

CO2 **Build (C3)** a NN using modern software tools and extend it by adding multiple layers to it to understand how Multi-layer NN works.

CO3 **Investigate (C4)** the results of a neural network.

CO4 **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√				√							
CO2	√	√	√		√				√			
CO3	√	√	√	√					√			
CO4										√		

Course Description:

SL No.	Course Contents	COs
1.	Environment Setup: General Issues, Setting-up Anaconda/CoLab, Recap of Python Containers and Collection Modules, NumPy Basics.	CO1, CO4
2.	Introduction to the different data sources, reading, preprocessing, and splitting the data into training and test set	CO1, CO2, CO4
3.	Building A Simple NN using latest frameworks. Compiling Multi-Layer NN and Parameter/Hyper Parameter Tuning, Regression using Neural Networks,	CO1 - CO4

	Classification using Neural Networks, plotting loss and accuracy curve of training and validation set, Performance Measure	

Text Books, Reference Books and Other Resources:

1. "Deep Learning with Python" – By Francois Chollet
2. "Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow", O'Reilly, 2nd Ed. (2019) – By Aurélien Géron
3. "Deep Learning from Scratch: Building with Python from First Principles" – By Seth Weidman

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO5	C3	Lecture, Lab Task	Project, Presentation, Assignment, Lab Performance, Viva
CO2	PO1, PO2, PO3, PO5, PO9	C3	Lecture, Lab Task	Project, Presentation, Assignment, Lab Performance, Viva
CO3	PO1, PO2, PO3, PO4, PO9	C4	Lecture, Lab Task	Project, Presentation, Assignment, Lab Performance, Viva
CO4	PO10	A3	Demonstration	Report

Course Title: Computer Graphics & Image Processing
Course Code: CSE 455

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: SP**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
Computer graphics concentrates on the fundamentals of computer graphics and addresses the knowledge and skill in computer graphics development which are essential for computing professionals. Image processing, on the other hand emphasizes on general principles of image processing and its application.

Course Objectives:
The objectives of this course are:
1. To develop a theoretical foundation of fundamental computer Graphics & Digital Image Processing concepts.
2. To learn basic algorithms for computer graphics and image processing.
3. To understand and imply various filters, Point processing, and Arithmetic operations in image processing.
4. To understand different applications of graphics and image processing.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Explain (C2)** the fundamental concepts of computer graphics and image processing.
CO2 **Demonstrate (C3)** the scan conversion algorithms for drawing various types of geometric shapes.
CO3 **Illustrate (C3)** the ideas of 2D and 3D transformation and clipping techniques.
CO4 **Use (C3)** different types of image processing techniques for transformation, filtering, smoothing and enhancement.
CO5 **Solve (C3)** different image compression algorithms, segmentation and feature extraction techniques.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√	√										
CO3	√	√										
CO4	√	√										
CO5	√	√										

Course Description:

SL No.	Course Contents	COs
1.	**Fundamental Concepts:** Image representation, color model, pixel, rasterization, lookup table, image types, graphical hardware, image fundamentals, sampling, quantization, mathematical tools.	CO1
2.	**Scan conversion:** Scan conversion algorithms for point, line, circle, ellipse, rectangle**,** Region filling, scan converting character, anti-aliasing	CO2
3.	**2D and 3D Transformation, viewing, clipping:** Geometric transformation, translation, rotation, scaling, mirror reflection, co-ordinate transformation, composite transformation, instance transformation, viewport mapping, shape clipping, clipping algorithms, curves	CO3
4.	**Image processing:** Fundamental concepts, color model conversion, smoothing, filtering, sharpening, fourier transformation, image enhancement, image restoration.	CO1, CO4
5.	**Compression and segmentation:** Huffman coding, Arithmetic coding, LZW coding, Bit-plane coding, Wavelet coding, watermarking, thresholding, segmentation, feature extraction.	CO5

Text Books, Reference Books and Other Resources:

1. Computer Graphics: Zhigang Xiang and Roy Plastock
2. Digital Image Processing: Rafael C. Gonzalez, Richard E. Woods
3. Computer Graphics: Principles and Practice: James D Foley, Andries Van Dan

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture, Slide	Class Test, Assignment, Midterm, Final
CO2	PO1, PO2	C3	Lecture, Slide	Class Test, Assignment, Midterm, Final
CO3	PO1, PO2	C3	Lecture, Slide	Class Test, Assignment, Midterm, Final
CO4	PO1, PO2	C3	Lecture, Slide	Class Test, Assignment, Midterm, Final
CO5	PO1, PO2	C3	Lecture, Slide	Class Test, Assignment, Midterm, Final

Course Title: Computer Graphics & Image Processing Laboratory
Course Code: CSE 456

Credits: 1 **Class Hours/Week: 2**
Course Type: Core **Pre-requisite: SPL**
CIE Marks: 40 **SEE Marks: 60**

Course Rationale:
In Computer Graphics & Image Processing Laboratory course, the students will learn various computer graphics algorithms. In this program, the students will have to implement the mostly used computer graphics algorithms like line drawing, polygon clipping, line clipping etc. For the image processing part, students will learn about different color models and conversion among them. In this part, binary image manipulation will be performed.

Course Objectives:
The objective of the course is to enable the students to
1. Contribute effectively in the research area of computer graphics and image processing.
2. Students will learn about the practical use of various computer graphics and image processing algorithms**.**
3. Students will be able to implement new algorithms to overcome the existing problems**.**

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Implement** (**C3**) various scan conversion algorithms using available graphical tools.

CO2 **Execute** (**C3**) various graphical transformation methods.

CO3 **Implement** (**C3**) various image processing techniques.

CO4 **Demonstrate** (**C3**) the skills of computer graphics and image processing to develop a mini project.

CO5 **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓				✓							
CO2	✓				✓							
CO3	✓				✓							
CO4	✓	✓	✓		✓							
CO5										✓		

Course Description:

SL No.	Course Contents	COs
1.	**Scan Conversion Algorithm:** Digital Differential Analyzer Algorithm, Bresenham Line Drawing Algorithm, Midpoint Circle Drawing Algorithm.	CO1, CO5
2.	**Transformation:** Geometric transformation, translation, rotation, window to viewport mapping.	CO2, CO5
3.	**Image Processing:** Color model, Binary image analysis, Image restoration techniques.	CO3, CO5
4.	**Mini Project Development:** Unity, OpenGL or any other platforms/tools.	CO4, CO5

Text Books, Reference Books and Other Resources:

1. Computer Graphics (Second Edition): Zhigang Xiang
2. Digital Image Processing: Rafael C. Gonzalez, Richard E. Woods
3. Computer Graphics: Principles and Practice: James D Foley, Andries Van Dan

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO5	C3	Lecture, Slide	Performance, Assignment
CO2	PO1, PO5	C3	Lecture, Slide	Performance, Assignment
CO3	PO1, PO5	C3	Lecture, Slide	Performance, Assignment
CO4	PO1, PO2, PO3, PO5	C3	Lecture, Slide	Performance, Assignment
CO5	PO10	A3	Demonstration	Report

Course Title: Control Systems
Course Code: EEE 313

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: SS**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale: This course aims to provide basic knowledge of different control techniques used in industries to control electrical and mechanical devices.

Course Objectives:
The objectives of this course are to:
1. Acquaint students with the basic knowledge of modern control systems.
2. Help the students develop their ability in developing mathematical models of electrical system.
3. Get idea about the principle of block diagram reduction and signal flow graphs.
4. Provide knowledge of the stability of electrical and mechanical systems.
5. Help them conceptualize the basic theories of the system analysis technique.
6. Make the students understand the basic knowledge of different types of controllers.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Describe (C2)** the fundamentals of control systems.
CO2 **Construct (C4, A4)** mathematical models of different Control systems satisfying desirable control objectives.
CO3 **Investigate (C4)** different control systems using block diagrams and signal flow graphs.
CO4 **Design (C4)** different control systems using classical approaches and modern techniques.
CO5 **Inspect (C4)** system stability.
CO6 **Explain (C1)** the basics of nonlinear control.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√	√	√									
CO3	√	√										
CO4	√	√	√									
CO5		√	√									
CO6	√											

Course Description:

SL No.	Course Contents	COs
1.	**Introduction:** Introduction to the control system, open loop and closed loop systems, the design process.	CO1
2.	**Mathematical Model:** Review of Laplace transform, initial and final value theorems, transfer functions: Open-loop stability, Poles Zeros state, space	CO2

	representation/transfer function/zero-pole of control system design; state space representation; solution of state equation.	
3.	**Block diagram approach:** Signal flow graph; block diagram theory; block diagram reduction method;	CO3
4.	**Classical Control System:** Phase lead and lag controllers, Linear control system design using state feedback, Closed-loop sensitivity functions, LQR design, pole placement, lead compensation, lag compensation, lead-lag compensation.	CO4
5.	**Modern Control System:** Application of Eigen value and Eigen vectors, state variable analysis, canonical forms, controllability and observability, controller, and observer design, Riccati equation, data driven control basics.	CO4
6.	**Stability analysis:** Analysis methods such as Nyquist stability criterion, root locus, routh's criteria, Stability margins, gain and phase margin, maximum magnitude, resonant frequency.	CO5
7.	**Controller Design:** P, I, PI, PD, and PID types, optimum controller design, robust controller design.	CO2
8.	**Non-linear control:** Introduction to nonlinear control, Lyapunov stability criteria. Introduction to neural and fuzzy control.	CO6

Textbooks, References and Other Resources:

1. Control Systems Engineering by Norman S. Nise
2. Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control by J. Nathan Kutz and Steven L. Brunton
3. Linear control system analysis and Design, John J. Azzo

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture, Slide, Problem Solving, and examples	Class Test, Midterm, Assignment, Final Exam
CO2	PO1, PO2, PO3	C4, A4	Lecture, Slide, Problem Solving, and examples	Class Test, Midterm, Assignment, Final Exam
CO3	PO1, PO2	C4	Lecture, Slide, Problem Solving, and examples	Class Test, Midterm, Assignment, Final Exam
CO4	PO1, PO2, PO3	C4	Lecture, Slide, Problem Solving, and examples	Class Test, Midterm, Assignment, Final Exam
CO5	PO2, PO3	C4	Lecture, Slide, Problem Solving, and examples	Class Test, Midterm, Assignment, Final Exam
CO6	PO1	C1	Lecture, Slide, Problem Solving, and examples	Class Test, Midterm, Assignment, Final Exam

Course Title: Control Systems Laboratory
Course Code: EEE 314

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: SSL**
CIE Marks: 30 **SEE Marks: 70**

Course Rationale:
This course aims to build foundation skills on designing and analyzing different industrial control systems using PLC and MATLAB.

Course Objectives:
The objectives of this course are to:
1. Help students demonstrate the controlling of systems using PLC
2. Improve the ability to present the results of investigations.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 Implement (**C3, P2**) ladder diagram for PLC based automation system and design the best controller based on the systems requirement using CAD tool as an individual or as a member of a team.
CO2 Evaluate (**C4**) the results from experimental data.
CO3 Write (**A3**) comprehensive reports on the work done in laboratory.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√			√							
CO2		√		√								
CO3										√		

Course Description:

SL No.	Course Contents	COs
1.	Laboratory work based on theory course EEE 313. It will cover hardware-based work in PLC system.	CO1
2.	Report writing based on laboratory work.	CO2
3.	Presentation on mini project work.	CO3

Text Books, Reference Books and Other Resources:
1. Lab manual

CO Delivery and Assessment:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2, PO5	C3, P2	Lecture, Slide, Problem Solving, and examples	Performance, Assignment, Quiz, Viva
CO2	PO2, PO4	C4	Lecture, Slide, Problem Solving, and examples	Performance, Assignment, Quiz, Viva
CO3	PO10	A3	Demonstration	Report

Course Title: Technical Writing & Presentation
Course Code: ENG 401

Credits: 2 **Class Hours/Week: 2**
Course Type: Language **Pre-requisite: GE**
CIE Marks: 60 **SEE: 40**

Course Rationale:
The course aims to equip students with the skills and knowledge necessary to effectively communicate technical information in a clear, concise, and professional manner. In today's highly interconnected and information-driven world, the ability to convey complex technical concepts to various audiences is crucial in fields such as engineering, computer science, business, and science.

Course Objectives:
The objectives of this course are:
1. To develop the skills necessary to communicate technical information clearly, concisely, and accurately to various audiences.
2. To develop skills in collaboration, communication, and negotiation to produce cohesive and unified documents and presentations.
3. To acquire skills in designing and delivering effective visual presentations using tools like PowerPoint or other presentation software.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:
CO1 **Use (C3)** and **select (A2**) appropriate words to make sentence and then paragraph, reports etc.
CO2 **Use** (**C3**) and **practice** (**A2**) modern tools to prepare technical and scientific documents.
CO3 **Use** (**C3)** and **explain (A4**) information effectively.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1									√	√		
CO2					√				√	√		
CO3									√	√		

Course Description:

SL. No.	Course Contents	COs
1.	Beginning to Write: a) Making sensible sentences. b) Joining and expanding sentences. c) Contracting sentences. d) Logical development of sentences in context using an idea. e) Clear and effective communication of information.	CO1, CO2, CO3
2.	Reading for Writing: Students will be required to comprehend modern prose-passages drawn from different disciplines with attention to their (a) context, (b) vocabulary, and (c) syntax, and deliver feedback in the form of précis,	CO1, CO2, CO3

	summaries, and comprehension answers. They will also be required to present their writings before the class for discussion and reactions by the peers.	
3.	Expanding Writing: a) Writing paragraphs on technical aspects. b) Writing short, free and guided compositions. c) Developing essays on technical issues. d) Writing reports, memos and business letters. e) Editing compositions for clarity and effectiveness.	CO1, CO2, CO3

Text Books, Reference Books and Other Resources:

1. Engineer's Guide to Technical Writing by Kenneth G. Budinski
2. Thinking on Paper by J.H. Barton and V.A. Howard

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO9, PO10	C3, A2	Lectures notes, PPT slides, Audio/Video content, Problem Solving	Group Work, Class Test, Assignment, Presentation, Viva
CO2	PO5, PO9, PO10	C3, A2	Lectures notes, PPT slides, Audio/Video content, Editing Software, Problem Solving	Group Work, Class Test, Assignment, Presentation, Viva
CO3	PO9, PO10	C3, A4	Lectures notes, PPT slides, Audio/Video content, Problem Solving	Group Work, Class Test, Assignment, Presentation, Viva

Course Title: Thesis/Project
Course Code: CSE 400

Credits: 4 **Class Hours/Week: 4**
Course Type: Core **Pre-requisite: As Set by Supervisor**

Course Rationale:

This course introduces students to computer science and engineering research through planning and managing an innovative capstone research project, using research methods, and considering broader professional issues related to the project.

Course Objectives:

The objectives of the course are to:

1. Provide an opportunity to integrate and apply their previous engineering knowledge acquired throughout their undergraduate program to the solution of complex computing problems.
2. Develop creativity in analyzing and solving complex and possibly real-world or industry-related computing problems.
3. Develop skills to systematically design, develop, and document the solution of research projects.
4. Develop and enhance self-learning and life-long learning capabilities.
5. Make students aware of professional and ethical norms, conducts, and responsibilities in designing solutions to computing problems.
6. Facilitate modern software and engineering tools necessary to meet customer requirements.
7. Build the capacities needed to plan and manage a real-time project.
8. Gauge the impact of an engineering solution on the community, cultural context, physical and mental health, safety and security, environment, and sustainability.
9. Promote teamwork to solve real-world problems.
10. Develop written and verbal communication skills among students to communicate to a variety of audiences.

Course Outcomes (COs):

Upon successful completion of this course, students will be able to:

CO1 **Plan (C6, A3)** a project considering the social, cultural, global, health, and environmental responsibilities of the professional Engineer and the principles of sustainable design and development of national and international perspective

CO2 **Apply (C3, P3, A3)** modern tools and techniques to manage projects under commitment to professional and ethical responsibilities

CO3 **Develop (A3)** the ability to communicate effectively, not only with engineers but also with the community at large as an individual and in multi-disciplinary and multi-cultural teams

CO4 **Apply (C4, A4)** robust project risk identification, feasibility test, assessment and treatment process

CO5 **Develop (C4, A3)** the capacity to undertake lifelong learning by analyzing real world situations

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		√	√			√	√					
CO2					√			√				
CO3									√	√		
CO4			√	√							√	
CO5												√

Course Description:

SL No.	Course Contents	COs
1	The Capstone Project enables students to practice what they've learned in previous courses to the solution of a real-world computing problem. The types of research projects may include research-intensive, design, and development of a product or software or system, and solution to an engineering problem for an industry partner. The research will contribute new knowledge to the relevant field and achieve social, economic, environmental, and cultural outcomes. Students will work independently and or in a group under the guidance of a supervisor to carry out different tasks needed to solve problems and design solutions.	CO1, CO2, CO3, CO4, CO5

Textbooks, Reference Books and Other Resources:

1. World-wide web
2. ACM Digital Library
3. Digital Contents

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy domain/level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO2, PO3, PO6, PO7	C6, A3	Weekly meeting in-person or virtual, Web links, Reference material	Performance, Report, and Presentation
CO2	PO5, PO8	C3, P3, A3		
CO3	PO9, PO10	A3		
CO4	PO3, PO4, PO11	C4, A4		
CO5	PO12	C4, A3	Weekly meeting in-person or virtual, Web links, Reference material	Performance, Report, and Presentation

Course Title: Compiler Construction
Course Code: CSE 453

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: ToC**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
Intended to enable the students the basic techniques which underlie the practice of compiler construction and use the acquired knowledge to build a compiler for a language.

Course Objectives:
The objectives of this course are to:
1. To introduce the major concept areas of language translation and compiler design.
2. Provide the foundation for understanding the theory and practice of compilers.
3. To learn programming language translation and compiler design concepts, language recognition,
symbol table management, semantic analysis and code generation.
4. To provide practical programming skills necessary for constructing a compiler.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

- **CO1** **Describe (C2)** the basic concepts, principles, tools, and techniques of a compiler construction.
- **CO2** **Apply (C3)** the knowledge of different phases of a compiler to undertake language translation.
- **CO3** **Evaluate (C5)** the similarities and differences among various parsing techniques and grammar transformation techniques.
- **CO4** **Construct (C3)** context-free grammar and regular expressions to implement lexical syntactic and semantic structures.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√											
CO3		√		√								
CO4	√	√										

Course Description:

SL No.	Course Contents	COs
1.	**Introduction to compilers:** Introductory concepts, Types, Applications, Phases, compiler construction tools	CO1
2.	**Syntax directed translation:** Construction of syntax trees, Top down translation Parser and its role, Context free grammars, Syntax analysis	CO1

3.	**Lexical analysis:** Role of the lexical analyzer, Token specification, Recognition of tokens, Symbol table, Finite automata	CO2, CO4
4.	**Context free grammar:** Syntax error handling, Ambiguity, left recursion, Left factoring, Different types of parser.	CO3, CO4
5.	**Intermediate code generator:** DAG, three address code, code generation, Addressing mode, instruction cost, basic block, Flow graph.	CO2, CO3
6.	**Code optimization**: Basic concepts, Code optimization techniques	CO2
7.	**Syntax directed translation:** Syntax directed definition, Semantic rules analysis, Dependency graph, Annotated parse tree, implementation of SDT	CO1

Text Books, Reference Books and Other Resources:

1. Aho, Ulman & Ravishethi: Principal of Compiler Design.
2. Philip: Compiler Design Theory.
3. Willam A. Barrette: Compiler Construction, Theory and Design.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	POs	Bloom's Taxonomy domain/level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C2	Lecture notes, Slide	Class Test, Midterm, Assignment, Final Exam
CO2	PO1	C3	Lecture notes, Slide	Class Test, Midterm, Assignment, Final Exam
CO3	PO2, PO4	C5	Lectures notes, Slide	Class Test, Midterm, Assignment, Final Exam
CO4	PO1, PO2	C3	Lectures notes, Slide	Class Test, Midterm, Assignment, Final Exam

Course Title: Compiler Construction Laboratory
Course Code: CSE 454

Credits: 1.5 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: ToC**
CIE Marks: 40 **SEE Marks: 60**

Course Rationale:
Intended to enable the students the basic techniques which underlie the practice of compiler construction and use the acquired knowledge to build a compiler for a language.

Course Objectives:
The objectives of this course are:
1. To introduce the major concept areas of language translation and compiler design.
2. Provide the foundation for understanding the theory and practice of compilers.
3. To learn programming language translation and compiler design concepts, language recognition, symbol table management, semantic analysis and code generation.
4. To provide practical programming skills necessary for constructing a compiler.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

- **CO1** **Apply (C3)** software tools and techniques to construct compiler
- **CO2** **Apply (C3)** syntax analyzer, parser, semantic analyzer and various parsing techniques in Compiler Design
- **CO3** **Apply (C3)** mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems.
- **CO4** **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√				√							
CO2	√	√			√							
CO3	√	√			√							
CO4										√		

Course Description:

SL No.	Course Contents	COs
1.	Laboratory work based on theory course **Compiler Construction**	CO1, CO2, CO3
2.	Report generation based on Lab work	CO4

Text Books, Reference Books and Other Resources:

1. Aho, Ulman & Ravishethi: Principal of Compiler Design.
2. Philip: Compiler Design Theory.
3. Willam A. Barrette: Compiler Construction, Theory and Design

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1, PO5	C3	Problem analysis, Demonstration, Problem solving	Performance, Report, Quiz
CO2	PO1, PO2, PO5	C3	Problem analysis, Demonstration, Problem solving	Performance, Quiz, Report
CO3	PO1, PO2, PO5	C3	Problem analysis, Demonstration, Problem solving	Performance, Quiz, Report
CO4	PO10	A3	Demonstration	Report

Course Title: Machine Learning
Course Code: CSE 457

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: AI**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course provides appropriate machine learning techniques, and learning algorithm to best suit the current need and enhance the learning parameters for maximum performance.

Course Objectives:
1. Providing the students with an in-depth introduction to two main areas of Machine Learning: supervised and unsupervised
2. Introducing some of the main models and algorithms for regression, classification, and clustering
3. Developing an understanding of the theoretical relationships between the learning algorithms
4. Enhancing the learning parameters to achieve maximum performance

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Describe (C2)** the key concepts such as supervised learning, unsupervised learning, reinforcement learning, feature engineering, model evaluation, and trade-off bias and variance.
CO2 **Apply (C3)** supervised learning, unsupervised learning and reinforcement learning algorithms to datasets.
CO3 **Analyze (C4)** data for preprocessing and cleaning datasets to prepare them for machine learning tasks, gain insights and identify patterns in the data.
CO4 **Evaluate (C5)** and optimize the performance of machine learning models using model selection, hyperparameter tuning, model validation, and metrics such as accuracy, precision, recall, F1 score, and area under the curve (AUC).

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√										
CO3		√		√								
CO4		√		√								

Course Description:

SL No.	Course Contents	COs
1.	Introduction to Machine Learning: Classification of learning, Unsupervised and supervised learning	CO1, CO2

2.	Linear Regression with One Variable: application of linear regression, cost function, and gradient descent method for learning	CO2, CO3, CO4
3.	Linear Regression with Multiple Variables, Logistic Regression, and Regularization;	CO2, CO3, CO4
4.	Neural Networks: Representation, Learning, Multilayer feed forward network, Back propagation algorithm for training a feed forward network	CO2, CO3, CO4
5.	Applications: Advice for Applying Machine Learning, Machine Learning System Design, Support Vector Machines	CO3, CO4

Text Books, Reference Books and Other Resources:

1. Pattern Recognition and Machine Learning by Christopher M. Bishop
2. Machine Learning: A Probabilistic Perspective, by Kevin Murphy.

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2	C2	Lecture, Problem solution, Video, Web Link	Class test, Midterm, Assignment, Final
CO2	PO1, PO2	C3	Lecture, Problem solution, Video, Web Link	Class test, Midterm, Assignment, Final
CO3	PO2, PO4	C4	Lecture, Problem solution, Video, Web Link	Class test, Midterm, Assignment, Final
CO4	PO2, PO4	C5	Lecture, Problem solution, Video, Web Link	Class test, Midterm, Assignment, Final

Course Title: Machine Learning Laboratory
Course Code: CSE 458

Credits: 1 **Class Hours/Week: 2**
Course Type: Core **Pre-requisite: AIL**
CIE Marks: 40 **SEE Marks: 60**

Course Rationale:
This subject provides most effective machine learning techniques, learning algorithm to best suit the current need and enhance the learning parameters for maximum performance.

Course Objectives:
The objectives of the course are:

1. To provide students with an in-depth introduction to two main areas of Machine Learning: supervised and unsupervised.
2. To apply some of the main models and algorithms for regression, classification, and clustering.
3. To enhance the learning parameters to achieve maximum performance.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

- **CO1** **Implement (C3)** different machine learning algorithms for classification, regression, clustering, and dimensionality reduction using modern tools and libraries such as scikit-learn, TensorFlow and/or PyTorch etc.
- **CO2** **Apply (C3)** appropriate methods for handling missing data and dealing with class imbalance using the above stated tools
- **CO3** **Apply (C3)** appropriate mitigation techniques to solve the issues of overfitting, underfitting, bias, and variance in machine learning models using the same tools
- **CO4** **Evaluate (C5)** the performance of different machine learning algorithms using cross-validation and other validation techniques using the mentioned tools
- **CO5** **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√			√							
CO2	√	√			√							
CO3	√	√			√							
CO4	√	√		√	√							
CO5										√		

Course Description:

SL No.	Course Contents	COs
1.	Data Preprocessing, Regression, SVM, KNN, Decision Tree, Naïve Bayes classification, K-Means Clustering	CO1 – CO4
2.	Association Rule Learning: Apriori, Eclat, PCA, Model Selection and Boosting	CO1 – CO4
3.	Report preparation based on lab work	CO5

Text and Reference books:

1. Introduction to Machine Learning with Python: A Guide for Data Scientists - Andreas C. Müller, Sarah Guido
2. Machine Learning in Action - Peter Harrington
3. Python Machine Learning - By Sebastian Raschka

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2, PO5	C3	Demonstration, Discussion, Experiment	Performance, Assignment, Report
CO2	PO1, PO2, PO5	C3	Demonstration, Discussion, Experiment	Performance, Assignment, Report
CO3	PO1, PO2, PO5	C3	Demonstration, Discussion, Experiment	Performance, Assignment, Report
CO4	PO1, PO2, PO4, PO5	C5	Demonstration, Discussion, Experiment	Performance, Assignment, Report
CO5	PO10	A3	Demonstration	Report

Course Title: Pattern Recognition
Course Code: CSE 459

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: AI**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course will study state-of-the-art techniques for analyzing data. The goal is to extract meaningful information from feature data. This includes statistical and information theoretic concepts relating to machine learning, data mining and pattern recognition.

Course Objectives:
The main objectives of this course are:

4. The students should have a clear understanding of the design and construction and a pattern recognition system.
5. To introduce the major approaches of pattern recognition.
6. Finally, the student will have a clear working knowledge of implementing pattern recognition techniques.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to

CO1 **Demonstrate** (**C2**) the essential principles, theories, and techniques underlying pattern recognition: statistical modeling, machine learning algorithms, and data analysis methods.

CO2 **Implement** (**C3**) pattern recognition systems considering data preprocessing, extract relevant features, and train models using various algorithms.

CO3 **Apply** (**C3**) pattern recognition algorithms and tools to real-world problems: computer vision, speech recognition, natural language processing, and bioinformatics.

CO4 **Assess** (**C5**) the performance of pattern recognition algorithms using appropriate evaluation metrics.

CO5` **Comprehend** (**C2**) the ethical and social implications of pattern recognition such as privacy, bias, and fairness.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√										
CO2	√	√										
CO3	√	√										
CO4		√		√								
CO5				√		√		√				

Course Description:

SL No.	Course Contents	COs
1.	Introduction: Definitions, Datasets for Pattern Recognition, Different Paradigms of Pattern Recognition and Machine Learning, Data Normalization, Hypothesis Evaluation, VC-Dimensions and Distribution, Bias - Variance Tradeoff, Regression (Linear)	CO1
2.	**Feature Selection and Dimensionality Reduction:** PCA, LDA, ICA, SFFS, SBFS **Additional Features and Template Matching**: Texture, Shape and Size Characterization, Fractals, Features for Audio	CO2
3.	**Discriminative Methods:** Distance-based methods, Linear Discriminant Functions, Decision Tree, Random Decision Forest and Boosting. **Bayes Decision Theory:** Bayes decision rule, Minimum error rate classification, Normal density and discriminant functions, Bayesian networks **Parameter Estimation:** Maximum Likelihood and Bayesian Parameter Estimation **Artificial Neural Networks:** MLP and Backpropagation **Kernel Machines:** Kernel Tricks, Support Vector Machines **Clustering:** k-means clustering, Gaussian Mixture Modeling	CO3
4.	Confusion Matrix in Machine learning, Receiver Operating Characteristic (ROC) Curve	CO4
5.	Legal and Ethical consideration in pattern recognition	CO5

Text and Reference Books:

1. Duda, R.O., Hart, P.E., and Stork, D.G. (2001), Pattern Classification. Wiley-Interscience. (2nd Edition)
2. Devi V.S.; Murty, M.N. (2011), Pattern Recognition: An Introduction, Universities Press, Hyderabad.
3. L. I. Kuncheva (2004.), Combining Pattern Classifiers: Methods and Algorithms, Wiley-Blackwell

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	Corresponding POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1, PO2	C2	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam

CO2	PO1, PO2	C3	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam
CO3	PO1, PO2	C3	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam
CO4	PO2, PO4	C5	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam
CO5	PO4, PO6, PO8	C2	Lecture notes, PPT slides, problem solving, web content	Class Test, Midterm, Assignment, Final Exam

Course Title: Pattern Recognition Laboratory
Course Code: CSE 460

Credits: 01 **Class Hours/Week: 2**
Course Type: Core **Pre-requisite: AIL**
CIE Marks: 30 **SEE Marks: 70**

Course Rationale:
This subject aims to practically implement as well analyze the various procedures, methods and algorithms related to pattern recognition for specific application.

Course Objectives:
The objective of the course is to enable the students to

1. Have a clear understanding of the design and construction and a pattern recognition system.
2. Introduce the students, the major approaches in template and statistical pattern recognition.
3. Have a clear working knowledge of implementing pattern recognition techniques and the scientific python computing environment.

Course Outcomes:
Upon successful completion of this course, students will be able to

CO1 **Apply (C3)** pattern recognition algorithms to real-world datasets: preprocess data, select appropriate features, train models, and evaluate the performance of the algorithms.

CO2 **Use (C3)** pattern recognition tools and software packages to implement and test different algorithms, analyze the results, and visualize the patterns discovered.

CO3 **Analyze (C4)** complex datasets to extract meaningful insights, identify relevant patterns, and make informed decisions based on the analysis.

CO4 **Report** (**A3**) project work both written and verbally.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√	√			√							
CO2	√	√			√							
CO3				√	√							
CO4									√	√		

Course Description:

SL No.	Course Contents	COs
1	Image Manipulation, Enhancement, Filtering, Convolution, Segmentation, Feature Extraction, Feature Selection	CO1, CO4

2	Supervised classification, Unsupervised clustering, Artificial Neural Network, Convolutional Neural Networks: *elements, architectures, convolution visualization;* Software Packages and tools: *Tensorflow, Keras, Sklearn, NumPy, Matpoltlib*	CO2, CO4
3	Design a project on open-source dataset using state-of-the-art models	CO3, CO4

Text and Reference Books:

1. Duda, R.O., Hart, P.E., and Stork, D.G. (2001), Pattern Classification. Wiley Interscience. 2nd Edition.
2. Devi V.S.; Murty, M.N. (2011), Pattern Recognition: An Introduction, Universities Press, Hyderabad.
3. L. I. Kuncheva (2004.), Combining Pattern Classifiers: Methods and Algorithms, Wiley-Blackwell

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**Corresponding POs**	**Bloom's taxonomy domain/level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1, PO2, PO5	C3	Demonstration, Discussion, Experiment	Report, Performance, Presentation, Viva
CO2	PO1, PO2, PO5	C3	Demonstration, Discussion, Experiment	Report, Performance, Presentation, Viva
CO3	PO4, PO5	C4	Demonstration, Discussion, Experiment	Report, Performance, Presentation, Viva
CO4	PO9, PO10	A3	Demonstration	Report

Course Title: Contemporary Courses of Computer Science
Course Code: CSE 481

Credits: 3 **Class Hours/Week: 3**
Course Type: Core **Pre-requisite: CN**
CIE Marks: 60 **SEE Marks: 40**

Course Rationale:
This course is intended for the students who seek an overall understanding of cloud computing core concepts, architecture, its history, popular cloud services, security, pricing, and the technologies enabled by cloud computing.

Course Objectives:
1. Identify and demonstrate different cloud computing and cloud deployment models.
2. Comprehend the compute, storage, database, networking, and content delivery facilities offered by the underlying cloud infrastructure.
3. Identify security and privacy issues in cloud computing.
4. Analyze the automatic scaling and monitoring performance of the underlying cloud technologies
5. Implement applications for different cloud platforms.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

CO1 **Define** (**C1**) the key concepts, standards, and technologies in cloud computing.

CO2 **Describe (C2)** the performance, scalability, and availability of the underlying cloud technologies.

CO3 **Compare (C4)** resource usage in cloud computing infrastructures for different scientific applications.

CO4 **Describe (C2)** privacy and security issues for cloud infrastructure and virtual environments.

CO5 **Design (C6)** solutions for complex cloud hosting problems.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2	√	√										
CO3		√		√								
CO4	√	√										
CO5			√	√	√							

Course Description:

SL No.	Course Contents	COs
1.	Introduction to Cloud Computing: Grid Computing, Distributed computing, Cloud computing models such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS)	CO1
2.	Cloud deployment models: public, private, and hybrid models. Example cloud platforms (e.g. AWS, Google Cloud Platform, Microsoft Azure), and Interoperability.	CO1
3.	Cloud infrastructure: Points of presence, Region, Availability Zones, Data Centers, Edge Locations	CO1
4.	Cloud Security: Shared responsibility in the cloud, Identity and Access Management, Securing Users, Securing Data	CO4
5.	Networking and Content Delivery: Virtual Private Cloud (VPC), VPC networking, VPC sharing, VPC Peering, Site-to-site VPN, VPC Transit Gateway, Connecting on-premises data center to cloud, VPC Security, Routing Tables, Security groups, ACLs, Routing and DNS, Content Delivery Network, Pricing Transit Gateway	CO2, CO4
6.	Compute: EC2, EC2 Cost Optimization, Types of EC2 instances, Container Services, Docker, Serverless Computing: AWS Lamda and Elastic Beanstalk, key management services	CO3, CO5
7.	Cloud Storage: Object Storage, Block Storage, Network File System, Long Term Storage, automatic scaling of storages, Lifecycle management, Types of Object Storage, Use cases of different types of storages	CO3
8.	Cloud Database: Relational/SQL Database, NoSQL databases, Data storage for OLTP systems, warehouse, automatic scaling database systems, Reliability, Availability, Replication.	CO2
9.	Cloud Architecture: Well-Architected Framework, Pillars of the framework, Reliability and Scalability, Cloud Resource Provisioning	CO2, CO3
10.	Load Balancing, Auto Scaling, Dynamic Scaling, Cloud Performance Monitoring	CO2

Text Books, Reference Books and Other Resources:

1. AWS Cloud Fundamentals Online Course Material
2. Cloud Computing: A Hands-On Approach, by Arshdeep Bahga and Vijay Madisetti.
3. Cloud Computing Theory and Practice, by Dan Marinescu, 2nd Edition.
4. Cloud Computing: Principles and Paradigms, by Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, 1st Edition, Wiley

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	POs	Bloom's Taxonomy Domain/Level *(C: Cognitive, P: Psychomotor A: Affective)*	Delivery Methods and Activities	Assessment Tools
CO1	PO1	C1	Lecture notes, PPT slides, problem solving, AWS web content	Class Test, Midterm, Assignment, Final Exam
CO2	PO1, PO2	C2	Lecture notes, PPT slides, problem solving, AWS web content	Class Test, Midterm, Assignment, Final Exam
CO3	PO2, PO4	C4	Lecture notes, PPT slides, problem solving, AWS web content	Class Test, Midterm, Assignment, Final Exam
CO4	PO1, PO2	C2	Lecture notes, PPT slides, problem solving, AWS web content	Class Test, Midterm, Assignment, Final Exam
CO5	PO3, PO4, PO5	C6	Lecture notes, PPT slides, problem solving, AWS web content	Class Test, Midterm, Assignment, Final Exam

Course Title: Contemporary Courses of Computer Science Laboratory
Course Code: CSE 482

Credits: 1 **Class Hours/Week: 2**
Course Type: Core **Pre-requisite: CNL**
CIE Marks: 70 **SEE Marks: 30**

Course Rationale:
This course enables the students to become familiar with web application implementation and hosting, and service configuration in real-world cloud platforms.

Course Objectives:
1. Become familiar with virtual machine configuration in different cloud platforms.
2. Demonstrate the advantage of managed cloud services.
3. Implement custom web applications in different cloud computing service models.
4. Use advanced techniques to scale cloud resources under dynamic requirements.

Course Outcomes (COs):
Upon successful completion of this course, students will be able to:

- **CO1** **Configure (C3)** virtual machines (instances) with underlying storage mechanisms, supporting security and access mechanisms.
- **CO2** **Leverage (C3)** managed database services for solving relational database needs.
- **CO3** **Implement (C3)** web applications in **IaaS** and **PaaS** environments.
- **CO4** **Use (C3)** load balancing and auto-scaling to automatically control virtual instances and distribute traffic across the instances.
- **CO5** **Design (C6)** a custom cloud hosting solution for a given problem.
- **CO6** **Report** (**A3**) lab activities and experimental results or findings.

Mapping of Course Outcomes to Program Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓				✓							
CO2	✓				✓							
CO3	✓				✓							
CO4	✓				✓							
CO5	✓	✓	✓	✓	✓							
CO6										✓		

Course Description:

SL No	Course Contents	COs
1.	Introduction to AWS IAM	CO1, CO5, CO6

2.	Network and Content Delivery: Build VPC and Launch a Web Server	CO1, CO5, CO6
3.	Introduction to Amazon EC2, AWS Lambda, AWS Elastic Beanstalk	CO2, CO3, CO5
4.	Working with EBS	CO1, CO5, CO6
5.	Build a DB Server and Interact with the DB Using an App	CO2, CO5, CO6
6.	Scale and Load Balance Cloud Architecture	CO4, CO5, CO6

Text Books, Reference Books and Other Resources:

1. AWS Cloud Fundamentals Online Course Material
2. Cloud Computing: A Hands-On Approach, by Arshdeep Bahga and Vijay Madisetti.
3. Cloud Computing Theory and Practice, by Dan Marinescu, 2nd Edition.
4. Cloud Computing: Principles and Paradigms, by Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, 1st Edition, Wiley

Mapping Course Outcomes with the Teaching-Learning and Assessment Strategy:

COs	**POs**	**Bloom's Taxonomy Domain/Level** ***(C: Cognitive, P: Psychomotor A: Affective)***	**Delivery Methods and Activities**	**Assessment Tools**
CO1	PO1, PO5	C3	Demonstration, discussion, experiment	Performance, Lab, Quiz
CO2	PO1, PO5	C3	Demonstration, discussion, experiment	Performance, Lab, Quiz
CO3	PO1, PO5	C3	Demonstration, discussion, experiment	Performance, Lab, Quiz
CO4	PO1, PO5	C3	Demonstration, discussion, experiment	Performance, Lab, Quiz
CO5	PO1, PO2, PO3, PO4, PO5	C6	Demonstration, discussion, experiment	Performance, Lab, Quiz
CO6	PO10	A3	Demonstration	Report

Annexure: 6.2
Flow Chart

1st Semester	CSE 110 - Introduction to Computer System (Laboratory) (2)	EEE 101- Electrical Circuits I (3)	ENG 101- General English (3)	MAT 105- Engineering Mathematics I (3)	PHY 101- Engineering Physics I (3)	ME 102- Mechanical Engineering Drawing & CAD (Laboratory)
2nd Semester	CSE 111- Structured Programming (2)	EEE 211- Electronics I (3)	ENG 103- Developing English Skills (2)	MAT 107- Engineering Mathematics II (3)	PHY 103- Engineering Physics II (3)	CSE 103- Discrete Mathematics (3)
3rd Semester	CSE 211- Object Oriented Programming (3)	CSE 221- Data Structure (3)	EEE 311- Digital Electronics (3)	MAT 201- Engineering Mathematics III (3)	ECO 202– Basic Economics (3)	
4th Semester	CSE 225- Algorithm Design and Analysis (3)	CSE 237- Database Management System (3)	EEE 201- Signals & Systems (3)	MAT 203- Engineering Mathematics IV (3)	MGT 203- Industrial & Business Management (3)	
5th Semester	CSE 305- Software Engineering & Information System Design (4)	EEE 371- Microprocessors & Microcontrollers (3)	CSE 301- Computational Methods for Engineering Problems (3)	MGT 251- Organizational Behavior (3)	EEE 309- Communication Engineering (3)	
6th Semester	CSE 317- Artificial Intelligence (3)	CSE 333- Operating Systems (3)	CSE 338 - Software Development (2)	CSE 337- Computer Organization & Architecture (3)	CSE 367- Computer Network (3)	CSE 364- Data Communication (3)
7th Semester	CSE 309- Theory of Computation (2)	CSE 451- Neural Network & Fuzzy Logic (3)	CSE 455– Computer Graphics and Image Processing (3)	CSE 437- Network and Computer Security (3)	EEE 313- Control Systems (3)	ENG 401- Technical Writing & Presentation (2)
8th Semester	CSE 453- Compiler Construction (3)	CSE 457– Machine Learning (3)	CSE 459- Pattern Recognition (3)	CSE 481– Contemporary Course of Computer Science (3)		

Annexure: 6.3
List of Offered Courses

Offer List Spring 2023

Sem.	Course Code	Course Name	Credit	Teacher
1st	BAN 1101	Functional Bengali Language (□□□□□□□□□□ □□□□□ □□□□) - Section A	2	Hasnaine Isteafaz (GT)
1st	BAN 1101	Functional Bengali Language (□□□□□□□□□□ □□□□□ □□□□) - Section B	2	Hasnaine Isteafaz (GT)
1st	CSE 1111(V4)	Computer Fundamentals and Ethics - Section A	1.5	Avisheak Das (CSE)
1st	CSE 1111(V4)	Computer Fundamentals and Ethics - Section B	1.5	Adiba Ibnat Hossain (CSE)
1st	CSE 1113(V4)	Programming Fundamentals - Section A	3	Adiba Ibnat Hossain (CSE)
1st	CSE 1113(V4)	Programming Fundamentals - Section B	3	Md. Hasan (CSE)
1st	CSE 1114(V4)	Programming Fundamentals Laboratory - Section A	1.5	Adiba Ibnat Hossain (CSE)
1st	CSE 1114(V4)	Programming Fundamentals Laboratory - Section B1	1.5	Md. Hasan (CSE)
1st	CSE 1114(V4)	Programming Fundamentals Laboratory - Section B2	1.5	Md. Hasan (CSE)
1st	EEE 1101(V4)	Introduction to Electrical Engineering - Section A	3	Kazi Md. Abrar Yeaser (CSE)
1st	EEE 1101(V4)	Introduction to Electrical Engineering - Section B	3	Akramul Haque (EEE)
1st	EEE 1102(V4)	Introduction to Electrical Engineering Laboratory - Section A	1.5	Kazi Md. Abrar Yeaser (CSE)
1st	EEE 1102(V4)	Introduction to Electrical Engineering Laboratory - Section B1	1.5	Mohammed Saifuddin Munna (EEE)
1st	EEE 1102(V4)	Introduction to Electrical Engineering Laboratory - Section B2	1.5	Akramul Haque (EEE)
1st	ENG 1101(V4)	General English - Section A	3	Shreyashi Paul (CSE)
1st	ENG 1101(V4)	General English - Section B	3	Shreyashi Paul (CSE)
1st	MAT 1203(V4)	Differential and Integral Calculus - Section A	3	Md. Mamun Sarkar (CSE)
1st	MAT 1203(V4)	Differential and Integral Calculus - Section B	3	Md. Mamun Sarkar (CSE)
1st	ME 1104(V4)	Mechanical Engineering Drawing - Section A	0.75	Dr. Md. Sanaul Rabbi (GT)

1st	ME 1104(V4)	Mechanical Engineering Drawing - Section B1	0.75	Dr. Sajal Chandra Banik (GT)
1st	ME 1104(V4)	Mechanical Engineering Drawing - Section B2	0.75	Dr. Sajal Chandra Banik (GT)
2nd	BAN 1101	Functional Bengali Language (□□□□□□□□□□ □□□□□ □□□□) - Section 2A	2	Hasnaine Isteafaz (GT)
2nd	BAN 1101	Functional Bengali Language (□□□□□□□□□□ □□□□□ □□□□) - Section 2B	2	Dr. Prakash Dasgupta (GT)
2nd	BAN 1101	Functional Bengali Language (□□□□□□□□□□ □□□□□ □□□□) - Section 2C	2	Dr. Prakash Dasgupta (GT)
2nd	BAN 1101	Functional Bengali Language (□□□□□□□□□□ □□□□□ □□□□) - Section 2D	2	Dr. Prakash Dasgupta (GT)
2nd	CSE 1110	Competitive Programming Laboratory - Section A	0.75	Md. Hasan (CSE)
2nd	CSE 1110	Competitive Programming Laboratory - Section B	0.75	Md. Hasan (CSE)
2nd	CSE 1110	Competitive Programming Laboratory - Section C	0.75	Md. Hasan (CSE)
2nd	CSE 1110	Competitive Programming Laboratory - Section D	0.75	Md. Hasan (CSE)
2nd	CSE 1411	Discrete Mathematics and Number Theory - Section A	3	Farhana Shirin Chowdhury (CSE)
2nd	CSE 1411	Discrete Mathematics and Number Theory - Section B	3	Farhana Shirin Chowdhury (CSE)
2nd	CSE 1411	Discrete Mathematics and Number Theory - Section C	3	Farhana Shirin Chowdhury (CSE)
2nd	CSE 1411	Discrete Mathematics and Number Theory - Section D	3	Farhana Shirin Chowdhury (CSE)
2nd	CSE 1413	Data Structures - Section A	3	Dhrubajyoti Das (CSE)
2nd	CSE 1413	Data Structures - Section B	3	Dhrubajyoti Das (CSE)
2nd	CSE 1413	Data Structures - Section C	3	MD. SHAFIUL ALAM FORHAD (GT)
2nd	CSE 1413	Data Structures - Section D	3	MD. SHAFIUL ALAM FORHAD (GT)
2nd	CSE 1414	Data Structures Laboratory - Section A	1.5	Dhrubajyoti Das (CSE)
2nd	CSE 1414	Data Structures Laboratory - Section B	1.5	Dhrubajyoti Das (CSE)

2nd	CSE 1414	Data Structures Laboratory - Section C	1.5	Syed Md. Minhaz Hossain (CSE)
2nd	CSE 1414	Data Structures Laboratory - Section D	1.5	Syed Md. Minhaz Hossain (CSE)
2nd	ECO 1101	Engineering Economics - Section A	3	Mahmuda Akhter Khanam (GT)
2nd	ECO 1101	Engineering Economics - Section B	3	Mahmuda Akhter Khanam (GT)
2nd	ECO 1101	Engineering Economics - Section C	3	Nazma Afroz (GT)
2nd	ECO 1101	Engineering Economics - Section D	3	Nazma Afroz (GT)
2nd	EEE 1201	Electronics Devices and Circuits - Section A	3	Asif Mohammed Siddiqee (EEE)
2nd	EEE 1201	Electronics Devices and Circuits - Section B	3	Mohammed Rezaur Rahman Chowdhury (CSE)
2nd	EEE 1201	Electronics Devices and Circuits - Section C	3	Mohammed Rezaur Rahman Chowdhury (CSE)
2nd	EEE 1201	Electronics Devices and Circuits - Section D	3	Aditta Chowdhury (GT)
2nd	EEE 1202	Electronics Device and Circuits Laboratory - Section A	1.5	Asif Mohammed Siddiqee (EEE)
2nd	EEE 1202	Electronics Device and Circuits Laboratory - Section B	1.5	Mohammed Rezaur Rahman Chowdhury (CSE)
2nd	EEE 1202	Electronics Device and Circuits Laboratory - Section C	1.5	Mohammed Rezaur Rahman Chowdhury (CSE)
2nd	EEE 1202	Electronics Device and Circuits Laboratory - Section D	1.5	Mohammed Saifuddin Munna (EEE)
2nd	PHY 1104	Physics Laboratory - Section A1	0.75	Nusrat Jahan Shirin (CSE)
2nd	PHY 1104	Physics Laboratory - Section B1	0.75	Nusrat Jahan Shirin (CSE)
2nd	PHY 1104	Physics Laboratory - Section C1	0.75	Nusrat Jahan Shirin (CSE)
2nd	PHY 1104	Physics Laboratory - Section D1	0.75	Nusrat Jahan Shirin (CSE)
2nd	PHY 1104	Physics Laboratory - Section A2	0.75	Nusrat Jahan Shirin (CSE)
2nd	PHY 1104	Physics Laboratory - Section B2	0.75	Nusrat Jahan Shirin (CSE)

2nd	PHY 1104	Physics Laboratory - Section C2	0.75	Nusrat Jahan Shirin (CSE)
2nd	PHY 1104	Physics Laboratory - Section D2	0.75	Nusrat Jahan Shirin (CSE)
3rd	CSE 211(V3)	Object Oriented Programming - Section A	3	Sheikh Md. Rukunuddin Osmani (CSE)
3rd	CSE 211(V3)	Object Oriented Programming - Section B	3	Dr. Abu Nowshed Chy (GT)
3rd	CSE 211(V3)	Object Oriented Programming - Section C	3	Dr. Abu Nowshed Chy (GT)
3rd	CSE 212(V3)	Object Oriented Programming Laboratory - Section A	1.5	Sheikh Md. Rukunuddin Osmani (CSE)
3rd	CSE 212(V3)	Object Oriented Programming Laboratory - Section B	1.5	Sourav Adhikary (CSE)
3rd	CSE 212(V3)	Object Oriented Programming Laboratory - Section C	1.5	Sourav Adhikary (CSE)
3rd	CSE 221(V3)	Data Structure - Section A	3	Mohammad Hasan (CSE)
3rd	CSE 221(V3)	Data Structure - Section B	3	Sourav Adhikary (CSE)
3rd	CSE 221(V3)	Data Structure - Section C	3	Adiba Ibnat Hossain (CSE)
3rd	CSE 222(V3)	Data Structure Laboratory - Section A	1.5	Mohammad Hasan (CSE)
3rd	CSE 222(V3)	Data Structure Laboratory - Section B	1.5	Sourav Adhikary (CSE)
3rd	CSE 222(V3)	Data Structure Laboratory - Section C	1.5	Adiba Ibnat Hossain (CSE)
3rd	ECO 201(V3)	Basic Economics - Section A	3	Sudip Dey (GT)
3rd	ECO 201(V3)	Basic Economics - Section B	3	Arpita Datta (GT)
3rd	ECO 201(V3)	Basic Economics - Section C	3	Badrul Hasan Awal (GT)
3rd	EEE 311(V3)	Digital Electronics - Section A	3	Dr. Sampad Ghosh (GT)
3rd	EEE 311(V3)	Digital Electronics - Section B	3	Dr. Sampad Ghosh (GT)
3rd	EEE 311(V3)	Digital Electronics - Section C	3	Aditta Chowdhury (GT)
3rd	EEE 312(V3)	Digital Electronics Laboratory - Section A	1.5	Sabrina Tarannum (CSE)

3rd	EEE 312(V3)	Digital Electronics Laboratory - Section B	1.5	Asma Joshita Trisha (CSE)
3rd	EEE 312(V3)	Digital Electronics Laboratory - Section C	1.5	Mohammed Rezaur Rahman Chowdhury (CSE)
3rd	MAT 201(V3)	Engineering Mathematics III - Section A	3	Md. Mamun Sarkar (CSE)
3rd	MAT 201(V3)	Engineering Mathematics III - Section B	3	Kollol Dey (DMath)
3rd	MAT 201(V3)	Engineering Mathematics III - Section C	3	Kollol Dey (DMath)
4th	CSE 225(V3)	Algorithm Design and Analysis - Section A	3	Fairuz Bilquis Khan (CSE)
4th	CSE 225(V3)	Algorithm Design and Analysis - Section B	3	Fairuz Bilquis Khan (CSE)
4th	CSE 225(V3)	Algorithm Design and Analysis - Section C	3	Md. Hasan (CSE)
4th	CSE 225(V3)	Algorithm Design and Analysis - Section D	3	Noortaz Rezoana (CSE)
4th	CSE 225(V3)	Algorithm Design and Analysis - Section E	3	Fairuz Bilquis Khan (CSE)
4th	CSE 226(V3)	Algorithm Design and Analysis Laboratory - Section A	1	Fairuz Bilquis Khan (CSE)
4th	CSE 226(V3)	Algorithm Design and Analysis Laboratory - Section B	1	Fairuz Bilquis Khan (CSE)
4th	CSE 226(V3)	Algorithm Design and Analysis Laboratory - Section C	1	Syed Md. Minhaz Hossain (CSE)
4th	CSE 226(V3)	Algorithm Design and Analysis Laboratory - Section D	1	Syed Md. Minhaz Hossain (CSE)
4th	CSE 226(V3)	Algorithm Design And Analysis Laboratory - Section E	1	Syed Md. Minhaz Hossain (CSE)
4th	CSE 237(V3)	Database Management System - Section A	3	Anik Sen (CSE)
4th	CSE 237(V3)	Database Management System - Section B	3	Anik Sen (CSE)
4th	CSE 237(V3)	Database Management System - Section C	3	Mohammad Hasan (CSE)
4th	CSE 237(V3)	Database Management System - Section D	3	Mohammad Hasan (CSE)
4th	CSE 237(V3)	Database Management System - Section E	3	Dhrubajyoti Das (CSE)
4th	CSE 238(V3)	Database Management System Laboratory - Section A	1.5	Anik Sen (CSE)
4th	CSE 238(V3)	Database Management System Laboratory - Section B	1.5	Anik Sen (CSE)

4th	CSE 238(V3)	Database Management System Laboratory - Section C	1.5	Mohammad Hasan (CSE)
4th	CSE 238(V3)	Database Management System Laboratory - Section D	1.5	Mohammad Hasan (CSE)
4th	CSE 238(V3)	Database Management System Laboratory - Section E	1.5	Dhrubajyoti Das (CSE)
4th	EEE 201(V3)	Signals & Systems - Section A	3	Sumit Majumder (GT)
4th	EEE 201(V3)	Signals & Systems - Section B	3	Sumit Majumder (GT)
4th	EEE 201(V3)	Signals & Systems - Section C	3	Sujan Chowdhury (EEE)
4th	EEE 201(V3)	Signals & Systems - Section D	3	Dr. Md. Jahedul Islam (GT)
4th	EEE 201(V3)	Signals & Systems - Section E	3	Dr. Md. Jahedul Islam (GT)
4th	EEE 202(V3)	Signals & Systems Laboratory - Section A	1	Mohammed Rezaur Rahman Chowdhury (CSE)
4th	EEE 202(V3)	Signals & Systems Laboratory - Section B	1	Mohammed Rezaur Rahman Chowdhury (CSE)
4th	EEE 202(V3)	Signals & Systems Laboratory - Section C	1	Sujan Chowdhury (EEE)
4th	EEE 202(V3)	Signals & Systems Laboratory - Section D	1	Kazi Md. Abrar Yeaser (CSE)
4th	EEE 202(V3)	Signals & Systems Laboratory - Section E	1	Kazi Md. Abrar Yeaser (CSE)
4th	MAT 203(V3)	Engineering Mathematics IV - Section A	3	Md. Mamun Sarkar (CSE)
4th	MAT 203(V3)	Engineering Mathematics IV - Section B	3	Md. Mamun Sarkar (CSE)
4th	MAT 203(V3)	Engineering Mathematics IV - Section C	3	Md. Mamun Sarkar (CSE)
4th	MAT 203(V3)	Engineering Mathematics IV - Section D	3	Kollol Dey (DMath)
4th	MAT 203(V3)	Engineering Mathematics IV - Section E	3	Kollol Dey (DMath)
4th	MGT 203(V3)	Industrial & Business Management - Section A	3	Md. Touhidul Islam Chowdhury (GT)
4th	MGT 203(V3)	Industrial & Business Management - Section B	3	Md. Touhidul Islam Chowdhury (GT)
4th	MGT 203(V3)	Industrial & Business Management - Section C	3	Md. Touhidul Islam Chowdhury (GT)

4th	MGT 203(V3)	Industrial & Business Management - Section D	3	Ela Saha (GT)
4th	MGT 203(V3)	Industrial & Business Management - Section E	3	Ela Saha (GT)
5th	CSE 301(V3)	Computational Methods for Engineering Problems	3	Kollol Dey (DMath)
5th	CSE 302(V3)	Computational Methods for Engineering Problems Laboratory - Section A1	1	Kollol Dey (DMath)
5th	CSE 302(V3)	Computational Methods for Engineering Problems Laboratory - Section A2	1	Kollol Dey (DMath)
5th	CSE 305(V3)	Software Engineering & Information System Design	4	Jannathul Maowa Hasi (CSE)
5th	CSE 306(V3)	Software Engineering & Information System Design Laboratory - Section A1	1.5	Jannathul Maowa Hasi (CSE)
5th	CSE 306(V3)	Software Engineering & Information System Design Laboratory - Section A2	1.5	Jannathul Maowa Hasi (CSE)
5th	EEE 309(V3)	Communication Engineering	3	Sujan Chowdhury (EEE)
5th	EEE 310(V3)	Communication Engineering Laboratory - Section A1	1.5	Sharith Dhar (EEE)
5th	EEE 310(V3)	Communication Engineering Laboratory - Section A2	1.5	Sujan Chowdhury (EEE)
5th	EEE 371(V3)	Microprocessors & Microcontrollers	3	Kazi Md. Abrar Yeaser (CSE)
5th	EEE 372(V3)	Microprocessors & Microcontrollers Laboratory - Section A1	1.5	Kazi Md. Abrar Yeaser (CSE)
5th	EEE 372(V3)	Microprocessors & Microcontrollers Laboratory - Section A2	1.5	Kazi Md. Abrar Yeaser (CSE)
5th	MGT 251(V3)	Organizational Behavior	3	Ela Saha (GT)
6th	CSE 317(V3)	Artificial Intelligence - Section A	3	Avisheak Das (CSE)
6th	CSE 317(V3)	Artificial Intelligence - Section B	3	Avisheak Das (CSE)
6th	CSE 318(V3)	Artificial Intelligence Laboratory - Section A	1.5	Avisheak Das (CSE)
6th	CSE 318(V3)	Artificial Intelligence Laboratory - Section B	1.5	Avisheak Das (CSE)
6th	CSE 333(V3)	Operating Systems - Section A	3	Asma Joshita Trisha (CSE)

6th	CSE 333(V3)	Operating Systems - Section B	3	Asma Joshita Trisha (CSE)
6th	CSE 334(V3)	Operating Systems Laboratory - Section A	1.5	Asma Joshita Trisha (CSE)
6th	CSE 334(V3)	Operating Systems Laboratory - Section B	1.5	Asma Joshita Trisha (CSE)
6th	CSE 337(V3)	Computer Organization & Architecture - Section A	3	Asma Joshita Trisha (CSE)
6th	CSE 337(V3)	Computer Organization & Architecture - Section B	3	Sabrina Tarannum (CSE)
6th	CSE 338(V3)	Software Development - Section A	2	Anik Sen (CSE)
6th	CSE 338(V3)	Software Development - Section B	2	Anik Sen (CSE)
6th	CSE 364(V3)	Data Communication - Section A	3	Rezaul Karim (GT)
6th	CSE 364(V3)	Data Communication - Section B	3	Rezaul Karim (GT)
6th	CSE 367(V3)	Computer Network - Section A	3	Sheikh Md. Rukunuddin Osmani (CSE)
6th	CSE 367(V3)	Computer Network - Section B	3	Sheikh Md. Rukunuddin Osmani (CSE)
6th	CSE 368(V3)	Computer Network Laboratory - Section A	1.5	Sheikh Md. Rukunuddin Osmani (CSE)
6th	CSE 368(V3)	Computer Network Laboratory - Section B	1.5	Sheikh Md. Rukunuddin Osmani (CSE)
7th	CSE 309(V3)	Theory of Computation - Section A	2	Sabrina Tarannum (CSE)
7th	CSE 309(V3)	Theory of Computation - Section B	2	Sabrina Tarannum (CSE)
7th	CSE 309(V3)	Theory of Computation - Section C	2	Adiba Ibnat Hossain (CSE)
7th	CSE 309(V3)	Theory of Computation - Section D	2	Adiba Ibnat Hossain (CSE)
7th	CSE 437(V3)	Network and Computer Security - Section A	3	Kingshuk Dhar (CSE)
7th	CSE 437(V3)	Network and Computer Security - Section B	3	Kingshuk Dhar (CSE)
7th	CSE 437(V3)	Network and Computer Security - Section C	3	Kingshuk Dhar (CSE)
7th	CSE 437(V3)	Network and Computer Security - Section D	3	Md. Neamul Haque (CSE)

7th	CSE 451(V3)	Neural Network & Fuzzy Logic - Section A	3	Farhana Shirin Chowdhury (CSE)
7th	CSE 451(V3)	Neural Network & Fuzzy Logic - Section B	3	Farhana Shirin Chowdhury (CSE)
7th	CSE 451(V3)	Neural Network & Fuzzy Logic - Section C	3	Sourav Adhikary (CSE)
7th	CSE 451(V3)	Neural Network & Fuzzy Logic - Section D	3	Sourav Adhikary (CSE)
7th	CSE 452(V3)	Neural Network & Fuzzy Logic Laboratory - Section A	1	Jannathul Maowa Hasi (CSE)
7th	CSE 452(V3)	Neural Network & Fuzzy Logic Laboratory - Section B	1	Jannathul Maowa Hasi (CSE)
7th	CSE 452(V3)	Neural Network & Fuzzy Logic Laboratory - Section C	1	Avisheak Das (CSE)
7th	CSE 452(V3)	Neural Network & Fuzzy Logic Laboratory - Section D	1	Avisheak Das (CSE)
7th	CSE 455(V3)	Computer Graphics & Image Processing - Section A	3	Jannathul Maowa Hasi (CSE)
7th	CSE 455(V3)	Computer Graphics & Image Processing - Section B	3	Md. Neamul Haque (CSE)
7th	CSE 455(V3)	Computer Graphics & Image Processing - Section C	3	Md. Neamul Haque (CSE)
7th	CSE 455(V3)	Computer Graphics & Image Processing - Section D	3	Md. Neamul Haque (CSE)
7th	CSE 456(V3)	Computer Graphics & Image Processing Laboratory - Section A	1	Md. Neamul Haque (CSE)
7th	CSE 456(V3)	Computer Graphics & Image Processing Laboratory - Section B	1	Md. Neamul Haque (CSE)
7th	CSE 456(V3)	Computer Graphics & Image Processing Laboratory - Section C	1	Md. Neamul Haque (CSE)
7th	CSE 456(V3)	Computer Graphics & Image Processing Laboratory - Section D	1	Md. Neamul Haque (CSE)
7th	EEE 313(V3)	Control Systems - Section A	3	Somen Dutta (EEE)
7th	EEE 313(V3)	Control Systems - Section B	3	Somen Dutta (EEE)
7th	EEE 313(V3)	Control Systems - Section C	3	Arif Istiaque (GT)
7th	EEE 313(V3)	Control Systems - Section D	3	Arif Istiaque (GT)
7th	EEE 314(V3)	Control Systems Laboratory - Section A	1.5	Somen Dutta (EEE)

7th	EEE 314(V3)	Control Systems Laboratory - Section B	1.5	Somen Dutta (EEE)
7th	EEE 314(V3)	Control Systems Laboratory - Section C	1.5	Somen Dutta (EEE)
7th	EEE 314(V3)	Control Systems Laboratory - Section D	1.5	Akramul Haque (EEE)
7th	ENG 401(V3)	Technical Writing & Presentation - Section A	2	Dr. Mohammad Kamrul Hassan (GT)
7th	ENG 401(V3)	Technical Writing & Presentation - Section B	2	Dr. Mohammad Kamrul Hassan (GT)
7th	ENG 401(V3)	Technical Writing & Presentation - Section C	2	Dr. Mohammad Kamrul Hassan (GT)
7th	ENG 401(V3)	Technical Writing & Presentation - Section D	2	Dr. Mohammad Kamrul Hassan (GT)
8th	CSE 453(V3)	Compiler Construction - Section A	3	Noortaz Rezoana (CSE)
8th	CSE 453(V3)	Compiler Construction - Section B	3	Noortaz Rezoana (CSE)
8th	CSE 453(V3)	Compiler Construction - Section C	3	Noortaz Rezoana (CSE)
8th	CSE 454(V3)	Compiler Construction Laboratory - Section A1	1.5	Tanni Dhoom (CSE)
8th	CSE 454(V3)	Compiler Construction Laboratory - Section A2	1.5	Tanni Dhoom (CSE)
8th	CSE 454(V3)	Compiler Construction Laboratory - Section B1	1.5	Tanni Dhoom (CSE)
8th	CSE 454(V3)	Compiler Construction Laboratory - Section B2	1.5	Tanni Dhoom (CSE)
8th	CSE 454(V3)	Compiler Construction Laboratory - Section C1	1.5	Tanni Dhoom (CSE)
8th	CSE 454(V3)	Compiler Construction Laboratory - Section C2	1.5	Tanni Dhoom (CSE)
8th	CSE 457(V3)	Machine Learning - Section A	3	Taufique Sayeed (CSE)
8th	CSE 457(V3)	Machine Learning - Section B	3	Taufique Sayeed (CSE)
8th	CSE 457(V3)	Machine Learning - Section C	3	Taufique Sayeed (CSE)
8th	CSE 458(V3)	Machine Learning Laboratory - Section A1	1	Fairuz Bilquis Khan (CSE)
8th	CSE 458(V3)	Machine Learning Laboratory - Section A2	1	Fairuz Bilquis Khan (CSE)
8th	CSE 458(V3)	Machine Learning Laboratory - Section B1	1	Noortaz Rezoana (CSE)
8th	CSE 458(V3)	Machine Learning Laboratory - Section B2	1	Noortaz Rezoana (CSE)

8th	CSE 458(V3)	Machine Learning Laboratory - Section C1	1	Jannathul Maowa Hasi (CSE)
8th	CSE 458(V3)	Machine Learning Laboratory - Section C2	1	Jannathul Maowa Hasi (CSE)
8th	CSE 459(V3)	Pattern Recognition - Section A	3	Dr. Kaushik Deb (GT)
8th	CSE 459(V3)	Pattern Recognition - Section B	3	Dr. Kaushik Deb (GT)
8th	CSE 459(V3)	Pattern Recognition - Section C	3	Dr. Kaushik Deb (GT)
8th	CSE 460(V3)	Pattern Recognition Laboratory - Section A1	1	Fairuz Bilquis Khan (CSE)
8th	CSE 460(V3)	Pattern Recognition Laboratory - Section A2	1	Fairuz Bilquis Khan (CSE)
8th	CSE 460(V3)	Pattern Recognition Laboratory - Section B1	1	Noortaz Rezoana (CSE)
8th	CSE 460(V3)	Pattern Recognition Laboratory - Section B2	1	Noortaz Rezoana (CSE)
8th	CSE 460(V3)	Pattern Recognition Laboratory - Section C1	1	Noortaz Rezoana (CSE)
8th	CSE 460(V3)	Pattern Recognition Laboratory - Section C2	1	Noortaz Rezoana (CSE)
8th	CSE 481(V3)	Contemporary Course of Computer Science - Section A	3	Shahid Md. Asif Iqbal (CSE)
8th	CSE 481(V3)	Contemporary Course of Computer Science - Section B	3	Shahid Md. Asif Iqbal (CSE)
8th	CSE 481(V3)	Contemporary Course of Computer Science - Section C	3	Kingshuk Dhar (CSE)
8th	CSE 482(V3)	Contemporary Courses of Computer Science Laboratory - Section A1	1	Shahid Md. Asif Iqbal (CSE)
8th	CSE 482(V3)	Contemporary Courses of Computer Science Laboratory - Section A2	1	Shahid Md. Asif Iqbal (CSE)
8th	CSE 482(V3)	Contemporary Courses of Computer Science Laboratory - Section B1	1	Shahid Md. Asif Iqbal (CSE)
8th	CSE 482(V3)	Contemporary Courses of Computer Science Laboratory - Section B2	1	Shahid Md. Asif Iqbal (CSE)
8th	CSE 482(V3)	Contemporary Courses of Computer Science Laboratory - Section C1	1	Kingshuk Dhar (CSE)
8th	CSE 482(V3)	Contemporary Courses of Computer Science Laboratory - Section C2	1	Kingshuk Dhar (CSE)

Offer List Fall 2022

Semester	Course Code	Course Name	Credit	Teacher
1st	ACC 101(V3)	Basic Accounting - Section (Only Retake)	3	Shafaitun Nahar (GT)
1st	CSE 110(V3)	Introduction to Computer System Laboratory	2	Mohammad Hasan (CSE)
1st	CSE 1111(V4)	Computer Fundamentals and Ethics - Section A	1.5	Taufique Sayeed (CSE)
1st	CSE 1111(V4)	Computer Fundamentals and Ethics - Section B	1.5	Taufique Sayeed (CSE)
1st	CSE 1111(V4)	Computer Fundamentals and Ethics - Section C	1.5	Taufique Sayeed (CSE)
1st	CSE 1111(V4)	Computer Fundamentals and Ethics - Section D	1.5	Taufique Sayeed (CSE)
1st	CSE 1111(V4)	Computer Fundamentals and Ethics - Section E	1.5	Taufique Sayeed (CSE)
1st	CSE 1113(V4)	Programming Fundamentals - Section A	3	Tanni Dhoom (CSE)
1st	CSE 1113(V4)	Programming Fundamentals - Section B	3	Tanni Dhoom (CSE)
1st	CSE 1113(V4)	Programming Fundamentals - Section C	3	Tanni Dhoom (CSE)
1st	CSE 1113(V4)	Programming Fundamentals - Section D	3	Mohammad Hasan (CSE)
1st	CSE 1113(V4)	Programming Fundamentals - Section E	3	Mohammad Hasan (CSE)
1st	CSE 1114(V4)	Programming Fundamentals Laboratory - Section A	1.5	Tanni Dhoom (CSE)
1st	CSE 1114(V4)	Programming Fundamentals Laboratory - Section B	1.5	Tanni Dhoom (CSE)
1st	CSE 1114(V4)	Programming Fundamentals Laboratory - Section C	1.5	Tanni Dhoom (CSE)
1st	CSE 1114(V4)	Programming Fundamentals Laboratory - Section D	1.5	Mohammad Hasan (CSE)
1st	CSE 1114(V4)	Programming Fundamentals Laboratory - Section E	1.5	Mohammad Hasan (CSE)
1st	EEE 1101(V4)	Introduction to Electrical Engineering - Section A	3	Kazi Md. Abrar Yeaser (CSE)

1st	EEE 1101(V4)	Introduction to Electrical Engineering - Section B	3	Kazi Md. Abrar Yeaser (CSE)
1st	EEE 1101(V4)	Introduction to Electrical Engineering - Section C	3	Adnan Hossain Khan (CSE)
1st	EEE 1101(V4)	Introduction to Electrical Engineering - Section D	3	Adnan Hossain Khan (CSE)
1st	EEE 1101(V4)	Introduction to Electrical Engineering - Section E	3	Adnan Hossain Khan (CSE)
1st	EEE 1102(V4)	Introduction to Electrical Engineering Laboratory - Section A	1.5	Kazi Md. Abrar Yeaser (CSE)
1st	EEE 1102(V4)	Introduction to Electrical Engineering Laboratory - Section B	1.5	Kazi Md. Abrar Yeaser (CSE)
1st	EEE 1102(V4)	Introduction to Electrical Engineering Laboratory - Section C	1.5	Adnan Hossain Khan (CSE)
1st	EEE 1102(V4)	Introduction to Electrical Engineering Laboratory - Section D	1.5	Adnan Hossain Khan (CSE)
1st	EEE 1102(V4)	Introduction to Electrical Engineering Laboratory - Section E	1.5	Adnan Hossain Khan (CSE)
1st	ENG 1101(V4)	General English - Section A	3	Shreyashi Paul (CSE)
1st	ENG 1101(V4)	General English - Section B	3	Shreyashi Paul (CSE)
1st	ENG 1101(V4)	General English - Section C	3	Shreyashi Paul (CSE)
1st	ENG 1101(V4)	General English - Section D	3	Shreyashi Paul (CSE)
1st	ENG 1101(V4)	General English - Section E	3	Shreyashi Paul (CSE)
1st	MAT 1203(V4)	Differential and Integral Calculus - Section A	3	DR. MOHAMMAD ALI (GT)
1st	MAT 1203(V4)	Differential and Integral Calculus - Section B	3	DR. MOHAMMAD ALI (GT)
1st	MAT 1203(V4)	Differential and Integral Calculus - Section C	3	Dr. Mohammed Aman Ullah (GT)
1st	MAT 1203(V4)	Differential and Integral Calculus - Section D	3	Dr. Mohammed Aman Ullah (GT)
1st	MAT 1203(V4)	Differential and Integral Calculus - Section E	3	Dr. Ujjwal Kumar Deb (GT)
1st	ME 1104(V4)	Mechanical Engineering Drawing - Section A	0.75	Dr. Md. Sanaul Rabbi (GT)

1st	ME 1104(V4)	Mechanical Engineering Drawing - Section B	0.75	Dr. Md. Sanaul Rabbi (GT)
1st	ME 1104(V4)	Mechanical Engineering Drawing - Section C	0.75	Dr. Md. Sanaul Rabbi (GT)
1st	ME 1104(V4)	Mechanical Engineering Drawing - Section D	0.75	Prof. Dr. Mahbubul Alam (GT)
1st	ME 1104(V4)	Mechanical Engineering Drawing - Section E	0.75	Prof. Dr. Mahbubul Alam (GT)
1st	PHY 101(V3)	Engineering Physics I - Section (Only Retake)	3	Dr. Mihir Kumar Roy (GT)
1st	PHY 1103(V4)	Introduction to Classical & Modern Physics - Section A	3	Nusrat Jahan Shirin (CSE)
1st	PHY 1103(V4)	Introduction to Classical & Modern Physics - Section B	3	Nusrat Jahan Shirin (CSE)
1st	PHY 1103(V4)	Introduction to Classical & Modern Physics - Section C	3	Dr. Mihir Kumar Roy (GT)
1st	PHY 1103(V4)	Introduction to Classical & Modern Physics - Section D	3	Dr. Mihir Kumar Roy (GT)
1st	PHY 1103(V4)	Introduction to Classical & Modern Physics - Section E	3	Dr. Mihir Kumar Roy (GT)
2nd	CSE 103(V3)	Discrete Mathematics - Section A	3	Farhana Shirin Chowdhury (CSE)
2nd	CSE 103(V3)	Discrete Mathematics - Section B	3	Farhana Shirin Chowdhury (CSE)
2nd	CSE 103(V3)	Discrete Mathematics - Section C	3	Farhana Shirin Chowdhury (CSE)
2nd	CSE 111(V3)	Structured Programming - Section A	2	Mohammad Hasan (CSE)
2nd	CSE 111(V3)	Structured Programming - Section B	2	Adiba Ibnat Hossain (CSE)
2nd	CSE 111(V3)	Structured Programming - Section C	2	Md. Hasan (CSE)
2nd	CSE 112(V3)	Structured Programming Laboratory - Section A	2	Mohammad Hasan (CSE)
2nd	CSE 112(V3)	Structured Programming Laboratory - Section B	2	Adiba Ibnat Hossain (CSE)
2nd	CSE 112(V3)	Structured Programming Laboratory - Section C1	2	Md. Hasan (CSE)
2nd	CSE 112(V3)	Structured Programming Laboratory - Section C2	2	Md. Hasan (CSE)
2nd	EEE 211(V3)	Electronics I - Section A	3	Kazi Md. Abrar Yeaser (CSE)
2nd	EEE 211(V3)	Electronics I - Section B	3	Md. Kamrul Hasan (GT)
2nd	EEE 211(V3)	Electronics I - Section C	3	Md. Kamrul Hasan (GT)

2nd	EEE 212(V3)	Electronics I Laboratory - Section A	1.5	Mohammed Rezaur Rahman Chowdhury (CSE)
2nd	EEE 212(V3)	Electronics I Laboratory - Section B	1.5	Somen Dutta (EEE)
2nd	EEE 212(V3)	Electronics I Laboratory - Section C1	1.5	Mohammed Rezaur Rahman Chowdhury (CSE)
2nd	EEE 212(V3)	Electronics I Laboratory - Section C2	1.5	Mohammed Rezaur Rahman Chowdhury (CSE)
2nd	ENG 103(V3)	Developing English Skills - Section A	2	Shreyashi Paul (CSE)
2nd	ENG 103(V3)	Developing English Skills - Section B	2	Shreyashi Paul (CSE)
2nd	ENG 103(V3)	Developing English Skills - Section C	2	Shreyashi Paul (CSE)
2nd	ENG 103(V3)	Developing English Skills - Section (Retake)	2	Shreyashi Paul (CSE)
2nd	MAT 107(V3)	Engineering Mathematics II - Section A	3	Kollol Dey (DMath)
2nd	MAT 107(V3)	Engineering Mathematics II - Section B	3	Kollol Dey (DMath)
2nd	MAT 107(V3)	Engineering Mathematics II - Section C	3	Kollol Dey (DMath)
2nd	PHY 103(V3)	Engineering Physics II - Section A	3	Nusrat Jahan Shirin (CSE)
2nd	PHY 103(V3)	Engineering Physics II - Section B	3	Nusrat Jahan Shirin (CSE)
2nd	PHY 103(V3)	Engineering Physics II - Section C	3	Nusrat Jahan Shirin (CSE)
3rd	CSE 211(V3)	Object Oriented Programming - Section A	3	Sheikh Md. Rukunuddin Osmani (CSE)
3rd	CSE 211(V3)	Object Oriented Programming - Section B	3	Sheikh Md. Rukunuddin Osmani (CSE)
3rd	CSE 211(V3)	Object Oriented Programming - Section C	3	Sourav Adhikary (CSE)
3rd	CSE 211(V3)	Object Oriented Programming - Section D	3	Dr. Abu Nowshed Chy (GT)
3rd	CSE 212(V3)	Object Oriented Programming Laboratory - Section A	1.5	Sheikh Md. Rukunuddin Osmani (CSE)

3rd	CSE 212(V3)	Object Oriented Programming Laboratory - Section B	1.5	Sheikh Md. Rukunuddin Osmani (CSE)
3rd	CSE 212(V3)	Object Oriented Programming Laboratory - Section C1	1.5	Sourav Adhikary (CSE)
3rd	CSE 212(V3)	Object Oriented Programming Laboratory - Section C2	1.5	Sourav Adhikary (CSE)
3rd	CSE 212(V3)	Object Oriented Programming Laboratory - Section D1	1.5	Dr. Abu Nowshed Chy (GT)
3rd	CSE 212(V3)	Object Oriented Programming Laboratory - Section D2	1.5	Dr. Abu Nowshed Chy (GT)
3rd	CSE 221(V3)	Data Structure - Section A	3	Dhrubajyoti Das (CSE)
3rd	CSE 221(V3)	Data Structure - Section B	3	Dhrubajyoti Das (CSE)
3rd	CSE 221(V3)	Data Structure - Section C	3	MD. SHAFIUL ALAM FORHAD (GT)
3rd	CSE 221(V3)	Data Structure - Section D	3	MD. SHAFIUL ALAM FORHAD (GT)
3rd	CSE 222(V3)	Data Structure Laboratory - Section A	1.5	Dhrubajyoti Das (CSE)
3rd	CSE 222(V3)	Data Structure Laboratory - Section B	1.5	Dhrubajyoti Das (CSE)
3rd	CSE 222(V3)	Data Structure Laboratory - Section C1	1.5	Syed Md. Minhaz Hossain (CSE)
3rd	CSE 222(V3)	Data Structure Laboratory - Section C2	1.5	Syed Md. Minhaz Hossain (CSE)
3rd	CSE 222(V3)	Data Structure Laboratory - Section D1	1.5	Syed Md. Minhaz Hossain (CSE)
3rd	CSE 222(V3)	Data Structure Laboratory - Section D2	1.5	Syed Md. Minhaz Hossain (CSE)
3rd	ECO 201(V3)	Basic Economics - Section A	3	Noushin Farjana Huda Chowdhury (GT)
3rd	ECO 201(V3)	Basic Economics - Section B	3	Noushin Farjana Huda Chowdhury (GT)
3rd	ECO 201(V3)	Basic Economics - Section C	3	Kazi Mezbah Uddin Ahamad (GT)
3rd	ECO 201(V3)	Basic Economics - Section D	3	Kazi Mezbah Uddin Ahamad (GT)
3rd	EEE 311(V3)	Digital Electronics - Section A	3	FAHIM MAHMUD (GT)

3rd	EEE 311(V3)	Digital Electronics - Section B	3	Aditta Chowdhury (GT)
3rd	EEE 311(V3)	Digital Electronics - Section C	3	Dr. Sampad Ghosh (GT)
3rd	EEE 311(V3)	Digital Electronics - Section D	3	Dr. Sampad Ghosh (GT)
3rd	EEE 312(V3)	Digital Electronics Laboratory - Section A	1.5	Sujan Chowdhury (EEE)
3rd	EEE 312(V3)	Digital Electronics Laboratory - Section B	1.5	Mohammed Saifuddin Munna (EEE)
3rd	EEE 312(V3)	Digital Electronics Laboratory - Section C1	1.5	Mohammed Rezaur Rahman Chowdhury (CSE)
3rd	EEE 312(V3)	Digital Electronics Laboratory - Section C2	1.5	Mohammed Rezaur Rahman Chowdhury (CSE)
3rd	EEE 312(V3)	Digital Electronics Laboratory - Section D1	1.5	Adiba Ibnat Hossain (CSE)
3rd	EEE 312(V3)	Digital Electronics Laboratory - Section D2	1.5	Sourav Adhikary (CSE)
3rd	MAT 201(V3)	Engineering Mathematics III - Section A	3	Kollol Dey (DMath)
3rd	MAT 201(V3)	Engineering Mathematics III - Section B	3	Kollol Dey (DMath)
3rd	MAT 201(V3)	Engineering Mathematics III - Section C	3	Kollol Dey (DMath)
3rd	MAT 201(V3)	Engineering Mathematics III - Section D	3	Kollol Dey (DMath)
4th	CSE 225(V3)	Algorithm Design And Analysis - Section A	3	Syed Md. Minhaz Hossain (CSE)
4th	CSE 226(V3)	Algorithm Design And Analysis Laboratory - Section A1	1	Syed Md. Minhaz Hossain (CSE)
4th	CSE 226(V3)	Algorithm Design And Analysis Laboratory - Section A2	1	Syed Md. Minhaz Hossain (CSE)
4th	CSE 237(V3)	Database Management System - Section A	3	Anik Sen (CSE)
4th	CSE 238(V3)	Database Management System Laboratory - Section A1	1.5	Anik Sen (CSE)
4th	CSE 238(V3)	Database Management System Laboratory - Section A2	1.5	Anik Sen (CSE)
4th	EEE 201(V3)	Signals & Systems - Section A	3	Mohammed Rezaur Rahman Chowdhury (CSE)
4th	EEE 202(V3)	Signals & Systems Laboratory - Section A1	1	Sujan Chowdhury (EEE)

4th	EEE 202(V3)	Signals & Systems Laboratory - Section A2	1	Sujan Chowdhury (EEE)
4th	MAT 203(V3)	Engineering Mathematics IV - Section A	3	Tania Noor (DMath)
4th	MGT 203(V3)	Industrial & Business Management - Section A	3	Ela Saha (GT)
5th	CSE 301(V3)	Computational Methods for Engineering Problems - Section A	3	Tania Noor (DMath)
5th	CSE 302(V3)	Computational Methods for Engineering Problems Laboratory - Section A1	1	Tania Noor (DMath)
5th	CSE 302(V3)	Computational Methods for Engineering Problems Laboratory - Section A2	1	Tania Noor (DMath)
5th	CSE 305(V3)	Software Engineering & Information System Design - Section A	4	Md. Neamul Haque (CSE)
5th	CSE 306(V3)	Software Engineering & Information System Design Laboratory - Section A1	1.5	Md. Neamul Haque (CSE)
5th	CSE 306(V3)	Software Engineering & Information System Design Laboratory - Section A2	1.5	Md. Neamul Haque (CSE)
5th	EEE 309(V3)	Communication Engineering - Section A	3	Samina Alam (EEE)
5th	EEE 310(V3)	Communication Engineering Laboratory - Section A1	1.5	Sujan Chowdhury (EEE)
5th	EEE 310(V3)	Communication Engineering Laboratory - Section A2	1.5	Sujan Chowdhury (EEE)
5th	EEE 371(V3)	Microprocessors & Microcontrollers - Section A	3	Kingshuk Dhar (CSE)
5th	EEE 372(V3)	Microprocessors & Microcontrollers Laboratory - Section A1	1.5	Kingshuk Dhar (CSE)
5th	EEE 372(V3)	Microprocessors & Microcontrollers Laboratory - Section A2	1.5	Kazi Md. Abrar Yeaser (CSE)
5th	MGT 251(V3)	Organizational Behavior - Section A	3	Tasnim Sultana (GT)
6th	CSE 317(V3)	Artificial Intelligence - Section A	3	Avisheak Das (CSE)
6th	CSE 317(V3)	Artificial Intelligence - Section B	3	Faisal Ahmed (CSE)
6th	CSE 317(V3)	Artificial Intelligence - Section C	3	Md. Hasan (CSE)

6th	CSE 318(V3)	Artificial Intelligence Laboratory - Section A1	1.5	Avisheak Das (CSE)
6th	CSE 318(V3)	Artificial Intelligence Laboratory - Section A2	1.5	Avisheak Das (CSE)
6th	CSE 318(V3)	Artificial Intelligence Laboratory - Section B1	1.5	Faisal Ahmed (CSE)
6th	CSE 318(V3)	Artificial Intelligence Laboratory - Section B2	1.5	Faisal Ahmed (CSE)
6th	CSE 318(V3)	Artificial Intelligence Laboratory - Section C	1.5	Md. Hasan (CSE)
6th	CSE 333(V3)	Operating Systems - Section A	3	Sabrina Tarannum (CSE)
6th	CSE 333(V3)	Operating Systems - Section B	3	Asma Joshita Trisha (CSE)
6th	CSE 333(V3)	Operating Systems - Section C	3	Asma Joshita Trisha (CSE)
6th	CSE 334(V3)	Operating Systems Laboratory - Section A1	1.5	Sabrina Tarannum (CSE)
6th	CSE 334(V3)	Operating Systems Laboratory - Section A2	1.5	Sabrina Tarannum (CSE)
6th	CSE 334(V3)	Operating Systems Laboratory - Section B1	1.5	Asma Joshita Trisha (CSE)
6th	CSE 334(V3)	Operating Systems Laboratory - Section B2	1.5	Asma Joshita Trisha (CSE)
6th	CSE 334(V3)	Operating Systems Laboratory - Section C	1.5	Asma Joshita Trisha (CSE)
6th	CSE 337(V3)	Computer Organization & Architecture - Section A	3	Sourav Adhikary (CSE)
6th	CSE 337(V3)	Computer Organization & Architecture - Section B	3	Sourav Adhikary (CSE)
6th	CSE 337(V3)	Computer Organization & Architecture - Section C	3	Avisheak Das (CSE)
6th	CSE 338(V3)	Software Development - Section A1	2	Dhrubajyoti Das (CSE)
6th	CSE 338(V3)	Software Development - Section A2	2	Dhrubajyoti Das (CSE)
6th	CSE 338(V3)	Software Development - Section B1	2	Anik Sen (CSE)
6th	CSE 338(V3)	Software Development - Section B2	2	Anik Sen (CSE)
6th	CSE 338(V3)	Software Development - Section C	2	Anik Sen (CSE)
6th	CSE 364(V3)	Data Communication - Section A	3	Sabrina Tarannum (CSE)
6th	CSE 364(V3)	Data Communication - Section B	3	Rezaul Karim (GT)

6th	CSE 364(V3)	Data Communication - Section C	3	Rezaul Karim (GT)
6th	CSE 367(V3)	Computer Network - Section A	3	Shahid Md. Asif Iqbal (CSE)
6th	CSE 367(V3)	Computer Network - Section B	3	Shahid Md. Asif Iqbal (CSE)
6th	CSE 367(V3)	Computer Network - Section C	3	Sheikh Md. Rukunuddin Osmani (CSE)
6th	CSE 368(V3)	Computer Network Laboratory - Section A1	1.5	Shahid Md. Asif Iqbal (CSE)
6th	CSE 368(V3)	Computer Network Laboratory - Section A2	1.5	Shahid Md. Asif Iqbal (CSE)
6th	CSE 368(V3)	Computer Network Laboratory - Section B1	1.5	Shahid Md. Asif Iqbal (CSE)
6th	CSE 368(V3)	Computer Network Laboratory - Section B2	1.5	Shahid Md. Asif Iqbal (CSE)
6th	CSE 368(V3)	Computer Network Laboratory - Section C	1.5	Sheikh Md. Rukunuddin Osmani (CSE)
7th	CSE 309(V3)	Theory of Computation - Section A	2	Avisheak Das (CSE)
7th	CSE 309(V3)	Theory of Computation - Section B	2	Adiba Ibnat Hossain (CSE)
7th	CSE 309(V3)	Theory of Computation - Section C	2	Adiba Ibnat Hossain (CSE)
7th	CSE 437(V3)	Network and Computer Security - Section A	3	Kingshuk Dhar (CSE)
7th	CSE 437(V3)	Network and Computer Security - Section B	3	Kingshuk Dhar (CSE)
7th	CSE 437(V3)	Network and Computer Security - Section C	3	Kingshuk Dhar (CSE)
7th	CSE 451(V3)	Neural Network & Fuzzy Logic - Section A	3	Farhana Shirin Chowdhury (CSE)
7th	CSE 451(V3)	Neural Network & Fuzzy Logic - Section B	3	Farhana Shirin Chowdhury (CSE)
7th	CSE 451(V3)	Neural Network & Fuzzy Logic - Section C	3	Faisal Ahmed (CSE)
7th	CSE 452(V3)	Neural Network & Fuzzy Logic Laboratory - Section A1	1	Farhana Shirin Chowdhury (CSE)
7th	CSE 452(V3)	Neural Network & Fuzzy Logic Laboratory - Section A2	1	Farhana Shirin Chowdhury (CSE)
7th	CSE 452(V3)	Neural Network & Fuzzy Logic Laboratory - Section B1	1	Faisal Ahmed (CSE)
7th	CSE 452(V3)	Neural Network & Fuzzy Logic Laboratory - Section B2	1	Faisal Ahmed (CSE)

7th	CSE 452(V3)	Neural Network & Fuzzy Logic Laboratory - Section C1	1	Faisal Ahmed (CSE)
7th	CSE 452(V3)	Neural Network & Fuzzy Logic Laboratory - Section C2	1	Faisal Ahmed (CSE)
7th	CSE 455(V3)	Computer Graphics & Image Processing - Section A	3	Dr. Kaushik Deb (GT)
7th	CSE 455(V3)	Computer Graphics & Image Processing - Section B	3	Dr. Kaushik Deb (GT)
7th	CSE 455(V3)	Computer Graphics & Image Processing - Section C	3	Dr. Kaushik Deb (GT)
7th	CSE 456(V3)	Computer Graphics & Image Processing Laboratory - Section A1	1	Md. Neamul Haque (CSE)
7th	CSE 456(V3)	Computer Graphics & Image Processing Laboratory - Section A2	1	Md. Neamul Haque (CSE)
7th	CSE 456(V3)	Computer Graphics & Image Processing Laboratory - Section B1	1	Md. Neamul Haque (CSE)
7th	CSE 456(V3)	Computer Graphics & Image Processing Laboratory - Section B2	1	Md. Neamul Haque (CSE)
7th	CSE 456(V3)	Computer Graphics & Image Processing Laboratory - Section C1	1	Md. Neamul Haque (CSE)
7th	CSE 456(V3)	Computer Graphics & Image Processing Laboratory - Section C2	1	Md. Neamul Haque (CSE)
7th	EEE 313(V3)	Control Systems - Section A	3	Arif Istiaque (GT)
7th	EEE 313(V3)	Control Systems - Section B	3	Arif Istiaque (GT)
7th	EEE 313(V3)	Control Systems - Section C	3	Somen Dutta (EEE)
7th	EEE 314(V3)	Control Systems Laboratory - Section A1	1.5	Muhammad Helal Uddin (EEE)
7th	EEE 314(V3)	Control Systems Laboratory - Section A2	1.5	Akramul Haque (EEE)
7th	EEE 314(V3)	Control Systems Laboratory - Section B1	1.5	Rahul Chowdhury (EEE)
7th	EEE 314(V3)	Control Systems Laboratory - Section B2	1.5	Asif Mohammed Siddiqee (EEE)
7th	EEE 314(V3)	Control Systems Laboratory - Section C1	1.5	Somen Dutta (EEE)
7th	EEE 314(V3)	Control Systems Laboratory - Section C2	1.5	Somen Dutta (EEE)

7th	ENG 401(V3)	Technical Writing & Presentation - Section A	2	Dr. Mohammad Kamrul Hassan (GT)
7th	ENG 401(V3)	Technical Writing & Presentation - Section B	2	Dr. Mohammad Kamrul Hassan (GT)
7th	ENG 401(V3)	Technical Writing & Presentation - Section C	2	Dr. Mohammad Kamrul Hassan (GT)
8th	CSE 453(V3)	Compiler Construction	3	Tanni Dhoom (CSE)
8th	CSE 459(V3)	Pattern Recognition	3	Dr. Kaushik Deb (GT)
8th	CSE 481(V3)	Contemporary Course of Computer Science	3	Shahid Md. Asif Iqbal (CSE)

Offer List Spring 2022

Semester	Course	Teacher
1st	ACC 101(V3) - Basic Accounting - Section A (3)	Nusrat Jahan
1st	ACC 101(V3) - Basic Accounting - Section B (3)	Naima Nazneen chowdhury
1st	ACC 101(V3) - Basic Accounting - Section C (3)	Nusrat Jahan
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section A (2)	Puja Chakraborty
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section B (2)	Puja Chakraborty
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section C1 (2)	Puja Chakraborty
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section C2 (2)	Puja Chakraborty
1st	EEE 101(V3) - Electrical Circuits I - Section A (3)	Arif Bhuyan
1st	EEE 101(V3) - Electrical Circuits I - Section B (3)	Arif Bhuyan
1st	EEE 101(V3) - Electrical Circuits I - Section C (3)	Arif Bhuyan
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section A (1.5)	Arif Bhuyan
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section B (1.5)	Arif Bhuyan
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section C1 (1.5)	Arif Bhuyan
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section C2 (1.5)	Arif Bhuyan
1st	ENG 101(V3) - General English - Section A (3)	Shreyashi Paul
1st	ENG 101(V3) - General English - Section B (3)	Shreyashi Paul
1st	ENG 101(V3) - General English - Section C (3)	Shreyashi Paul
1st	MAT 105(V3) - Engineering Mathematics I - Section A (3)	Dr. Ujjwal Kumar Deb
1st	MAT 105(V3) - Engineering Mathematics I - Section B (3)	Dr. Ujjwal Kumar Deb
1st	MAT 105(V3) - Engineering Mathematics I - Section C (3)	Dr. Ujjwal Kumar Deb
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section A (1)	Dr. Sajal Chandra Banik
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section B (1)	Dr. Sajal Chandra Banik

1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section C1 (1)	Dr. Sajal Chandra Banik
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section C2 (1)	Dr. Sajal Chandra Banik
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section D (1)	Dr. Md. Sanaul Rabbi
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section E (1)	A.S.M. Sayem
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section F (1)	A.S.M. Sayem
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section G (1)	A.S.M. Sayem
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section H (1)	A.S.M. Sayem
1st	PHY 101(V3) - Engineering Physics I - Section A (3)	Nusrat Jahan Shirin
1st	PHY 101(V3) - Engineering Physics I - Section B (3)	Nusrat Jahan Shirin
1st	PHY 101(V3) - Engineering Physics I - Section C (3)	Nusrat Jahan Shirin
2nd	CSE 103(V3) - Discrete Mathematics - Section A (3)	Syed Md. Minhaz Hossain
2nd	CSE 103(V3) - Discrete Mathematics - Section B (3)	Syed Md. Minhaz Hossain
2nd	CSE 103(V3) - Discrete Mathematics - Section C (3)	Farhana Shirin Chowdhury
2nd	CSE 103(V3) - Discrete Mathematics - Section D (3)	Farhana Shirin Chowdhury
2nd	CSE 111(V3) - Structured Programming - Section C (2)	Mohammad Hasan
2nd	CSE 111(V3) - Structured Programming - Section D (2)	Dhrubajyoti Das
2nd	CSE 111(V3) - Structured Programming - Section A (2)	Mohammad Hasan
2nd	CSE 111(V3) - Structured Programming - Section B (2)	Dhrubajyoti Das
2nd	CSE 112(V3) - Structured Programming Laboratory - Section A (2)	Mohammad Hasan
2nd	CSE 112(V3) - Structured Programming Laboratory - Section B (2)	Dhrubajyoti Das
2nd	CSE 112(V3) - Structured Programming Laboratory - Section C1 (2)	Mohammad Hasan
2nd	CSE 112(V3) - Structured Programming Laboratory - Section C2 (2)	Mohammad Hasan
2nd	CSE 112(V3) - Structured Programming Laboratory - Section D1 (2)	Farhana Shirin Chowdhury
2nd	CSE 112(V3) - Structured Programming Laboratory - Section D2 (2)	Dhrubajyoti Das

2nd	EEE 211(V3) - Electronics I - Section C (3)	Kazi Md. Abrar Yeaser
2nd	EEE 211(V3) - Electronics I - Section D (3)	Kazi Md. Abrar Yeaser
2nd	EEE 211(V3) - Electronics I - Section A (3)	Adnan Hossain Khan
2nd	EEE 211(V3) - Electronics I - Section B (3)	Adnan Hossain Khan
2nd	EEE 212(V3) - Electronics I Laboratory - Section A (1.5)	Adnan Hossain Khan
2nd	EEE 212(V3) - Electronics I Laboratory - Section B (1.5)	Adnan Hossain Khan
2nd	EEE 212(V3) - Electronics I Laboratory - Section C1 (1.5)	Kazi Md. Abrar Yeaser
2nd	EEE 212(V3) - Electronics I Laboratory - Section C2 (1.5)	Kazi Md. Abrar Yeaser
2nd	EEE 212(V3) - Electronics I Laboratory - Section D1 (1.5)	Kazi Md. Abrar Yeaser
2nd	EEE 212(V3) - Electronics I Laboratory - Section D2 (1.5)	Somen Dutta
2nd	ENG 103(V3) - Developing English Skills - Section C (2)	Shreyashi Paul
2nd	ENG 103(V3) - Developing English Skills - Section D (2)	Shreyashi Paul
2nd	ENG 103(V3) - Developing English Skills - Section A (2)	Shreyashi Paul
2nd	ENG 103(V3) - Developing English Skills - Section B (2)	Shreyashi Paul
2nd	MAT 107(V3) - Engineering Mathematics II - Section A (3)	Kollol Dey
2nd	MAT 107(V3) - Engineering Mathematics II - Section B (3)	Kollol Dey
2nd	MAT 107(V3) - Engineering Mathematics II - Section C (3)	Kollol Dey
2nd	MAT 107(V3) - Engineering Mathematics II - Section D (3)	Kollol Dey
2nd	PHY 103(V3) - Engineering Physics II - Section C (3)	Nusrat Jahan Shirin
2nd	PHY 103(V3) - Engineering Physics II - Section D (3)	Nusrat Jahan Shirin
2nd	PHY 103(V3) - Engineering Physics II - Section A (3)	Dr. AKM Moinul Haque Meaze
2nd	PHY 103(V3) - Engineering Physics II - Section B (3)	Dr. AKM Moinul Haque Meaze
3rd	CSE 211(V3) - Object Oriented Programming - Section A (3)	Dr. Abu Nowshed Chy
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section A1 (1.5)	Dr. Abu Nowshed Chy

3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section A2 (1.5)	Dr. Abu Nowshed Chy
3rd	CSE 221(V3) - Data Structure - Section A (3)	Farhana Shirin Chowdhury
3rd	CSE 222(V3) - Data Structure Laboratory - Section A1 (1.5)	Farhana Shirin Chowdhury
3rd	CSE 222(V3) - Data Structure Laboratory - Section A2 (1.5)	Farhana Shirin Chowdhury
3rd	ECO 201(V3) - Basic Economics - Section A (3)	Sudip Dey
3rd	EEE 311(V3) - Digital Electronics - Section A (3)	Somen Dutta
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section A1 (1.5)	Somen Dutta
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section A2 (1.5)	Somen Dutta
3rd	MAT 201(V3) - Engineering Mathematics III - Section A (3)	Kollol Dey
4th	CSE 225(V3) - Algorithm Design And Analysis - Section A (3)	Syed Md. Minhaz Hossain
4th	CSE 225(V3) - Algorithm Design And Analysis - Section B (3)	Tahmina Khanam
4th	CSE 226(V3) - Algorithm Design And Analysis Laboratory - Section B (1)	Tahmina Khanam
4th	CSE 226(V3) - Algorithm Design And Analysis Laboratory - Section A1 (1)	Syed Md. Minhaz Hossain
4th	CSE 226(V3) - Algorithm Design And Analysis Laboratory - Section A2 (1)	Syed Md. Minhaz Hossain
4th	CSE 237(V3) - Database Management System - Section A (3)	Md. Neamul Haque
4th	CSE 238(V3) - Database Management System Laboratory - Section A1 (1.5)	Mohammad Hasan
4th	CSE 238(V3) - Database Management System Laboratory - Section A2 (1.5)	Mohammad Hasan
4th	EEE 201(V3) - Signals & Systems - Section A (3)	Sujan Chowdhury
4th	EEE 202(V3) - Signals & Systems Laboratory - Section A1 (1)	Sujan Chowdhury
4th	EEE 202(V3) - Signals & Systems Laboratory - Section A2 (1)	Sujan Chowdhury
4th	MAT 203(V3) - Engineering Mathematics IV - Section A (3)	Kollol Dey
4th	MGT 203(V3) - Industrial & Business Management - Section A (3)	Tasnim Sultana
5th	CSE 301(V3) - Computational Methods for Engineering Problems - Section A (3)	Tania Noor
5th	CSE 301(V3) - Computational Methods for Engineering Problems - Section B (3)	Tania Noor

5th	CSE 301(V3) - Computational Methods for Engineering Problems - Section C (3)	Tania Noor
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section A1 (1)	Tanni Dhoom
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section A2 (1)	Tanni Dhoom
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section B1 (1)	Tanni Dhoom
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section B2 (1)	Tanni Dhoom
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section C (1)	Tanni Dhoom
5th	CSE 305(V3) - Software Engineering & Information System Design - Section A (4)	Md. Neamul Haque
5th	CSE 305(V3) - Software Engineering & Information System Design - Section B (4)	Md. Neamul Haque
5th	CSE 305(V3) - Software Engineering & Information System Design - Section C (4)	Md. Neamul Haque
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section A1 (1.5)	Md. Neamul Haque
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section A2 (1.5)	Md. Neamul Haque
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section B1 (1.5)	Dhrubajyoti Das
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section B2 (1.5)	Dhrubajyoti Das
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section C (1.5)	Md. Neamul Haque
5th	EEE 309(V3) - Communication Engineering - Section A (3)	Sujan Chowdhury
5th	EEE 309(V3) - Communication Engineering - Section B (3)	Sujan Chowdhury
5th	EEE 309(V3) - Communication Engineering - Section C (3)	Sujan Chowdhury
5th	EEE 310(V3) - Communication Engineering Laboratory - Section A1 (1.5)	Khaleque Md. Aashiq Kamal
5th	EEE 310(V3) - Communication Engineering Laboratory - Section A2 (1.5)	Khaleque Md. Aashiq Kamal
5th	EEE 310(V3) - Communication Engineering Laboratory - Section B1 (1.5)	Khaleque Md. Aashiq Kamal
5th	EEE 310(V3) - Communication Engineering Laboratory - Section B2 (1.5)	Khaleque Md. Aashiq Kamal
5th	EEE 310(V3) - Communication Engineering Laboratory - Section C (1.5)	Khaleque Md. Aashiq Kamal
5th	EEE 371(V3) - Microprocessors & Microcontrollers - Section A (3)	Kingshuk Dhar

5th	EEE 371(V3) - Microprocessors & Microcontrollers - Section B (3)	Kingshuk Dhar
5th	EEE 371(V3) - Microprocessors & Microcontrollers - Section C (3)	Kingshuk Dhar
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section A1 (1.5)	Kingshuk Dhar
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section A2 (1.5)	Kingshuk Dhar
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section B1 (1.5)	Adnan Hossain Khan
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section B2 (1.5)	Adnan Hossain Khan
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section C (1.5)	Kazi Md. Abrar Yeaser
5th	MGT 251(V3) - Organizational Behavior - Section A (3)	Md. Touhidul Islam Chowdhury
5th	MGT 251(V3) - Organizational Behavior - Section B (3)	Md. Touhidul Islam Chowdhury
5th	MGT 251(V3) - Organizational Behavior - Section C (3)	Ela Saha
6th	CSE 317(V3) - Artificial Intelligence - Section A (3)	Faisal Ahmed
6th	CSE 317(V3) - Artificial Intelligence - Section B (3)	Faisal Ahmed
6th	CSE 317(V3) - Artificial Intelligence - Section C (3)	Faisal Ahmed
6th	CSE 318(V3) - Artificial Intelligence Laboratory - Section A1 (1.5)	Faisal Ahmed
6th	CSE 318(V3) - Artificial Intelligence Laboratory - Section A2 (1.5)	Faisal Ahmed
6th	CSE 318(V3) - Artificial Intelligence Laboratory - Section B1 (1.5)	Syed Md. Minhaz Hossain
6th	CSE 318(V3) - Artificial Intelligence Laboratory - Section B2 (1.5)	Syed Md. Minhaz Hossain
6th	CSE 318(V3) - Artificial Intelligence Laboratory - Section C1 (1.5)	Faisal Ahmed
6th	CSE 318(V3) - Artificial Intelligence Laboratory - Section C2 (1.5)	Faisal Ahmed
6th	CSE 333(V3) - Operating Systems - Section A (3)	Sabrina Tarannum
6th	CSE 333(V3) - Operating Systems - Section B (3)	Sabrina Tarannum
6th	CSE 333(V3) - Operating Systems - Section C (3)	Sabrina Tarannum
6th	CSE 334(V3) - Operating Systems Laboratory - Section A1 (1.5)	Sabrina Tarannum
6th	CSE 334(V3) - Operating Systems Laboratory - Section A2 (1.5)	Sabrina Tarannum
6th	CSE 334(V3) - Operating Systems Laboratory - Section B1 (1.5)	Sabrina Tarannum

6th	CSE 334(V3) - Operating Systems Laboratory - Section B2 (1.5)	Sabrina Tarannum
6th	CSE 334(V3) - Operating Systems Laboratory - Section C1 (1.5)	Sabrina Tarannum
6th	CSE 334(V3) - Operating Systems Laboratory - Section C2 (1.5)	Sabrina Tarannum
6th	CSE 337(V3) - Computer Organization & Architecture - Section A (3)	Taufique Sayeed
6th	CSE 337(V3) - Computer Organization & Architecture - Section B (3)	Taufique Sayeed
6th	CSE 337(V3) - Computer Organization & Architecture - Section C (3)	Taufique Sayeed
6th	CSE 338(V3) - Software Development - Section A1 (2)	Anik Sen
6th	CSE 338(V3) - Software Development - Section A2 (2)	Anik Sen
6th	CSE 338(V3) - Software Development - Section B1 (2)	Anik Sen
6th	CSE 338(V3) - Software Development - Section B2 (2)	Anik Sen
6th	CSE 338(V3) - Software Development - Section C1 (2)	Anik Sen
6th	CSE 338(V3) - Software Development - Section C2 (2)	Anik Sen
6th	CSE 364(V3) - Data Communication - Section A (3)	Rezaul Karim
6th	CSE 364(V3) - Data Communication - Section B (3)	Rezaul Karim
6th	CSE 364(V3) - Data Communication - Section C (3)	Rezaul Karim
6th	CSE 367(V3) - Computer Network - Section A (3)	Shahid Md. Asif Iqbal
6th	CSE 367(V3) - Computer Network - Section B (3)	Sheikh Md. Rukunuddin Osmani
6th	CSE 367(V3) - Computer Network - Section C (3)	Sheikh Md. Rukunuddin Osmani
6th	CSE 368(V3) - Computer Network Laboratory - Section A1 (1.5)	Shahid Md. Asif Iqbal
6th	CSE 368(V3) - Computer Network Laboratory - Section A2 (1.5)	Shahid Md. Asif Iqbal
6th	CSE 368(V3) - Computer Network Laboratory - Section B1 (1.5)	Sheikh Md. Rukunuddin Osmani
6th	CSE 368(V3) - Computer Network Laboratory - Section B2 (1.5)	Sheikh Md. Rukunuddin Osmani
6th	CSE 368(V3) - Computer Network Laboratory - Section C1 (1.5)	Sheikh Md. Rukunuddin Osmani
6th	CSE 368(V3) - Computer Network Laboratory - Section C2 (1.5)	Sheikh Md. Rukunuddin Osmani
8th	CSE 400(V3) - Thesis - Section A (4)	Tanni Dhoom

8th	CSE 400(V3) - Thesis - Section B (4)	Taufique Sayeed
8th	CSE 400(V3) - Thesis - Section C (4)	Dhrubajyoti Das
8th	CSE 400(V3) - Thesis - Section D (4)	Farhana Shirin Chowdhury
8th	CSE 400(V3) - Thesis - Section E (4)	Syed Md. Minhaz Hossain
8th	CSE 400(V3) - Thesis - Section F (4)	Sabrina Tarannum
8th	CSE 400(V3) - Thesis - Section G (4)	Anik Sen
8th	CSE 400(V3) - Thesis - Section H (4)	Syed Md. Minhaz Hossain
8th	CSE 400(V3) - Thesis - Section I (4)	Tanni Dhoom
8th	CSE 400(V3) - Thesis - Section J (4)	Mohammad Hasan
8th	CSE 400(V3) - Thesis - Section K (4)	Mohammad Hasan
8th	CSE 400(V3) - Thesis - Section L (4)	Md. Neamul Haque
8th	CSE 400(V3) - Thesis - Section M (4)	Arif Bhuyan
8th	CSE 400(V3) - Thesis - Section N (4)	Sheikh Md. Rukunuddin Osmani
8th	CSE 400(V3) - Thesis - Section O (4)	Faisal Ahmed
8th	CSE 400(V3) - Thesis - Section P (4)	Mohammad Hasan
8th	CSE 400(V3) - Thesis - Section Q (4)	Anik Sen
8th	CSE 400(V3) - Thesis - Section R (4)	Shahid Md. Asif Iqbal
8th	CSE 400(V3) - Thesis - Section S (4)	Asma Joshita Trisha
8th	CSE 400(V3) - Thesis - Section T (4)	Md. Neamul Haque
8th	CSE 400(V3) - Thesis - Section U (4)	Kingshuk Dhar
8th	CSE 400(V3) - Thesis - Section V (4)	Syed Md. Minhaz Hossain
8th	CSE 400(V3) - Thesis - Section W (4)	Asma Joshita Trisha
8th	CSE 400(V3) - Thesis - Section X (4)	Md. Neamul Haque
8th	CSE 453(V3) - Compiler Construction - Section A (3)	Tanni Dhoom
8th	CSE 454(V3) - Compiler Construction Laboratory - Section A1 (1.5)	Tanni Dhoom
8th	CSE 454(V3) - Compiler Construction Laboratory - Section A2 (1.5)	Tanni Dhoom
8th	CSE 457(V3) - Machine Learning - Section A (3)	Taufique Sayeed
8th	CSE 458(V3) - Machine Learning Laboratory - Section A1 (1)	Taufique Sayeed
8th	CSE 458(V3) - Machine Learning Laboratory - Section A2 (1)	Taufique Sayeed
8th	CSE 459(V3) - Pattern Recognition - Section A (3)	Dr. Kaushik Deb
8th	CSE 460(V3) - Pattern Recognition Laboratory - Section A1 (1)	Dr. Kaushik Deb

8th	CSE 460(V3) - Pattern Recognition Laboratory - Section A2 (1)	Dr. Kaushik Deb
8th	CSE 481(V3) - Contemporary Courses of Computer Science - Section A (3)	Shahid Md. Asif Iqbal
8th	CSE 482(V3) - Contemporary Courses of Computer Science Laboratory - Section A1 (1)	Shahid Md. Asif Iqbal
8th	CSE 482(V3) - Contemporary Courses of Computer Science Laboratory - Section A2 (1)	Shahid Md. Asif Iqbal

Offer List Fall 2021

Semester	Course	Teacher
1st	ACC 101(V3) - Basic Accounting - Section A (3)	Sumana Podder
1st	ACC 101(V3) - Basic Accounting - Section B (3)	Sumana Podder
1st	ACC 101(V3) - Basic Accounting - Section C (3)	Sumana Podder
1st	ACC 101(V3) - Basic Accounting - Section D (3)	Nusrat Jahan
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section C1 (2)	Sheikh Md. Rukunuddin Osmani
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section C2 (2)	Sheikh Md. Rukunuddin Osmani
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section D1 (2)	Khaleque Md. Aashiq Kamal
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section D2 (2)	Khaleque Md. Aashiq Kamal
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section A (2)	Puja Chakraborty
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section B (2)	Puja Chakraborty
1st	EEE 101(V3) - Electrical Circuits I - Section A (3)	Arif Bhuyan
1st	EEE 101(V3) - Electrical Circuits I - Section B (3)	Arif Bhuyan
1st	EEE 101(V3) - Electrical Circuits I - Section C (3)	Somen Dutta
1st	EEE 101(V3) - Electrical Circuits I - Section D (3)	Somen Dutta
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section D1 (1.5)	Asma Joshita Trisha
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section D2 (1.5)	Asma Joshita Trisha
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section A (1.5)	Arif Bhuyan
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section B (1.5)	Arif Bhuyan
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section C1 (1.5)	Asma Joshita Trisha
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section C2 (1.5)	Asma Joshita Trisha
1st	ENG 101(V3) - General English - Section A (3)	Shreyashi Paul
1st	ENG 101(V3) - General English - Section B (3)	Shreyashi Paul
1st	ENG 101(V3) - General English - Section C (3)	Shreyashi Paul
1st	ENG 101(V3) - General English - Section D (3)	Shreyashi Paul

1st	MAT 105(V3) - Engineering Mathematics I - Section C (3)	Dr. Ujjwal Kumar Deb
1st	MAT 105(V3) - Engineering Mathematics I - Section D (3)	Dr. Ujjwal Kumar Deb
1st	MAT 105(V3) - Engineering Mathematics I - Section A (3)	Dr. Ujjwal Kumar Deb
1st	MAT 105(V3) - Engineering Mathematics I - Section B (3)	Dr. Ujjwal Kumar Deb
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section A (1)	Dr. Md. Sanaul Rabbi
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section B (1)	Dr. Md. Sanaul Rabbi
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section C1 (1)	Dr. Md. Sanaul Rabbi
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section C2 (1)	Dr. Sajal Chandra Banik
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section D1 (1)	Dr. Sajal Chandra Banik
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section D2 (1)	Dr. Sajal Chandra Banik
1st	PHY 101(V3) - Engineering Physics I - Section C (3)	Nusrat Jahan Shirin
1st	PHY 101(V3) - Engineering Physics I - Section D (3)	Nusrat Jahan Shirin
1st	PHY 101(V3) - Engineering Physics I - Section A (3)	Nusrat Jahan Shirin
1st	PHY 101(V3) - Engineering Physics I - Section B (3)	Nusrat Jahan Shirin
2nd	CSE 103(V3) - Discrete Mathematics - Section A (3)	Syed Md. Minhaz Hossain
2nd	CSE 111(V3) - Structured Programming - Section A (2)	Puja Chakraborty
2nd	CSE 112(V3) - Structured Programming Laboratory - Section A1 (2)	Puja Chakraborty
2nd	CSE 112(V3) - Structured Programming Laboratory - Section A2 (2)	Puja Chakraborty
2nd	EEE 211(V3) - Electronics I - Section A (3)	Adnan Hossain Khan
2nd	EEE 212(V3) - Electronics I Laboratory - Section A1 (1.5)	M. A. Hafiz
2nd	EEE 212(V3) - Electronics I Laboratory - Section A2 (1.5)	M. A. Hafiz
2nd	ENG 103(V3) - Developing English Skills - Section A (2)	Shreyashi Paul
2nd	MAT 107(V3) - Engineering Mathematics II - Section A (3)	Tania Noor

2nd	PHY 103(V3) - Engineering Physics II - Section A (3)	Nusrat Jahan Shirin
3rd	CSE 211(V3) - Object Oriented Programming - Section A (3)	Sheikh Md. Rukunuddin Osmani
3rd	CSE 211(V3) - Object Oriented Programming - Section B (3)	Sheikh Md. Rukunuddin Osmani
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section A1 (1.5)	Sheikh Md. Rukunuddin Osmani
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section A2 (1.5)	Sheikh Md. Rukunuddin Osmani
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section B (1.5)	Sheikh Md. Rukunuddin Osmani
3rd	CSE 221(V3) - Data Structure - Section A (3)	Farhana Shirin Chowdhury
3rd	CSE 221(V3) - Data Structure - Section B (3)	Syed Md. Minhaz Hossain
3rd	CSE 222(V3) - Data Structure Laboratory - Section A1 (1.5)	Farhana Shirin Chowdhury
3rd	CSE 222(V3) - Data Structure Laboratory - Section A2 (1.5)	Farhana Shirin Chowdhury
3rd	CSE 222(V3) - Data Structure Laboratory - Section B (1.5)	Syed Md. Minhaz Hossain
3rd	ECO 201(V3) - Basic Economics - Section A (3)	Sudip Dey
3rd	EEE 311(V3) - Digital Electronics - Section A (3)	Sabrina Tarannum
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section A1 (1.5)	Asma Joshita Trisha
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section A2 (1.5)	Asma Joshita Trisha
3rd	MAT 201(V3) - Engineering Mathematics III - Section A (3)	Tania Noor
4th	CSE 225(V3) - Algorithm Design And Analysis - Section A (3)	Farhana Shirin Chowdhury
4th	CSE 225(V3) - Algorithm Design And Analysis - Section B (3)	Syed Md. Minhaz Hossain
4th	CSE 225(V3) - Algorithm Design And Analysis - Section C (3)	Syed Md. Minhaz Hossain
4th	CSE 226(V3) - Algorithm Design And Analysis Laboratory - Section A1 (1)	Farhana Shirin Chowdhury
4th	CSE 226(V3) - Algorithm Design And Analysis Laboratory - Section A2 (1)	Farhana Shirin Chowdhury
4th	CSE 226(V3) - Algorithm Design And Analysis Laboratory - Section B1 (1)	Dhrubajyoti Das
4th	CSE 226(V3) - Algorithm Design And Analysis Laboratory - Section B2 (1)	Dhrubajyoti Das
4th	CSE 226(V3) - Algorithm Design And Analysis Laboratory - Section C (1)	Dhrubajyoti Das

4th	CSE 237(V3) - Database Management System - Section A (3)	Mohammad Hasan
4th	CSE 237(V3) - Database Management System - Section B (3)	Dhrubajyoti Das
4th	CSE 237(V3) - Database Management System - Section C (3)	Anik Sen
4th	CSE 238(V3) - Database Management System Laboratory - Section A1 (1.5)	Mohammad Hasan
4th	CSE 238(V3) - Database Management System Laboratory - Section A2 (1.5)	Mohammad Hasan
4th	CSE 238(V3) - Database Management System Laboratory - Section B1 (1.5)	Dhrubajyoti Das
4th	CSE 238(V3) - Database Management System Laboratory - Section B2 (1.5)	Dhrubajyoti Das
4th	CSE 238(V3) - Database Management System Laboratory - Section C (1.5)	Anik Sen
4th	EEE 201(V3) - Signals & Systems - Section A (3)	Kazi Md. Abrar Yeaser
4th	EEE 201(V3) - Signals & Systems - Section B (3)	Adnan Hossain Khan
4th	EEE 201(V3) - Signals & Systems - Section C (3)	Kazi Md. Abrar Yeaser
4th	EEE 202(V3) - Signals & Systems Laboratory - Section A1 (1)	Khaleque Md. Aashiq Kamal
4th	EEE 202(V3) - Signals & Systems Laboratory - Section A2 (1)	Khaleque Md. Aashiq Kamal
4th	EEE 202(V3) - Signals & Systems Laboratory - Section B1 (1)	Khaleque Md. Aashiq Kamal
4th	EEE 202(V3) - Signals & Systems Laboratory - Section B2 (1)	Khaleque Md. Aashiq Kamal
4th	EEE 202(V3) - Signals & Systems Laboratory - Section C (1)	Khaleque Md. Aashiq Kamal
4th	MAT 203(V3) - Engineering Mathematics IV - Section A (3)	Kollol Dey
4th	MAT 203(V3) - Engineering Mathematics IV - Section B (3)	Kollol Dey
4th	MAT 203(V3) - Engineering Mathematics IV - Section C (3)	Kollol Dey
4th	MGT 203(V3) - Industrial & Business Management - Section A (3)	Tasnim Sultana
4th	MGT 203(V3) - Industrial & Business Management - Section B (3)	Tasnim Sultana
4th	MGT 203(V3) - Industrial & Business Management - Section C (3)	Tasnim Sultana
5th	CSE 301(V3) - Computational Methods for Engineering Problems - Section A (3)	Kollol Dey
5th	CSE 301(V3) - Computational Methods for Engineering Problems - Section B (3)	Kollol Dey

5th	CSE 301(V3) - Computational Methods for Engineering Problems - Section C (3)	Kollol Dey
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section A1 (1)	Sabrina Tarannum
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section A2 (1)	Sabrina Tarannum
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section B1 (1)	Sabrina Tarannum
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section B2 (1)	Sabrina Tarannum
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section C1 (1)	Sabrina Tarannum
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section C2 (1)	Sabrina Tarannum
5th	CSE 305(V3) - Software Engineering & Information System Design - Section A (4)	Mohammad Hasan
5th	CSE 305(V3) - Software Engineering & Information System Design - Section B (4)	Faisal Ahmed
5th	CSE 305(V3) - Software Engineering & Information System Design - Section C (4)	Anik Sen
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section A1 (1.5)	Md. Neamul Haque
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section A2 (1.5)	Md. Neamul Haque
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section B1 (1.5)	Md. Neamul Haque
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section B2 (1.5)	Md. Neamul Haque
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section C1 (1.5)	Anik Sen
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section C2 (1.5)	Anik Sen
5th	EEE 309(V3) - Communication Engineering - Section A (3)	Sharith Dhar
5th	EEE 309(V3) - Communication Engineering - Section B (3)	Muhammad Helal Uddin
5th	EEE 309(V3) - Communication Engineering - Section C (3)	Muhammad Helal Uddin
5th	EEE 310(V3) - Communication Engineering Laboratory - Section A1 (1.5)	Rahul Chowdhury

5th	EEE 310(V3) - Communication Engineering Laboratory - Section A2 (1.5)	Rahul Chowdhury
5th	EEE 310(V3) - Communication Engineering Laboratory - Section B1 (1.5)	Samina Alam
5th	EEE 310(V3) - Communication Engineering Laboratory - Section B2 (1.5)	Samina Alam
5th	EEE 310(V3) - Communication Engineering Laboratory - Section C1 (1.5)	Asif Mohammed Siddiqee
5th	EEE 310(V3) - Communication Engineering Laboratory - Section C2 (1.5)	Asif Mohammed Siddiqee
5th	EEE 371(V3) - Microprocessors & Microcontrollers - Section A (3)	Kingshuk Dhar
5th	EEE 371(V3) - Microprocessors & Microcontrollers - Section B (3)	Asma Joshita Trisha
5th	EEE 371(V3) - Microprocessors & Microcontrollers - Section C (3)	Asma Joshita Trisha
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section A1 (1.5)	Kingshuk Dhar
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section A2 (1.5)	Kingshuk Dhar
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section B1 (1.5)	Kingshuk Dhar
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section B2 (1.5)	Kingshuk Dhar
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section C1 (1.5)	Arif Bhuyan
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section C2 (1.5)	Arif Bhuyan
5th	MGT 251(V3) - Organizational Behavior - Section A (3)	Ela Saha
5th	MGT 251(V3) - Organizational Behavior - Section B (3)	Ela Saha
5th	MGT 251(V3) - Organizational Behavior - Section C (3)	Md. Touhidul Islam Chowdhury
7th	CSE 309(V3) - Theory of Computation - Section A (2)	Sabrina Tarannum
7th	CSE 309(V3) - Theory of Computation - Section B (2)	Sabrina Tarannum
7th	CSE 437(V3) - Network and Computer Security - Section A (3)	Khaleque Md. Aashiq Kamal
7th	CSE 437(V3) - Network and Computer Security - Section B (3)	Khaleque Md. Aashiq Kamal
7th	CSE 451(V3) - Neural Network & Fuzzy Logic - Section A (3)	Faisal Ahmed
7th	CSE 451(V3) - Neural Network & Fuzzy Logic - Section B (3)	Faisal Ahmed

7th	CSE 452(V3) - Neural Network & Fuzzy Logic Laboratory - Section A (1)	Faisal Ahmed
7th	CSE 452(V3) - Neural Network & Fuzzy Logic Laboratory - Section B (1)	Faisal Ahmed
7th	CSE 455(V3) - Computer Graphics & Image Processing - Section A (3)	Md. Neamul Haque
7th	CSE 455(V3) - Computer Graphics & Image Processing - Section B (3)	Md. Neamul Haque
7th	CSE 456(V3) - Computer Graphics & Image Processing Laboratory - Section A (1)	Md. Neamul Haque
7th	CSE 456(V3) - Computer Graphics & Image Processing Laboratory - Section B (1)	Md. Neamul Haque
7th	EEE 313(V3) - Control Systems - Section A (3)	Somen Dutta
7th	EEE 313(V3) - Control Systems - Section B (3)	Somen Dutta
7th	EEE 314(V3) - Control Systems Laboratory - Section A (1.5)	Somen Dutta
7th	EEE 314(V3) - Control Systems Laboratory - Section B (1.5)	Somen Dutta
7th	ENG 401(V3) - Technical Writing & Presentation - Section A (2)	Dr. Mohammad Kamrul Hassan
7th	ENG 401(V3) - Technical Writing & Presentation - Section B (2)	Dr. Mohammad Kamrul Hassan
8th	CSE 453(V3) - Compiler Construction - Section A (3)	Tanni Dhoom
8th	CSE 453(V3) - Compiler Construction - Section B (3)	Tanni Dhoom
8th	CSE 453(V3) - Compiler Construction - Section C (3)	Tanni Dhoom
8th	CSE 454(V3) - Compiler Construction Laboratory - Section A (1.5)	Tanni Dhoom
8th	CSE 454(V3) - Compiler Construction Laboratory - Section B (1.5)	Tanni Dhoom
8th	CSE 454(V3) - Compiler Construction Laboratory - Section C (1.5)	Tanni Dhoom
8th	CSE 457(V3) - Machine Learning - Section A (3)	Taufique Sayeed
8th	CSE 457(V3) - Machine Learning - Section B (3)	Taufique Sayeed
8th	CSE 457(V3) - Machine Learning - Section C (3)	Taufique Sayeed
8th	CSE 458(V3) - Machine Learning Laboratory - Section A (1)	Taufique Sayeed
8th	CSE 458(V3) - Machine Learning Laboratory - Section B (1)	Taufique Sayeed
8th	CSE 458(V3) - Machine Learning Laboratory - Section C (1)	Taufique Sayeed
8th	CSE 459(V3) - Pattern Recognition - Section A (3)	Dr. Kaushik Deb

8th	CSE 459(V3) - Pattern Recognition - Section B (3)	Dr. Kaushik Deb
8th	CSE 459(V3) - Pattern Recognition - Section C (3)	Dr. Kaushik Deb
8th	CSE 460(V3) - Pattern Recognition Laboratory - Section A (1)	Syed Md. Minhaz Hossain
8th	CSE 460(V3) - Pattern Recognition Laboratory - Section B (1)	Syed Md. Minhaz Hossain
8th	CSE 460(V3) - Pattern Recognition Laboratory - Section C (1)	Syed Md. Minhaz Hossain
8th	CSE 481(V3) - Contemporary Courses of Computer Science - Section A (3)	Shahid Md. Asif Iqbal
8th	CSE 481(V3) - Contemporary Courses of Computer Science - Section B (3)	Shahid Md. Asif Iqbal
8th	CSE 481(V3) - Contemporary Courses of Computer Science - Section C (3)	Shahid Md. Asif Iqbal
8th	CSE 482(V3) - Contemporary Courses of Computer Science Laboratory - Section A (1)	Shahid Md. Asif Iqbal
8th	CSE 482(V3) - Contemporary Courses of Computer Science Laboratory - Section B (1)	Shahid Md. Asif Iqbal
8th	CSE 482(V3) - Contemporary Courses of Computer Science Laboratory - Section C (1)	Shahid Md. Asif Iqbal

Offer List Spring 2021

Semester	Course	Teacher
1st	ACC 101(V3) - Basic Accounting (3)	Nusrat Jahan
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section A (2)	Mohammad Hasan
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) - Section B (2)	Mohammad Hasan
1st	EEE 101(V3) - Electrical Circuits I (3)	Arif Bhuyan
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section A (1.5)	Arif Bhuyan
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section B (1.5)	Arif Bhuyan
1st	ENG 101(V3) - General English (3)	Md Nurul Ahad
1st	MAT 105(V3) - Engineering Mathematics I (3)	Kollol Dey
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section A1 (1)	A.S.M. Sayem
1st	ME 102(V3) - Mechanical Engineering Drawing & CAD (Laboratory) - Section A2 (1)	A.S.M. Sayem
1st	PHY 101(V3) - Engineering Physics I (3)	Nusrat Jahan Shirin
2nd	CSE 103(V3) - Discrete Mathematics (3)	Syed Md. Minhaz Hossain
2nd	CSE 111(V3) - Structured Programming (2)	Puja Chakraborty
2nd	CSE 112(V3) - Structured Programming Laboratory (2)	Puja Chakraborty
2nd	EEE 211(V3) - Electronics I (3)	Arif Bhuyan
2nd	EEE 212(V3) - Electronics I Laboratory (1.5)	Arif Bhuyan
2nd	ENG 103(V3) - Developing English Skills (2)	Md Nurul Ahad
2nd	MAT 107(V3) - Engineering Mathematics II (3)	Dr. Ujjwal Kumar Deb
2nd	PHY 103(V3) - Engineering Physics II (3)	Nusrat Jahan Shirin
3rd	CSE 211(V3) - Object Oriented Programming - Section A (3)	Sheikh Md. Rukunuddin Osmani
3rd	CSE 211(V3) - Object Oriented Programming - Section B (3)	Sheikh Md. Rukunuddin Osmani
3rd	CSE 211(V3) - Object Oriented Programming - Section C (3)	Kingshuk Dhar
3rd	CSE 211(V3) - Object Oriented Programming - Section D (3)	Kingshuk Dhar
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section A (1.5)	Sheikh Md. Rukunuddin Osmani
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section B (1.5)	Sheikh Md. Rukunuddin Osmani

3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section C (1.5)	Kingshuk Dhar
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section D (1.5)	Kingshuk Dhar
3rd	CSE 221(V3) - Data Structure - Section A (3)	Shahid Md. Asif Iqbal
3rd	CSE 221(V3) - Data Structure - Section B (3)	Farhana Shirin Chowdhury
3rd	CSE 221(V3) - Data Structure - Section C (3)	Farhana Shirin Chowdhury
3rd	CSE 221(V3) - Data Structure - Section D (3)	Farhana Shirin Chowdhury
3rd	CSE 222(V3) - Data Structure Laboratory - Section A (1.5)	Shahid Md. Asif Iqbal
3rd	CSE 222(V3) - Data Structure Laboratory - Section B (1.5)	Farhana Shirin Chowdhury
3rd	CSE 222(V3) - Data Structure Laboratory - Section C (1.5)	Farhana Shirin Chowdhury
3rd	CSE 222(V3) - Data Structure Laboratory - Section D (1.5)	Farhana Shirin Chowdhury
3rd	ECO 201(V3) - Basic Economics - Section A (3)	Ms. Faria Hossain Borsha
3rd	ECO 201(V3) - Basic Economics - Section B (3)	Ms. Faria Hossain Borsha
3rd	ECO 201(V3) - Basic Economics - Section C (3)	Arpita Datta
3rd	ECO 201(V3) - Basic Economics - Section D (3)	Arpita Datta
3rd	EEE 311(V3) - Digital Electronics - Section A (3)	Asma Joshita Trisha
3rd	EEE 311(V3) - Digital Electronics - Section B (3)	Asma Joshita Trisha
3rd	EEE 311(V3) - Digital Electronics - Section C (3)	Puja Chakraborty
3rd	EEE 311(V3) - Digital Electronics - Section D (3)	Faisal Ahmed
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section A (1.5)	Sabrina Tarannum
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section B (1.5)	Sabrina Tarannum
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section C (1.5)	Tanni Dhoom
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section D (1.5)	Puja Chakraborty
3rd	MAT 201(V3) - Engineering Mathematics III - Section A (3)	Tania Noor
3rd	MAT 201(V3) - Engineering Mathematics III - Section B (3)	Tania Noor
3rd	MAT 201(V3) - Engineering Mathematics III - Section C (3)	Tania Noor
3rd	MAT 201(V3) - Engineering Mathematics III - Section D (3)	Tania Noor
4th	CSE 225(V3) - Algorithm Design And Analysis - Section A (3)	Syed Md. Minhaz Hossain

4th	CSE 225(V3) - Algorithm Design And Analysis - Section B (3)	Syed Md. Minhaz Hossain
4th	CSE 225(V3) - Algorithm Design And Analysis - Section C (3)	Dhrubajyoti Das
4th	CSE 226(V3) - Algorithm Design And Analysis Laboratory - Section A (1)	Syed Md. Minhaz Hossain
4th	CSE 226(V3) - Algorithm Design And Analysis Laboratory - Section B (1)	Syed Md. Minhaz Hossain
4th	CSE 226(V3) - Algorithm Design And Analysis Laboratory - Section C (1)	Dhrubajyoti Das
4th	CSE 237(V3) - Database Management System - Section A (3)	Mohammad Hasan
4th	CSE 237(V3) - Database Management System - Section B (3)	Dhrubajyoti Das
4th	CSE 237(V3) - Database Management System - Section C (3)	Anik Sen
4th	CSE 238(V3) - Database Management System Laboratory - Section A (1.5)	Mohammad Hasan
4th	CSE 238(V3) - Database Management System Laboratory - Section B (1.5)	Dhrubajyoti Das
4th	CSE 238(V3) - Database Management System Laboratory - Section C (1.5)	Anik Sen
4th	EEE 201(V3) - Signals & Systems - Section A (3)	M. A. Hafiz
4th	EEE 201(V3) - Signals & Systems - Section B (3)	M. A. Hafiz
4th	EEE 201(V3) - Signals & Systems - Section C (3)	M. A. Hafiz
4th	EEE 202(V3) - Signals & Systems Laboratory - Section A (1)	M. A. Hafiz
4th	EEE 202(V3) - Signals & Systems Laboratory - Section B (1)	M. A. Hafiz
4th	EEE 202(V3) - Signals & Systems Laboratory - Section C (1)	M. A. Hafiz
4th	MAT 203(V3) - Engineering Mathematics IV - Section A (3)	Kollol Dey
4th	MAT 203(V3) - Engineering Mathematics IV - Section B (3)	Kollol Dey
4th	MAT 203(V3) - Engineering Mathematics IV - Section C (3)	Kollol Dey
4th	MGT 203(V3) - Industrial & Business Management - Section A (3)	Tasnim Sultana
4th	MGT 203(V3) - Industrial & Business Management - Section B (3)	Ela Saha
4th	MGT 203(V3) - Industrial & Business Management - Section C (3)	Falguni Shome
6th	CSE 317(V3) - Artificial Intelligence - Section A (3)	Faisal Ahmed
6th	CSE 317(V3) - Artificial Intelligence - Section B (3)	Faisal Ahmed

6th	CSE 318(V3) - Artificial Intelligence Laboratory - Section A (1.5)	Faisal Ahmed
6th	CSE 318(V3) - Artificial Intelligence Laboratory - Section B (1.5)	Faisal Ahmed
6th	CSE 333(V3) - Operating Systems - Section A (3)	Asma Joshita Trisha
6th	CSE 333(V3) - Operating Systems - Section B (3)	Asma Joshita Trisha
6th	CSE 334(V3) - Operating Systems Laboratory - Section A (1.5)	Asma Joshita Trisha
6th	CSE 334(V3) - Operating Systems Laboratory - Section B (1.5)	Asma Joshita Trisha
6th	CSE 337(V3) - Computer Organization & Architecture - Section A (3)	Tanni Dhoom
6th	CSE 337(V3) - Computer Organization & Architecture - Section B (3)	Tanni Dhoom
6th	CSE 338(V3) - Software Development - Section A (2)	Anik Sen
6th	CSE 338(V3) - Software Development - Section B (2)	Anik Sen
6th	CSE 364(V3) - Data Communication - Section A (3)	Sabrina Tarannum
6th	CSE 364(V3) - Data Communication - Section B (3)	Sabrina Tarannum
6th	CSE 367(V3) - Computer Network - Section A (3)	Sheikh Md. Rukunuddin Osmani
6th	CSE 367(V3) - Computer Network - Section B (3)	Sheikh Md. Rukunuddin Osmani
6th	CSE 368(V3) - Computer Network Laboratory - Section A (1.5)	Khaleque Md. Aashiq Kamal
6th	CSE 368(V3) - Computer Network Laboratory - Section B (1.5)	Khaleque Md. Aashiq Kamal
7th	CSE 309(V3) - Theory of Computation - Section A (2)	Sabrina Tarannum
7th	CSE 309(V3) - Theory of Computation - Section B (2)	Sabrina Tarannum
7th	CSE 309(V3) - Theory of Computation - Section C (2)	Sabrina Tarannum
7th	CSE 437(V3) - Network and Computer Security - Section A (3)	Khaleque Md. Aashiq Kamal
7th	CSE 437(V3) - Network and Computer Security - Section B (3)	Khaleque Md. Aashiq Kamal
7th	CSE 437(V3) - Network and Computer Security - Section C (3)	Khaleque Md. Aashiq Kamal
7th	CSE 451(V3) - Neural Network & Fuzzy Logic - Section A (3)	Md. Ataur Rahman
7th	CSE 451(V3) - Neural Network & Fuzzy Logic - Section B (3)	Md. Ataur Rahman

7th	CSE 451(V3) - Neural Network & Fuzzy Logic - Section C (3)	Md. Ataur Rahman
7th	CSE 452(V3) - Neural Network & Fuzzy Logic Laboratory - Section A (1)	Md. Ataur Rahman
7th	CSE 452(V3) - Neural Network & Fuzzy Logic Laboratory - Section B (1)	Md. Ataur Rahman
7th	CSE 452(V3) - Neural Network & Fuzzy Logic Laboratory - Section C (1)	Md. Ataur Rahman
7th	CSE 455(V3) - Computer Graphics & Image Processing - Section A (3)	Md. Neamul Haque
7th	CSE 455(V3) - Computer Graphics & Image Processing - Section B (3)	Md. Neamul Haque
7th	CSE 455(V3) - Computer Graphics & Image Processing - Section C (3)	Md. Neamul Haque
7th	CSE 456(V3) - Computer Graphics & Image Processing Laboratory - Section A (1)	Md. Neamul Haque
7th	CSE 456(V3) - Computer Graphics & Image Processing Laboratory - Section B (1)	Md. Neamul Haque
7th	CSE 456(V3) - Computer Graphics & Image Processing Laboratory - Section C (1)	Md. Neamul Haque
7th	EEE 313(V3) - Control Systems - Section A (3)	Somen Dutta
7th	EEE 313(V3) - Control Systems - Section B (3)	Somen Dutta
7th	EEE 313(V3) - Control Systems - Section C (3)	Somen Dutta
7th	EEE 314(V3) - Control Systems Laboratory - Section A (1.5)	Somen Dutta
7th	EEE 314(V3) - Control Systems Laboratory - Section B (1.5)	Somen Dutta
7th	EEE 314(V3) - Control Systems Laboratory - Section C (1.5)	Somen Dutta
7th	ENG 401(V3) - Technical Writing & Presentation - Section A (2)	Dr. Mohammad Kamrul Hassan
7th	ENG 401(V3) - Technical Writing & Presentation - Section B (2)	Dr. Mohammad Kamrul Hassan
7th	ENG 401(V3) - Technical Writing & Presentation - Section C (2)	Dr. Mohammad Kamrul Hassan
8th	CSE 453(V3) - Compiler Construction (3)	Tanni Dhoom
8th	CSE 454(V3) - Compiler Construction Laboratory (1.5)	Tanni Dhoom
8th	CSE 457(V3) - Machine Learning (3)	Taufique Sayeed
8th	CSE 458(V3) - Machine Learning Laboratory (1)	Taufique Sayeed
8th	CSE 459(V3) - Pattern Recognition (3)	Dr. Kaushik Deb
8th	CSE 460(V3) - Pattern Recognition Laboratory (1)	Dr. Kaushik Deb
8th	CSE 481(V3) - Contemporary Courses of Computer Science (3)	Shahid Md. Asif Iqbal

8th	CSE 482(V3) - Contemporary Courses of Computer Science Laboratory (1)	Shahid Md. Asif Iqbal

Offer List Fall 2020

Semester	Course	Teacher
1st	ACC 101(V3) - Basic Accounting (3)	Shafaitun Nahar
1st	CSE 110(V3) - Introduction to Computer System (Laboratory) (2)	Mohammad Hasan
1st	EEE 101(V3) - Electrical Circuits I (3)	Arif Bhuyan
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section (OLD) A1 (1.5)	Rahul Chowdhury
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section (OLD) A2 (1.5)	Asif Mohammed Siddiqee
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section (OLD) B1 (1.5)	Mohammed Saifuddin Munna
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section (OLD) B2 (1.5)	Asif Mohammed Siddiqee
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section (OLD) C1 (1.5)	Rahul Chowdhury
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section (OLD) C2 (1.5)	Akramul Haque
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section (OLD) D1 (1.5)	Mohammed Saifuddin Munna
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section (OLD) D2 (1.5)	Akramul Haque
1st	EEE 102(V3) - Electrical Circuits I Laboratory - Section (NEW) (1.5)	Dhrubajyoti Das
1st	ENG 101(V3) - General English (3)	Md Nurul Ahad
1st	MAT 105(V3) - Engineering Mathematics I (3)	Dr. Ujjwal Kumar Deb
1st	PHY 101(V3) - Engineering Physics I (3)	Dr. Mihir Kumar Roy
2nd	CSE 103(V3) - Discrete Mathematics - Section A (3)	Syed Md. Minhaz Hossain
2nd	CSE 103(V3) - Discrete Mathematics - Section B (3)	Puja Chakraborty
2nd	CSE 103(V3) - Discrete Mathematics - Section C (3)	Syed Md. Minhaz Hossain
2nd	CSE 103(V3) - Discrete Mathematics - Section D (3)	Puja Chakraborty
2nd	CSE 111(V3) - Structured Programming - Section A (2)	Puja Chakraborty
2nd	CSE 111(V3) - Structured Programming - Section B (2)	Syed Md. Minhaz Hossain
2nd	CSE 111(V3) - Structured Programming - Section C (2)	Dhrubajyoti Das
2nd	CSE 111(V3) - Structured Programming - Section D (2)	Mohammad Hasan

2nd	CSE 112(V3) - Structured Programming Laboratory - Section A1 (2)	Puja Chakraborty
2nd	CSE 112(V3) - Structured Programming Laboratory - Section A2 (2)	Puja Chakraborty
2nd	CSE 112(V3) - Structured Programming Laboratory - Section B1 (2)	Syed Md. Minhaz Hossain
2nd	CSE 112(V3) - Structured Programming Laboratory - Section B2 (2)	Syed Md. Minhaz Hossain
2nd	CSE 112(V3) - Structured Programming Laboratory - Section C1 (2)	Dhrubajyoti Das
2nd	CSE 112(V3) - Structured Programming Laboratory - Section C2 (2)	Dhrubajyoti Das
2nd	CSE 112(V3) - Structured Programming Laboratory - Section D1 (2)	Mohammad Hasan
2nd	CSE 112(V3) - Structured Programming Laboratory - Section D2 (2)	Mohammad Hasan
2nd	EEE 211(V3) - Electronics I - Section A (3)	Somen Dutta
2nd	EEE 211(V3) - Electronics I - Section B (3)	Muhammad Helal Uddin
2nd	EEE 211(V3) - Electronics I - Section C (3)	Arif Bhuyan
2nd	EEE 211(V3) - Electronics I - Section D (3)	Arif Bhuyan
2nd	EEE 212(V3) - Electronics I Laboratory - Section A1 (1.5)	Rahul Chowdhury
2nd	EEE 212(V3) - Electronics I Laboratory - Section A2 (1.5)	Asif Mohammed Siddiqee
2nd	EEE 212(V3) - Electronics I Laboratory - Section B1 (1.5)	Mohammed Saifuddin Munna
2nd	EEE 212(V3) - Electronics I Laboratory - Section B2 (1.5)	Asif Mohammed Siddiqee
2nd	EEE 212(V3) - Electronics I Laboratory - Section C1 (1.5)	Rahul Chowdhury
2nd	EEE 212(V3) - Electronics I Laboratory - Section C2 (1.5)	Akramul Haque
2nd	EEE 212(V3) - Electronics I Laboratory - Section D1 (1.5)	Mohammed Saifuddin Munna
2nd	EEE 212(V3) - Electronics I Laboratory - Section D2 (1.5)	Akramul Haque
2nd	ENG 103(V3) - Developing English Skills - Section A (2)	Md Nurul Ahad
2nd	ENG 103(V3) - Developing English Skills - Section B (2)	Md Nurul Ahad
2nd	ENG 103(V3) - Developing English Skills - Section C (2)	Md Nurul Ahad
2nd	ENG 103(V3) - Developing English Skills - Section D (2)	Md Nurul Ahad

2nd	MAT 107(V3) - Engineering Mathematics II - Section A (3)	Kollol Dey
2nd	MAT 107(V3) - Engineering Mathematics II - Section B (3)	Kollol Dey
2nd	MAT 107(V3) - Engineering Mathematics II - Section C (3)	Kollol Dey
2nd	MAT 107(V3) - Engineering Mathematics II - Section D (3)	Kollol Dey
2nd	PHY 103(V3) - Engineering Physics II - Section (OLD) E (3)	Dr. Mihir Kumar Roy
2nd	PHY 103(V3) - Engineering Physics II - Section (NEW) A (3)	Dr. AKM Moinul Haque Meaze
2nd	PHY 103(V3) - Engineering Physics II - Section (NEW) B (3)	Dr. AKM Moinul Haque Meaze
2nd	PHY 103(V3) - Engineering Physics II - Section (NEW) C (3)	Dil Afroze Begum
2nd	PHY 103(V3) - Engineering Physics II - Section (NEW) D (3)	Dil Afroze Begum
3rd	CSE 211(V3) - Object Oriented Programming - Section A (3)	Sheikh Md. Rukunuddin Osmani
3rd	CSE 211(V3) - Object Oriented Programming - Section B (3)	Tanni Dhoom
3rd	CSE 211(V3) - Object Oriented Programming - Section C (3)	Faisal Ahmed
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section B1 (1.5)	Tanni Dhoom
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section B2 (1.5)	Tanni Dhoom
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section C1 (1.5)	Faisal Ahmed
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section C2 (1.5)	Faisal Ahmed
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section A1 (1.5)	Sheikh Md. Rukunuddin Osmani
3rd	CSE 212(V3) - Object Oriented Programming Laboratory - Section A2 (1.5)	Sheikh Md. Rukunuddin Osmani
3rd	CSE 221(V3) - Data Structure - Section A (3)	Farhana Shirin Chowdhury
3rd	CSE 221(V3) - Data Structure - Section B (3)	Farhana Shirin Chowdhury
3rd	CSE 221(V3) - Data Structure - Section C (3)	Kingshuk Dhar
3rd	CSE 222(V3) - Data Structure Laboratory - Section B1 (1.5)	Farhana Shirin Chowdhury
3rd	CSE 222(V3) - Data Structure Laboratory - Section B2 (1.5)	Farhana Shirin Chowdhury
3rd	CSE 222(V3) - Data Structure Laboratory - Section C1 (1.5)	Kingshuk Dhar

3rd	CSE 222(V3) - Data Structure Laboratory - Section C2 (1.5)	Kingshuk Dhar
3rd	CSE 222(V3) - Data Structure Laboratory - Section A1 (1.5)	Farhana Shirin Chowdhury
3rd	CSE 222(V3) - Data Structure Laboratory - Section A2 (1.5)	Farhana Shirin Chowdhury
3rd	ECO 201(V3) - Basic Economics - Section A (3)	Ms. Faria Hossain Borsha
3rd	ECO 201(V3) - Basic Economics - Section B (3)	Ms. Faria Hossain Borsha
3rd	ECO 201(V3) - Basic Economics - Section C (3)	Sudip Dey
3rd	EEE 311(V3) - Digital Electronics - Section B (3)	Md. Neamul Haque
3rd	EEE 311(V3) - Digital Electronics - Section C (3)	Md. Neamul Haque
3rd	EEE 311(V3) - Digital Electronics - Section A (3)	Mohammad Hasan
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section A1 (1.5)	Md. Neamul Haque
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section A2 (1.5)	Md. Neamul Haque
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section B1 (1.5)	Md. Neamul Haque
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section B2 (1.5)	Md. Neamul Haque
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section C1 (1.5)	Faisal Ahmed
3rd	EEE 312(V3) - Digital Electronics Laboratory - Section C2 (1.5)	Md. Neamul Haque
3rd	MAT 201(V3) - Engineering Mathematics III - Section B (3)	N. U. M. Akramul Kabir Khan
3rd	MAT 201(V3) - Engineering Mathematics III - Section C (3)	N. U. M. Akramul Kabir Khan
3rd	MAT 201(V3) - Engineering Mathematics III - Section D (3)	Tania Noor
3rd	MAT 201(V3) - Engineering Mathematics III - Section E (3)	Tania Noor
3rd	MAT 201(V3) - Engineering Mathematics III - Section A (3)	N. U. M. Akramul Kabir Khan
4th	MAT 203(V3) - Engineering Mathematics IV - Section D (3)	Kollol Dey
4th	MAT 203(V3) - Engineering Mathematics IV - Section E (3)	Kollol Dey
4th	MAT 203(V3) - Engineering Mathematics IV - Section A (3)	N. U. M. Akramul Kabir Khan
4th	MAT 203(V3) - Engineering Mathematics IV - Section B (3)	N. U. M. Akramul Kabir Khan
4th	MAT 203(V3) - Engineering Mathematics IV - Section C (3)	N. U. M. Akramul Kabir Khan

5th	CSE 301(V3) - Computational Methods for Engineering Problems - Section B (3)	Tania Noor
5th	CSE 301(V3) - Computational Methods for Engineering Problems - Section A (3)	Tania Noor
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section A (1)	Sabrina Tarannum
5th	CSE 302(V3) - Computational Methods for Engineering Problems Laboratory - Section B (1)	Sabrina Tarannum
5th	CSE 305(V3) - Software Engineering & Information System Design - Section B (4)	Anik Sen
5th	CSE 305(V3) - Software Engineering & Information System Design - Section A (4)	Anik Sen
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section A (1.5)	Dhrubajyoti Das
5th	CSE 306(V3) - Software Engineering & Information System Design Laboratory - Section B (1.5)	Mohammad Hasan
5th	EEE 309(V3) - Communication Engineering - Section B (3)	Somen Dutta
5th	EEE 309(V3) - Communication Engineering - Section A (3)	Somen Dutta
5th	EEE 310(V3) - Communication Engineering Laboratory - Section (OLD) A (1.5)	Sharith Dhar
5th	EEE 310(V3) - Communication Engineering Laboratory - Section (OLD) B (1.5)	Sanzida Akter Tunaz
5th	EEE 310(V3) - Communication Engineering Laboratory - Section (OLD) C (1.5)	Sharith Dhar
5th	EEE 310(V3) - Communication Engineering Laboratory - Section (NEW) (1.5)	Sanzida Akter Tunaz
5th	EEE 371(V3) - Microprocessors & Microcontrollers - Section A (3)	Kingshuk Dhar
5th	EEE 371(V3) - Microprocessors & Microcontrollers - Section B (3)	Arif Bhuyan
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section B (1.5)	Arif Bhuyan
5th	EEE 372(V3) - Microprocessors & Microcontrollers Laboratory - Section A (1.5)	Kingshuk Dhar
5th	MGT 251(V3) - Organizational Behavior (3)	Kainatun Nur
6th	CSE 317(V3) - Artificial Intelligence - Section A (3)	Faisal Ahmed
6th	CSE 317(V3) - Artificial Intelligence - Section B (3)	Md. Ataur Rahman
6th	CSE 317(V3) - Artificial Intelligence - Section C (3)	Md. Ataur Rahman

6th	CSE 318(V3) - Artificial Intelligence Laboratory - Section A (1.5)	Faisal Ahmed
6th	CSE 318(V3) - Artificial Intelligence Laboratory - Section B (1.5)	Md. Ataur Rahman
6th	CSE 318(V3) - Artificial Intelligence Laboratory - Section C (1.5)	Md. Ataur Rahman
6th	CSE 333(V3) - Operating Systems - Section A (3)	Asma Joshita Trisha
6th	CSE 333(V3) - Operating Systems - Section B (3)	Asma Joshita Trisha
6th	CSE 333(V3) - Operating Systems - Section C (3)	Asma Joshita Trisha
6th	CSE 334(V3) - Operating Systems Laboratory - Section A (1.5)	Asma Joshita Trisha
6th	CSE 334(V3) - Operating Systems Laboratory - Section B (1.5)	Asma Joshita Trisha
6th	CSE 334(V3) - Operating Systems Laboratory - Section C (1.5)	Asma Joshita Trisha
6th	CSE 337(V3) - Computer Organization & Architecture - Section A (3)	Tanni Dhoom
6th	CSE 337(V3) - Computer Organization & Architecture - Section B (3)	Tanni Dhoom
6th	CSE 337(V3) - Computer Organization & Architecture - Section C (3)	Tanni Dhoom
6th	CSE 338(V3) - Software Development - Section A (2)	Anik Sen
6th	CSE 338(V3) - Software Development - Section B (2)	Anik Sen
6th	CSE 338(V3) - Software Development - Section C (2)	Anik Sen
6th	CSE 364(V3) - Data Communication - Section A (3)	Sabrina Tarannum
6th	CSE 364(V3) - Data Communication - Section B (3)	Sabrina Tarannum
6th	CSE 364(V3) - Data Communication - Section C (3)	Sabrina Tarannum
6th	CSE 367(V3) - Computer Network - Section A (3)	Sheikh Md. Rukunuddin Osmani
6th	CSE 367(V3) - Computer Network - Section B (3)	Khaleque Md. Aashiq Kamal
6th	CSE 367(V3) - Computer Network - Section C (3)	Khaleque Md. Aashiq Kamal
6th	CSE 368(V3) - Computer Network Laboratory - Section A (1.5)	Sheikh Md. Rukunuddin Osmani
6th	CSE 368(V3) - Computer Network Laboratory - Section B (1.5)	Khaleque Md. Aashiq Kamal
6th	CSE 368(V3) - Computer Network Laboratory - Section C (1.5)	Khaleque Md. Aashiq Kamal

7th	CSE 309(V3) - Theory of Computation (2)	Sabrina Tarannum
7th	CSE 437(V3) - Network and Computer Security (3)	Khaleque Md. Aashiq Kamal
7th	CSE 451(V3) - Neural Network & Fuzzy Logic (3)	Md. Ataur Rahman
7th	CSE 452(V3) - Neural Network & Fuzzy Logic Laboratory (1)	Md. Ataur Rahman
7th	CSE 455(V3) - Computer Graphics & Image Processing (3)	Md. Neamul Haque
7th	CSE 456(V3) - Computer Graphics & Image Processing Laboratory (1)	Md. Neamul Haque
7th	EEE 313(V3) - Control Systems (3)	MD.AZIM KHAN
7th	EEE 314(V3) - Control Systems Laboratory (1.5)	MD.AZIM KHAN
7th	ENG 401(V3) - Technical Writing & Presentation - Section (OLD) A (2)	Dr. Mohammad Kamrul Hassan
7th	ENG 401(V3) - Technical Writing & Presentation - Section (OLD) B (2)	Dr. Mohammad Kamrul Hassan
7th	ENG 401(V3) - Technical Writing & Presentation - Section (NEW) (2)	Dr. Mohammad Kamrul Hassan
8th	CSE 453(V3) - Compiler Construction - Section A (3)	Effat Binte Fazal
8th	CSE 453(V3) - Compiler Construction - Section B (3)	Effat Binte Fazal
8th	CSE 454(V3) - Compiler Construction Laboratory - Section A (1.5)	Effat Binte Fazal
8th	CSE 454(V3) - Compiler Construction Laboratory - Section B (1.5)	Effat Binte Fazal
8th	CSE 457(V3) - Machine Learning - Section A (3)	Taufique Sayeed
8th	CSE 457(V3) - Machine Learning - Section B (3)	Taufique Sayeed
8th	CSE 458(V3) - Machine Learning Laboratory - Section A (1)	Taufique Sayeed
8th	CSE 458(V3) - Machine Learning Laboratory - Section B (1)	Taufique Sayeed
8th	CSE 459(V3) - Pattern Recognition - Section A (3)	Dr. Kaushik Deb
8th	CSE 459(V3) - Pattern Recognition - Section B (3)	Dr. Kaushik Deb
8th	CSE 460(V3) - Pattern Recognition Laboratory - Section A (1)	Dr. Kaushik Deb
8th	CSE 460(V3) - Pattern Recognition Laboratory - Section B (1)	Dr. Kaushik Deb
8th	CSE 481(V3) - Contemporary Courses of Computer Science - Section A (3)	Shahid Md. Asif Iqbal
8th	CSE 481(V3) - Contemporary Courses of Computer Science - Section B (3)	Shahid Md. Asif Iqbal
8th	CSE 482(V3) - Contemporary Courses of Computer Science Laboratory - Section A1 (1)	Shahid Md. Asif Iqbal

8th	CSE 482(V3) - Contemporary Courses of Computer Science Laboratory - Section B1 (1)	Shahid Md. Asif Iqbal
8th	CSE 482(V3) - Contemporary Courses of Computer Science Laboratory - Section B2 (1)	Shahid Md. Asif Iqbal
8th	CSE 482(V3) - Contemporary Courses of Computer Science Laboratory - Section A2 (1)	Shahid Md. Asif Iqbal

Annexure: 6.4
List of Laboratories Activities

Course Title: Introduction to Computer System
Course Code: CSE 110

Lab	Topic
1	Generation of Computer, Types of Computers, Block diagram of a Computer System, Functions of the Different Computer Units, Input & Output Devices.
2	Memories and Software.
3	Number Systems: Binary, Octal, Decimal and Hexadecimal.
4	MS Word, MS Excel, MS Power Point.
5	MS Word, MS Excel, MS Power Point.
6	Introduction to C Language: Background, Identifiers, Data Types, Variables, Constants.
7	Input/output Statements, Creating and Running Programs, Flow Charts.
8	Flow Charts, Arithmetic Operators and Expressions: Evaluating Expressions, Precedence and Associativity of Operators, Type Conversions.
9	Flow Charts, Conditional Control Statements: Bitwise Operators, Relational and Logical Operators.
10	Flow Charts, Conditional Control Statements: If, else If.
11	Flow Charts, Conditional Control Statements: Switch Statement and Examples.
12	Flow Charts, Looping Statements: For, While and Do-While Statements with Examples.

Course Title: Electrical Circuit I
Course Code: EEE 1012

Lab	Topic
1	Familiarization with the component and devices used in electrical circuit I laboratory,
2	Verification of Ohm's law
3	Verification of KVL and Voltage Divider Rule
4	Verification of KCL and Current Divide Rule
5	Branch Current Analysis
6	Verification of Thevenin's theorem
7	Verification of Norton's theorem
8	Verification of Superposition theorem
9	Demonstration of Maximum power transfer theorem
10	Demonstration of the transient behavior of capacitor and inductor

Course Title: Mechanical Engineering Drawing & CAD
Course Code: ME 101

Lab	Topic
1	Introduction & importance of engineering drawing, Drawing techniques - manual drawing and computer-aided drawing, Manual drawing instruments and their uses
2	Conventions, Layout of drawing sheets, Border lines, Title block, Folding of drawing sheets, Lines, Lettering and dimensioning.

3	Orthographic projections, Theories of projection, multi-view (orthographic) projection
4	Projection, Projection on profile plane First and third angle projections.
5	Auxiliary Views Projections of lines and planes by
6	Auxiliary plane method.
7	Sectional Views and Conventions Introduction, Conventions in sectional view drawings,
8	True shape of a section, Sections of prisms and cylinders cones pyramids and spheres.
9	Pictorial Drawing
10	Pictorial Drawing (cont.)
11	Computer Aided Design
12	Computer Aided Design (cont.)

Course Title: Programming Fundamentals Laboratory (V4)
Course Code: CSE 1114

Lab	Topic
1	Introduction to C, IDE installation.
2	Data types and variables, Operators, Expression & Statement.
3	Standard input and output, formatted input and output, File Handling
4	Problems related to Conditional Statements
5	Problems related to Looping Statements
6	Problems related to Nested loops
7	Problems related to Arrays
8	Problems related to Strings
9	Problems related to user defined functions
10	Problems related to Pointers
11	Problems related to Structure and Union

Course Title: Introduction to Electrical Engineering Laboratory (V4)
Course Code: EEE 1102

Lab	Topic
1	Familiarization with the component and devices used in electrical circuit I lab
2	Verification of Ohm's law
3	Verification of KVL and Voltage Divider Rule
4	Verification of KCL and Current Divider Rule
5	Branch Current Analysis
6	Verification of Thevenin's theorem
7	Verification of Reciprocity theorem

8	Verification of Superposition theorem
9	Demonstration of Maximum power transfer theorem
10	Demonstration of RLC circuit
11, 12	Project Preparation
13	Project Showcasing, Presentation

Course Title: Structured Programming Lab
Course Code: CSE 112

Lab	Topic
1	Overview of C programming, Compiler/IDE setup
2	Basic C program without using conditional and loop statements
3	Basic C program without using conditional and loop statements
4	Problem solving using Control Structures and loops (if-else, nested if-else, switch-case, for loop)
5	Problem solving using Control Structures and loops(if-else, nested if-else, switch-case, for loop)
6	Problem solving using loops (nested for loops, while loops, do-while loops)
7	Problem solving using loops (nested for loops, while loops, do-while loops)
8	Problem solving using loops (nested for loops, while loops, do-while loops)
9	Programming problems solving using one dimensional array and multi-dimensional array
10	Programming problems solving using one dimensional array and multi-dimensional array
11	Programming problems solving using one dimensional array and multi-dimensional array
12	Programming problems solving using one dimensional array and multi-dimensional array
13	Programming problem on string processing
14	Programming problem on string processing
15	Programming solving using functions, passing array to function
16	Programming solving using functions, passing array to function
17	Problem solving using recursive functions
18	Problem solving using pointer
19	Problem solving using structure and unions
20	Problem solving using file
21	Introduction to competitive programming (NCPC, ICPC, Online Judge, Contest criteria etc.)
22	Lab Exam

Course Title: Electronics I Lab
Course Code: EEE 212

Lab	Topic
1	Familiarization with the components & devices used in Electronics Lab.
2	Find out the v-i characteristics of a semiconductor diode under forward & reverse-biased conditions.
3	Design a Half wave rectifier.
4	Design a Full wave rectifier using bridge and center-tapped network.
5	Find out the transistor characteristics curve for common emitter configuration,
6	Find out the transistor characteristics curve for I_B vs V_{BE}.
7	Find out the transistor characteristics curve for common base configuration.
8	Single stage Common emitter configured NPN transistor amplifier.
9	Observation of Common collector configured NPN transistor amplifier
10	Observation of FET performance
11	Project Work

Course Title: Object Oriented Programming Lab
Course Code: CSE 212

Lab	Topic
1	**Introduction** Basics of Java programming language, equation solving, conditional statement-related problem solving, and loop
2	**Class, objects, and constructor:** Programs related to Class, Objects and Constructor
3	**Method basics and method overloading** Programs related to Java Methods, Overloading Methods, Using Objects as Parameters, Returning Objects, and constructor overloading
4	**Inheritance** Programs related to Method overriding, Inheritance, explain the use of Super, Final, Nested, abstract, wrapper, and Inner classes.
5	**Interface** Programs related to interface, multiple inheritances, and dynamic initialization
6	**Multithreaded Programming** Programs related to multithreading, creating a new thread, implementing the Runnable interface, synchronization, and setting the thread's priority.
7	**Managing Errors and Exceptions** Programs related to errors and exceptions, common Java exceptions, exception handling mechanism, and finally block, illustrating the usage of throw and throws, user-defined exception.
8	**Java Applet** HTML and applet program integration, graphics programming with the applet, repaint () method, Swing package
9	**Event Handling and Layout Managers**

	Programs related to AWT (Abstract Window Toolkit), Event Classes and Listener Interfaces, and Adapter Classes.
10	**File Handling (Input/Output)** Java file-related programs to check the I/O.
11	**Java Collections Framework** Programs related to Java Map methods e.g., tree map, hash map, collections, iterator, list, and set
12	**Java Collections Framework** Programs related to Java Map methods e.g., tree map, hash map, collections, iterator, list, and set

Course Title: Data Structures Lab
Course Code: CSE 222

Lab	Topic
1	Overview (Python)
2	Pattern Matching Algorithm
3	Linear Search, Binary Search
4	Bubble Sort, Insertion Sort, Selection Sort
5	List in python
6	Linked List: Insertion (Sorted and Unsorted), Linked List: Deletion, Searching, Display, Double Linked List: Insertion, Deletion (python)
7	Mapping, Dictionary (python)
8	To write a menu driven program to perform following operations on the stack- (i)Push (ii)Pop (iii) Peek (python)
9	To perform queue operations on the menu- (i)Insert (ii)Delete (iii)Peek (iv)Display Priority queue (python)
10	Recursion: Infix to postfix (python)
11	To implement binary search tree traversal (python)
12	BFS and DFS (python)

Course Title: Digital Electronics Lab
Course Code: EEE 312

Lab	Topic
1	Verification of Digital logic gates ICs and truth tables
2	Boolean function implementation
3	Boolean function simplification and implementation using k-map
4	Verification of universality of NAND & NOR gates
5	Verification of half adder and its different types of implementations.
6	Verification of the basic operation of Full adder circuit.
7	Verification of the basic operation of Decoder
8	Verification of the basic operation of Multiplexer
9	Verification of the basic operation of De-Multiplexer.
10	Verification of the basic operation of flip flops

11	Overview and Problem Solving.

Course Title: Algorithm Design and Analysis Lab
Course Code: CSE 226

Lab	Topic
1	Using Open, implement a parallelized Merge Sort algorithm to sort a given set of elements and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.
2	Sort a given set of elements using the Quick sort method and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the lIst to be sorted and plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator. **(Performance)**
3	Implement 0/1 Knapsack problem using Dynamic and Fractional Knapsack using Greedy Method.
4	a. Print all the nodes reachable from a given starting node in a digraph using BFS method. b. Check whether a given graph is connected or not using DFS method.
5	Find Minimum Cost Spanning Tree of a given undirected graph using Prims algorithm. **(Performance)**
6	Find Minimum Cost Spanning Tree of a given undirected graph using Krushkals algorithm.
7	From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijikstras algorithm.
8	a. Obtain the Topological ordering of vertices in a given digraph. b. Compute the transitive closure of a given directed graph using Warshall's algorithm.
9	Implement any scheme to find the optimal solution for the Traveling Salesperson problem and then solve the same problem instance using any approximation algorithm and determine the error in the approximation.
10	Design and implement the presence of Hamiltonian Cycle in an undirected Graph **G** of ***n*** vertices.

Course Title: Database Management Systems Lab
Course Code: CSE 238

Lab	Topic
1	Introduction to development tools, HTML. Task 1: Mini website development (Own Profile)
2	Use of CSS Task 2: Redesign the Own Profile site using CSS.
3	Basic database implementation in MYSQL using XAMPP server.

	Task 3: Design your own project database.
4	Database connection with User Interface and System login Task 4: Develop System login of your project.
5	Development of insert panels of Class project. Task 4: Developing insert panels of group project.
6	Development of select/data view panels of Class project. Task 5: Developing select/data view panels of group project.
7	Development of update panels of Class project. Task 6: Developing update panels of group project.
8	Development of delete panels of Class project. Task 7: Developing delete panels of group project.
9	Making ER- diagram of class project on MS Visio/ Other Tools Task 8: Making ER- diagram of group project on MS Visio/ Other Tools

Course Title: Signals & Systems Laboratory
Course Code: EEE 202

Lab	Topic
1	Familiarization with the MATLAB environment and running some basic MATLAB commands in MATLAB
2	Observation of Matrix operations in MATLAB
3	Plotting of various signals in MATLAB
4	Understanding the Basic Signals using MATLAB
5	Solving of differential equation in MATLAB
6	Evaluation of convolution integral in MATLAB
7	Implementing Fourier transform in MATLAB
8	Implementing Laplace transform in MATLAB
9	Implementing Z-transform in MATLAB

Course Title: Computational Methods for Engineering Problems Laboratory
Course Code: CSE 302

Lab	Topic
1	Introduction to numerical computational program with MATLAB.
2	Input data, Drawing graph using data in MATLAB
3	Find the roots of non-linear equations using the bisection method in MATLAB
4	Find the roots of non-linear equations using the False Position method in MATLAB
5	Find the root of non-linear equation using Newton-Raphson Method in MATLAB.
6	Solving the linear system equation using Gaussian elimination method in MATLAB
7	Application of Lagrange's interpolation Method in MATLAB
8	Application of trapezoidal rule in MATLAB

Course Title: Software Engineering & Information System Design Laboratory
Course Code: CSE 306

Lab	Topic
1	Introduction
2	Forming group
3	Proposal Presentation with report and slides
4	Software development model
5	Presentation on selected software development model with report and slides
6	UML Diagram(Use case diagram, Activity diagram,ER diagram)
7	Presentation on UML Diagram(Use case diagram, Activity diagram, ER diagram) with report and slides
8	UML Diagram(Sequence digram, Class diagram)
9	Presentation on UML Diagram(Sequence digram, Class diagram) with report and slides
10	Costing and Testing
11	Presentation on Costing and Testing with report and slides

Course Title: Communication Engineering Laboratory
Course Code: EEE 310

Lab	Topic
1	Familiarization with various instruments of electrical circuit II lab
2	Observation of Amplitude Modulation of a signal
3	Observation of Amplitude Demodulation of a signal
4	Observation of Frequency Modulation of a signal
5	Observation of Frequency Demodulation of a signal
6	Observation of Phase Modulation of a signal
7	Observation of Phase Demodulation of a signal
8	Generation of an amplitude shift keyed (ASK) modulated signal
9	Generation of an frequency shift keyed (ASK) modulated signal.
10	Simulation In MATLAB

Course Title: Microprocessors & Microcontrollers Laboratory
Course Code: EEE 372

Lab	Topic
1	Familiarization with various types of microcontroller, embedded systems and tools.
2	Design a sequential LED looping circuit.
3	Digital I/O operation in an embedded system
4	Analog I/O operation in an embedded system
5	Interfacing with different types of digital sensor with arduino
6	Interfacing with different types of analog sensor with arduino
7	LCD display interfacing with arduino
8	Keypad interfacing with microcontroller

9	PWM and its use in Arduino using Servo Motor
10	Wireless message transfer using microcontroller

Course Title: Artificial Intelligence Laboratory
Course Code: CSE 318

Lab	Topic
1	**Python Introduction:** literals, Operators, Variables, Loops, Logic and Bit Operations, Lists, Sorting, Functions in Python, Tuples, and dictionaries, Objects, properties, methods, and inheritance.
2	**Machine Learning tools:** Scikit-learn, Pandas, NumPy, Matplotlib
3	Performance Evaluation on **Python Introduction** and **Machine Learning tools**
4	Implementation of unformed search techniques
5	Implementation of informed search techniques
6	Text data reading and preprocessing, Project Supervision
7	Text representation using TFIDF, Project Supervision
8	Implementation of the KNN algorithm, Project Supervision
9	Implementation of the KNN algorithm, Project Supervision
10	Implementation of the KNN algorithm, Project Supervision

Course Title: Operating Systems Laboratory
Course Code: CSE 334

Lab	Topic
1	Introduction to various Operating Systems
2	Basic Overview of Boot Process
3	User management and Basic Shell commands
4	Essential Commands
5	Advanced commands & tools for System Administration
6	Shell script and Scheduling
7	Configuring a basic server

Course Title: Software Development Project (Laboratory)
Course Code: CSE 338

Lab	Topic
1	Understanding the purpose of the course, overview of learning outcomes
2	Introduction to Laravel and Model-View-Controller (MVC), Installation via composer, Data Passing, Routing
3	CRUD operation
4	Layout an Application
5	Authentication and Authorization
6	Migration, Eloquent ORM
7	3rd Party Package Integration

8	API Development and Client-Side Interaction

Course Title: Computer Networks Laboratory
Course Code: CSE 368

Lab	Topic
1	Introduction to Cisco Packet Tracer: An innovative network configuration simulation tool.
2	Introduction to different types of network cables
3	Application Layer Services
4	Basic Router Configuration
5	Subnetting and VLSM
6	Inter-VLAN routing.
7	Routing protocol configuration
8	Network Address translation
9	Access control list
10	Wireless router configuration
11	IPv6
12	Implement a custom topology

Course Title: Control Systems Laboratory
Course Code: EEE 314

Lab	Topic
1	Introduction to Programmable Logic Controller (PLC)
2	Introduction to ladder diagram and experiment on it using normally open and close switch
3	To observe the latching operation
4	To observe the operation of up-counter
5	To observe the operation of the down-counter circuit
6	To design ladder diagram of on-timer using PLC
7	To observe the ladder diagram of traffic light system using PLC
8	To observe step response, nyquist,bode plot and root locus using MATLAB
9	To observe the different types of controllers using SIMULINK

Course Title: Neural Network & Fuzzy Logic Laboratory
Course Code: CSE 452

Lab	Topic
1	Introduction to python and important terminology
2	Implementation of Fuzzy operations
3	Implementation of Fuzzy relation
4	Implementation of Defuzzification methods
5	Implementation of Fuzzy controller (part 1)
6	Implementation of Fuzzy controller (part 2)
7	Implementation of perceptron network (perceptron learning)

8	Implementation of Adaline network (Delta Rule)
9	Implementation of Radial basis function network (part 1)
10	Implementation of Radial basis function network (part 2)
11	Implementation of Multilayer neural network
12	Implementation of Backpropagation Algorithm

Course Title: Contemporary Courses of Computer Science Laboratory
Course Code: CSE 482

Lab	Topic
1	Introduction to AWS IAM
2	Build VPC and Launch a Web Server
3	Introduction to Amazon EC2
4	AWS Lambda
5	AWS Elastic Beanstalk
6	Working with EBS
7	Build a DB Server and Interact with the DB Using an App
8	Scale and Load Balance Cloud Architecture

Course Title: Compiler Construction Laboratory
Course Code: CSE 454

Lab	Topic
1	A program to identify whether a given line is a comment or not and find the number of line where the comments are written.
2	A program to implement regular expression matching
3	A program to simulate lexical analyzer for validating operators.
4	Identification of Tokens in a given Program
5	A program to check weather a string belongs to a grammar or not
6	A Program to construct symbol table
7	A program to check whether a grammar is left recursive and remove left recursion
8	A Program to construct a NFA from a regular expression
9	A program to compute first of non-terminal
10	A program to compute follow of non-terminal and construct predictive parsing table
11	Implement the lexical analyzer using flex or other lexical analyzer generating tools.

Course Title: Machine Learning Laboratory
Course Code: CSE 458

Lab	Topic
1	Data preprocessing
2	Regression
3	SVM, KNN

4	KNN,
5	Decision Tree, Naïve Bayes classification
6	K-Means Clustering,
7	Association Rule learning: Apriori, Eclat
8	PCA
9	Model Selection and Boosting

Course Title: Computer Graphics & Image Processing Laboratory
Course Code: CSE 456

Lab	Topic
1	Introduction
2	Bresenham's Line Drawing Algorithm
3	Scan-Converting a Circle
4	Midpoint Circle Generation Algorithm
5	Digital Differential Analyzer Algorithm
6	Cohen-Sutherland Line Clipping
7	Sutherland–Hodgman Polygon Clipping
8	Geometric Transformations
9	Window-to-Viewport Mapping
10	Basic Image Processing
11	Color Model Transformation
12	Binary Image Analysis

Annexure: 6.5
Final Year Design Project List

Project/Thesis List: 2023

S/L	Title	Supervisor	Roll	Students Name
1	House Rental Web Application Learning	Sabrina Tarannum	1703310201399	Abdullah Hamid Sakib
			1703310201403	Nusrat Jahan
			1703310201423	Md. Mehedi Hasan Ovi
2	PUAIS Attendance System	Kingshuk Dhar	1703310201473	Angana Barua
			1703310201445	Fariha Chowdhury Bristy
			1703310201452	Tasnim Sultana Chowdhury
3	Web Based Cloud Kitchen System	Syed Md. Minhaz Hossain	1703310201447	Anamika Dev
			1703310201453	Nishat Naher
			1703310201454	Swapnanila Dey
4	Image Caption Generator Using CNN and LSTM	Syed Md. Minhaz Hossain	1703310201390	Md. Mujibur Rahman
			1703310201394	Fahad Abdullah Shanto
			1502910201005	Bilash Das
5	IPL Cricket Data Visualization & Best Player Selection	Syed Md. Minhaz Hossain	1703310201379	Pallab Dey
6	Restaurant Management System	Sheikh Md Rukunuddin Osmani	1703310201500	Nazrana Rahman
			1703310201520	Tani Barua
			1703310201521	Beauty Akter
7	Yoga Pose Detection & Training System	Syed Md. Minhaz Hossain	1703310201443	Nazifa Siddqua
			1703310201486	Monira Akter Chowdhury
			1703310201469	Md Shamsuddin
8	Shop Management System	Asma Joshita Trisha	1703310201530	Rashma Dev
			1703310201539	Bibi Khadija Sadia

9	Alumni Association Application	Dr. Tauifque Sayeed	1703210201352	Pulak Dasgupta
			1703210201360	Sabrina Sultana
			1603110201178	Tanjina Nasrin Tanji
10	Application on Super Shop Guide for Visually Disable Persons	Tanni Dhoom	1603110201196	Hossain Md Abid
			1603110201215	Anowar Hossain
11	Image Caption Generation using CNN and LSTM	Syed Md. Minhaz Hossain	1803410201553	Arpita Mallick
			1803410201555	Partha Das Gupta
			1803410201564	Ummay Aymon
12	Online Learning System	Mr. Kingshuk Dhar	1703210201328	Shipatul Hasan Shakib
			1703210201331	Md. Nura Hasan
13	Offline handwritten bengali District Name Recognition Using Convolutional Neural Network	Mohammad hasan	1803410201557	Md. Ahsan hossen
			1803410201565	Anik Barua
			1803410201578	Avishek chowdhury
14	A Machine Learning Based Automatic System to Detect Polycystic Ovary Syndrome	Puja Chakraborty	1703310201433	Tasnia Islam
			1703310201438	Nandini Das
			1703310201449	Asif Khan
15	"Food Bari"- An online Food Order and Management System	Mrs Sabrina tarannum	1703310201503	Rahatil Ashkeen Rishta
			1703310201523	Md. Abdullah
			1703310201544	Shariful Islam
16	Wild Animal Classification using Convolutional Neural Network	Anik Sen	1803410201581	Pranta Roy
			1803410201582	Mohammad Sarfaraj Rahman
			1803410201588	Kowshik Chowdhury
17	Cricket Shot Classification Using Convolutional Neural Network	Anik Sen	1703210201359	Aditi talukdar
			1703210201421	Pallabi Rudra
			1703210201594	Md. Mahfizur Rahman

Project/Thesis List: 2022

S/L	Title	Supervisor	Roll	Name
1	Web Based Inventory Management System	Faisal Ahmed	1603110201225	Asif Jalal
			1603110201256	Sayed Salman Faiz
			1603110201261	Abdullah Al Salman
2	Smart Health Care : An Android Application	Mohammad Hasan	1603110201269	Annesha Talukder
			1603110201270	Anika Sultana
3	University Bus Tracking Application	Tanni Dhoom	1603110201172	Imtazul Hoque
			1603110201182	Monirul Islam Chowdhury
			1603110201212	Tanvir Nawoaz
4	Traffic Case Management System	Khaleque Md. Aashiq Kamal	1202310200431	Ashek Mahmood Siddique
			1302410200508	Jov Raj Barua
			1302410200517	Chinmoy Khastagir
5	Transfer Learning Based 2D Classification for lung CT Scan SARS-Cov-2 Disease	Khaleque Md. Aashiq Kamal	1603110201218	Eshrat Jahan
6	Face Recognition System	Dhrubajyoti Das	1603010201125	Songita Das
			1603010201126	Yeasmin Akter
7	Payroll Management System	Md. Neamul Haque	1502910201103	Sinto Das
			1603010201155	Mohammad Nur Sultan Badsha
8	Web Fingerprinting Attack on Different Browser Using Machine Learning	Khaleque Md. Aashiq Kamal	1603110201204	Md. Masudul Islam
			1603110201299	Nafisa Anjum
			1603110201302	Primanjita Muhury
9	Comparative Study of Transfer Learning algorithms in Recpgnition Handwritten Metropolitan Ploice Stations	Mohammad Hasan	1603110201245	Farheen Mahmood Khan
			1603110201248	Sumi Das
			1603110201263	Samiha Israt

10	An Android Application for Finding Tutor	Md. Neamul Haque	1703310201385	Md. Jahedul Alam
			1703310201385	Sourav Dev Nath
			1703310201414	Pronamika Davi
11	Bangla Kathon	Taufique Sayeed	1603110201170	Muhammad Raihan Uddin
			1603110201194	Sadekunnur Sikdar
			1603110201217	Shovik Chowdhury
12	KenaBecha : An E-Commerce Site with Recommendation System	Syed Md. Minhaz Hossain	1703210201324	Mohammad Ariful Islam
			1703210201325	Mohammad Ashek Bin Nezam
			1703210201332	Tahiat Tabassum
13	Ecommerce Web Application	Md. Neamul Haque	1603110201208	Farhana Akther Pinky
			1703310201495	Mohammad Salah Uddin
			1703310201512	Sushmita Debi Shanta
14	Web Based Event Management System	Faisal Ahmed	1603110201224	Ashim Shil
			1603110201258	Abdullah Bin Malek
			1603110201259	Sanjida Azim Sumaiya
15	ASL Gesture Words and Alphabets Recognition by Binary Video Transformation with Convolution Neural Network	Syed Md. Minhaz Hossain	1402710200710	Nayan Krishna Bal
			1402710200729	Md Shahidul Alam
			1603110201192	Md. Omer Faruq
16	P.U.C. Network Upgradations on IPv4 and Proposed IPv6	Taufique Sayeed	1703310201465	Sadia Chowdhury Dola
			1703310201493	Tajiba Salam
			1703310201525	Munem Sahriar
17	Smart Travelling	Dhrubajyoti Das	1603110201310	Dipa Sharma
			1603110201300	Aritry Das

18	A Benchmark Dataset for Document Level Offline Bangla Handwritten Text Recognition	Kingshuk Dhar	1603110201267	Md. Osman Gony
			1703210201326	Sheikh Mohammad Jubaer
			1703210201337	Bipon Das
19	Hospital Management System	Md. Neamul Haque	1603110201183	Mohammad Hassibul Hossain
			1603110201186	Mokaddis Borhan Chowdhury
			1603110201190	Jonayed Ahmed
20	Deep Learning & Machine Learning Methods To Develop An Ultraviolet Index Prediction Model	Tanni Dhoom	1703310201404	Md Ekram Uddin
			1703310201479	Shakil Ahmed
			1703310201481	Safana Quyum Chowdhury
21	Automation of Vaccine Center	Anik Sen	1603110201203	Antar Nandi
22	Dairy farm Management System	Mohammad Hasan	1502810200891	Mir Md Imdadul Islam
			1502910200952	Mohammad Faizul Hassan
			1	Yasin Arafath Chowdhury
23	Image Steganography Using LSB Technique	Md. Neamul Haque	1703310201487	Md. Saimon Nesar
			1703310201488	Farjana Yeasmin Rafi
			1703310201489	Tajmin Nafisa
24	Movie Genre Classification from Bangla Plot Summery Using Machine Learning	Faisal Ahmed	1502910201049	Sudip Dey
			1603110201305	Natif Bil Hoque
			1603110201314	Safin Siraj Chowdhury
25	Grocery Cart	Dhrubajyoti Das	1603110201281	Harun Ar Rashid
			1603110201164	Sabuj Nath
			1703210201322	Khadizatul Kobra Abir

26	Bangla Crime News Classification Using Machine Learning	Faisal Ahmed	1703310201386	Mainul Islam Forhad
			1703310201418	Sachin Biswas
			1703310201422	Arpita Das
27	Plant Leaf Diseases Detection Using Yolo Models	Syed Md. Minhaz Hossain	1703310201384	Murshed Alam
			1703310201464	Joya Acharjee
			1703310201484	Md. Mahtab Uddin

Project/Thesis List: 2021

S/L	Title	Supervisor	Roll	Name
1	Stock Management System	Farhana Shirin Chowdhury	1402710200810	Md Tiabul Islam
			1402710200841	Faria Alam
2	Image Scrambling With Color and Index Shuffling	Farhana Shirin Chowdhury	1402610200632	Tonmoyee Chowdhury
			1402610200664	Sadia Afrin Tumpa
3	A Donation and Volunteering System	Asma Joshita Trisha	1603110201273	Farhana Binte Karim Oishy
			1603110201274	Farhana Kohinur Navila
			1603110201315	Tasmin Sultana Reshmi
4	Onilne Doctor Appoinment System	Anik Sen	1502910201082	Arfan Chowdhury
			1502910201089	Ankan Das
			1502910201095	Mohammad Ridwan Hossain
5	An Efficieient Approach of Teaching Scheduling System	Mohammad Hasan	1402710200834	Iblu Hasan
7	Machine Learnung Methods for SMS Spam Detection of Bangladesh Cellular Networks	Khaleque Md. Aashiq Kamal	1402710200778	Magnolia Biswas
			1402710200781	Urmila Barua
8	Web Based Alumni Management System for University	Anik Sen	1603010201150	Kutlug Niger

9	Online Book Sharing System	Sheikh Md. Rukunuddin Osmani	1202310200453	Josna Aktar
11	Smart Attendance Information System	Shahid Md. Asif Iqbal	1202310200452	Mohammad Khaled Mosharaf
12	Supervision System: A Tool for Monitoring Final Year Project/ Thesis	Anik Sen	1603110201226	Fahad Bin Munir
			1603110201235	Sefat Tanjim
			1603110201275	Ishmam Bin Azim
15	Watage Review Web Application	Farhana Shirin Chowdhury	1603110201227	Manisha Dey
			1603110201229	Moitree Das Moni
			1603110201241	Fardina Hosen
17	Nur Ecommerce	Dhrubajyoti Das	1402710200805	Md Fahad Hasan
			1502810200948	Mohammad Ibrahim Nur
			1502810200880	Kawlkhum Thang Amlai
18	PUC Virtual Classroom Based on Web	Asma Joshita Trisha	1603110201222	Tanzina Sultana
			1603110201254	Susmita Nath
			1603110201255	Moammer Farshid Enan
19	Development of an Institutional Online Exam Management System	Tanni Dhoom	1502910201039	Monirul Islam
			1502910201037	Mohammad. Abdur Rahman
			1502910201051	Khairul Bashar
20	Covid-19 Fake News Classification Using Machine and Deep Learning Classifiers	Syed Md. Minhaz Hossain	1603110201207	Srabanti Paul
			1603110201213	Mahabuba Akhter
			1603110201216	Rizma Sijana Nigar
21	Smart Health Care: Symptoms Based Disese Prediction Using Machine Learning	Asma Joshita Trisha	1502910201070	Mohammad Jalal Uddin
			1502910201094	Kaisher Hamid Adar

22	Bangla Handwritten Word Recognition By Localizing & Classification Characters	Sheikh Md. Rukunuddin Osmani	1502910200971	A.T.M Abdullah Bin Quader
			1502910201044	Wahid Shakib
			1502910201057	Mohammad Ibrahim Khalil
24	A Freelancing Web Application	Dhrubajyoti Das	1603110201201	Mohiminul Islam
			1603110201306	Sinthi Sen
25	Papaya Diseases Recognition Using Convolution Neural Naetwork	Syed Md. Minhaz Hossain	1502910201023	Tanvir Ahmed Talukder
			1603010201115	Thuhidul Islam
26	Web Based School Management System	Faisal Ahmed	1603110201223	Mohammad Moin Uddin
			1603110201233	Abdul Al Joboyer Chy
			1603110201278	Muhammad Azazul Islam
27	Dairy Farm Mangement System	Mohammad Hasan	1603010201152	Md. Ariful Islam
			1603110201292	Md Asif Iqbal
			1603110201293	Pranta Chowdhury
28	Online Book Share: Web Application	Mohammad Hasan	1502910201045	Abdul Hakim Rabby
			1502910201047	Md. Sirajul Munir
29	School Management System	Arif Bhuyan	1603010201116	Bashir Azgar Abir
			1603010201130	Arpita Sarker
30	Student Tracking System With RFID and Google Maps	Dhrubajyoti Das	1603110201206	Aseer Ishraqul Hoque
			1603110201221	Fatama Zannat Nury
31	Fish & Rice Cultivation	Sabrina Tarannum	1502910200955	Mohammad Irfan Khan
			1502910201072	Sakib-Ul-Hoque
32		Mohammad Hasan	1603110201308	Md. Ibrahim Siddiquee

	Handwritten Bangla District Anme Recognition Using Deep Learning Techniques		1603110201307	Sultan Md. Ayman
			1603110201311	Md. Hosen Zisad
33	Smart Irrigation System Based on Internet of Thing	Arif Bhuyan	1502810200870	Poly Mohajan Priya
			1502810200899	Mohammed Rasel
35	Movie Streaming website MongoBD	Farhana Shirin Chowdhury	1603110201173	Mollah Mohammed Irfan Uddin
			1603110201174	Sadia Chowdhury
36	CRM-Integrated Web E-Commerce Platform	Anik Sen	1703210201330	Afifa Tasmina Rina
			1703210201349	Meskatul Islam
			1703310201460	Sanjoy Kumar Dhar
37	A Comparative Study Potato Leaf Diseases Recognition Using	Mohammad Hasan	1603010201110	Tapa Chowdhury
			1603010201112	Kazi Samha Shukaria
38	Bangla Numerical Sign Language Recognition on Based Dataset	Mohammad Hasan	1603010201134	Priota Banik Dipa
			1603010201156	Pratim Chowdhury
39	Smart Shop	Dhrubajyoti Das	1502810200908	Nasrin Akter Nayan
			1502810200929	Nazma Akter
39	Vehicles Management System	Asma Joshita Trisha	1603110201168	Dipra Guha
			1603110201211	Sadita Sakhawat
41	Hospital Management System	Md. Neamul Haque	1603110201189	Mohammad Mirajul Hoque
			1603110201219	Sanjoy Das
			1603110201220	Mohammad Sajibul Hasan Chowdhury
42	Book Recycling Management System	Tanni Dhoom	1402610200666	Mohammad Abdul Goni Osmani
			1402610200667	Abdullah Al Jubaer

			1502910200951	Amitav Ghosh
44	Track My Subscription	Anik Sen	1603110201187	Mohammad Arafat Ali
			1603110201188	Abdul Karim
			1603110201209	Sucitra Das
45	Robot Car Controlled by Android Apps	Kingshuk Dhar	1502810200911	Md Imran Chowdhury
			1502910201092	Pujaree das
			1502910201107	Roushan Akter
46	Krishi Sheba: A Web Based Marketplace for Rural Farmers	Khaleque Md. Aashiq Kamal	1502910200983	Raihan Uddin
			1502910200985	Md. Emtiyaz Uddin
			1502910200999	Syed Md Shorfuddin
47	Restaurant Food Waste Reduction	Anik Sen	1402710200858	Chy. Hossain Shahid Shohrowardy
			1502910201078	Towhidul Hoque
			1502910201073	Mohammad Sojatul Islam
48	Sentiment Polarity Detection Using Machine Learning Deep Learning	Puja Chakraborty	1703210201319	Tajri Tabassum
			1703210201338	Syed Montasir Mahmood
			1703310201408	Ahasanur Rahman Mehul
49	90's Kids Dream : Mobile Game	Sheikh Md. Rukunuddin Osmani	1502910201079	Abid Al Aqib
			1502910201091	Maherin Sanjida Ema
50	Potato Leaf Diseases Regognition Using Transfer Learning And Convolution Neural Network	Asma Joshita Trisha	1603010201137	Kazi Mahmud Reza Abid
			1603010201138	Susan Iqbal Sunny
51	Stock Market Prediction Using Machine Learning and Deep Learning Algorithms	Puja Chakraborty	1703310201398	Anik Dash
			1703310201416	Sheikh Robayet Ahmed

			1703310201419	Mohammad Minhaz Uddin
52	Donation Management System	Kingshuk Dhar	1603010201136	Mohammad Jobair
			1603010201158	Mohammad Ishtiaque Ahmed Shuvo
53	Shop Management System	Md. Neamul Haque	1703210201335	Mohammad Ekhtier Uddin
			1703210201350	Gazi Md. Akbar Hossain
			1703210201353	Jawad Bin Alam
54	Fighting Game: A PC Based Game	Kingshuk Dhar	1502910200975	Bishnu Nath
			1603010201122	Pappu Barua
55	Detection of Breast Cancer Using Mammography Based on Machine Learning	Asma Joshita Trisha	1603010201113	Tafsirul Alam Chowdhury
			1603010201148	Shuvra Dav Majumder
57	Project & Thesis Exhibition (Prothesis)	Syed Md. Minhaz Hossain	1402710200745	Md Shahidul Alam
			1402710200753	Md Shahidul Islam
58	Soil Monitoring and Automatic Irrigation System Based on IOT	Kingshuk Dhar	1603010201139	Atika Yesfat Tanjum
			1603010201140	Anika Hossain
59	Online Platform for Event Management	Arif Bhuyan	1603110201249	Muhammad Muktiar Hossen
			1603110201250	Emon Paul
			1603110201253	Mohammad Karim
60	A study on Web Application Vulnerabilities	Khaleque Md. Aashiq Kamal	1502910201077	Jainab Akther Afiya
			1502910201083	Umme Akhtarun Nusaiba Tuba
			1502910201097	Md. Hamidul Haque
61			1502810200904	Joyasree Das

	PUC Final Thesis/Project Handling System	Asma Joshita Trisha	1502810200905	Israt Jahan
64	Automated Auction System	Anik Sen	1502810200909	Shimul Chakraborty
			1502810200917	Sukanta Golder
			1502810200936	Mohammad Saddam Hossen
65	Travel Guide	Farhana Shirin Chowdhury	1603110201231	Tanvir Hasan Sakib
			1603110201264	Gazi Mohammad Tajuddin Islam
			1603110201268	Mohammad Mishkat Alam
67	Krishi Bangla	Puja Chakraborty	1603110201236	Tahera Alam
			1603110201239	Sadia Salim
			1603110201252	Syeda Sanjida Farin
68	Super Mall Shopping Guide: An Android Application	Tanni Dhoom	1603010201147	Khatijatul Islam
			1603010201153	Sabina Islam Srabonti
69	Bangla Sign Alphabet Recognition Using Deep Learning Method	Dhrubajyoti Das	1502910201032	Mohammad Saiful Islam
			1603010201131	Mohammad Nahid Ullah
			1603110201312	Saurav Das
70	Online News Portal	Mohammad Hasan	1502810200915	Saiyed Md. Akib
			1402710200791	Shibly Siraj Chowdhury
71	Income Tax Calculation System	Farhana Shirin Chowdhury	1502810200914	Md. Towhidul Islam
			1502810200916	Preash Roy

Project/Thesis List: 2020

S/L	Project Name	Supervisor	Student Name	Student ID
1	University Admission System	Farhana Shirin Chowdhury	Sanjida Kamal Chowdhury Rumky	1402710200832
			Ayrin Akter	1402710200838
2	A Comparative Study on Image Encryption Using Different Scrambling Methods	Farhana Shirin Chowdhury	Mega Dhar	1402710200802
			Russel Mohajan	1402710200772
3	Analysis of Mango Leaf Disease Classification Using Image Recognition	Syed Md. Minhaz Hossain	Priya Ghosh	1402710200762
			Susmita Barua	1402710200794
4	Automating Winodows Operation Through Bangla Voice Command	Faisal Ahmed	Mohammad Mishbab Ul Haq	1402710200689
			Asuk Elahi	1402710200707
5	Web Based Class Evaluation & Feedback System	Anik Sen	Farjana Habib Tuhin	1502810200886
			Tasbin Khair	1502810200884
6	Competitive Exam Quiz	Md. Neamul Haque	Sharmin Sultana	1402710200822
7	An Explicit Platform for Purchasing Assorted Products	Md. Neamul Haque	Shovan Das	1502810200868
			Mohammad Abdul hai	1502810200890
8	An Android Application for Finding Services	Anik Sen	Riya Tripathi	1121120335
9	Smart Attendance System	Kingshuk Dhar	Md Nur Hossen	1402710200734
			Shamim ferdous	1402710200854
10	Smart Home Design	Senjuthi Bhattacharjee	Uthowai Sing Marma	1402710200723
			Niloy Barua	1402710200725
11	Smart Air-Ticketing	Tanni Dhoom	Shahrear Haque Riyadh	1402710200700
			Dipta Dutta	1702710200696
12	Diagnosis of Pneumonia from Chest X-Ray Using Deep Learning	Faisal Ahmed	Joy Das Shanta	1402610200644
			Tasfia Azim	1402610200668
13	Mobile Doctor	Mrs. Asma Joshita Trisha	Arman Hossain	1402710200683
			Rajeshwari Dhar	1402710200698
14			Arijit Das	1502810200889

	A Comprehensive Solution to Conduct Your Event	Asma Joshita Trisha	Arpita Dey Trina	1502810200895
15	SMART MENU CARD (Web applications)	Senjuthi Bhattacharjee	Sheuli Debi	1402710200807
			Nandita Chowdhury	1702710200808
16	Business Sell Buy System Apps	Sabrina Tarannum	Priyam Biswas	1121120332
17	Tomato Leaf Diseases Detection Using CNN	Faisal Ahmed	Md. Muhit Chowdhury	1402610200638
			MD. Reyad	1402610200642
18	Cold storage management System	Mohammad Hasan	Md. Zahidul Islam	1402710200845
			Md. Iqbal Mahmud	1402710200863
19	Blind Shopping: An Android Application	Mr. Kingshuk Dhar	Sharmin Akter	1402710200717
			Mitra Saha	1702710200679
20	ÓPvlxÓ: An Android Application	Mr. Kingshuk Dhar	Shatilekha Barua	1402710200754
			Aishwariya Dey	1402710200758
21	Web Based SMS Portal System	Mr. Anik Sen	Md. Saiful Islam	1502810200930
			Md. Moinuddin Morshed	1502810200913
22	Palm Vein Feature Extraction and Recognition Using Convolutional Neurak Network	Syed Md. Minhaz Hossain	Sourav Chakraborty	1402610200652
			Animesh Bhowmik Anoy	1402610200641
23	Tika Barta: An Android App for Mother and Baby care	Mohammad Hasan	Syed Tangelul Islam Masud	1202210200374
24	A Farmwoek for Managing Different Educational Institute's Admission System by Using Laravel's MVC Format	Md. Neamul Haque	Sajib Chowdhury	1502910200997
25	Smart Power Controlling System Using IOT and Blockchain	Md. Neamul Haque	Omar Sani Chisty	1402710200790
			Mafroja Nawaj	1402710200817
			Fahmida Nazmin	1402710200757
26	IOT Based Automated Fire Regulate and Response System With Data Validation	MD. Neamul Haque	Ahsan Habib	1502910201060
			Srejon Sharma	1502910201056
			Mohammad Riduanur Rahman	1502910201041
27	Health Care	Senjuthi Bhattacharjee	Iffat Hossain Bithi	1502810200867
			Sweety Bhattacharjee	1302410200510
28	Women Safety Device	Senjuthi Bhattacharjee	Umme Taslima Nadia	1402710200823

	Smart Diagnostic System for Diabetes	Prof. Dr. Taufique Sayeed	Tamiz Uddin Ahmed	1402710200818
			Abu Sufian	1402710200828
29	IOT Based Garbage Monitoring Using Arduino	Effat Binte Fazal	Subrata Das	1502810200877
			Robin Dey	1502810200896
			Probal Chowdhury	1603010201159
30	An Android Application To Find Lawyer	Effat Binte Fazal	Tanjimul Alam	1302510200572
			Tuli Dey	1302510200660
31	CTG House Rental Service	Mr. Dhrubajyoti Das	Ranjit Kumar Dey	1502810200907
			Mohammad Akib Newaj	1502810200878
32	Blood2Support System	Mr. Dhrubajyoti Das	Dhiman Nath	1502810200887
			Atanu Barua	1502810200888
33	Design & Implementation of a Fire Fighting Robot	Syed Md. Minhaz Hossain	Md. Abul Kalam Azad	1402610200635
			Ruhul Saha	1402610200553
34	Car Race: A PC Based Game	Mr. Kingshuk Dhar	Nowrin Afrin	1402710200796
			Aungkita Sen	1402710200800
35	"Online Book Collection App"	Mr. Kingshuk Dhar	Kazi Satila Arefin	1202310200401
			Mahjabin Bushra	1202310200408
36	Home To Stay: Home Rental App	Mr. Khaeque Md. Aashiq Kamal	Prosenjit Datta Shovo	1402710200776
			Anik Dey	1702710200777
37	Multivendor E-Commerce Website	Mr. Kingshuk Dhar	Md. Farhadul Islam Rana	1502910200956
			Md. Ikramur Rashid	1502910200967
			Pranta Chowdhury	1502910200977
38	Smart Shopping Mall	Mrs. Tanni Dhoom	Mohammad Mizanur Rahman	1202310200457
39	Internet Service Provider Management System	Faisal Ahmed	1502910201035	Sheikh Rahbar Mainuddin
			1502910201055	S.M Shakhawat Hosen
			1502910201068	Sumaiya Akter
40	Automated Routine Generation System	Anik Sen	1502910200993	Partha Chowdhury
			1502910200995	Jitun Mohajan
			1502910201003	Joy Mozumder
41	Ecommerce Product Classification Using	Anik Sen	1502910200986	Nazmun Naher Kalam

	Convolutional Neural Network		1502910200994	Joyshree Mozumder
			1502910201031	Humyra Ibnath Ibna
42	Project/Thesis Management System : A Module of PUAIS	Khaleque Md. Aashiq Kamal	1402710200682	Arka Muhury
			1402710200691	Khairul Aman Sourav
43	Frequent Items Set Mining of Super Shop Using Apriori Algorithms	Faisal Ahmed	1402710200684	Mashratul Mowla
			1402710200718	Md. Arfat

Project/Thesis List: 2019

S/L	Project name	Supervisor	Student Name	Student ID
1	Detection of a Resistor & to find its resistance	Asma Joshita Trisha	Abdur Rahman	1302410200513
			Md. Abu Yousuf Nayim	1302410200502
2	Crowd Sourcing- A Web Application	Mr. Khaeque Md. Aashiq Kamal	Moinul Alam Joy	1202310200430
			Akash Sen	1202310200438
			Mafia Khanom	1202310200559
3	Tic-Tac-Toe An Android Bassed Game	Mr. Khaeque Md. Aashiq Kamal	Sathi Barua	1202210200383
4	An Image recognition system for crop disease identification of paddy fields in Bangladesh	Syed Md. Minhaz Hossain	Tashfia Hoque	1302510200582
			Snighda Barua	1402610200665
			Mahuda Kutub	1402610200669
5	Deep Learning Based Paddy Disease Detection	Faisal Ahmed	Md. Abdullah Sabit	1302410200524
			Md. Asif Uddin	1302410200526
6	“ZAMS: A Pipeline for Accurate Variant Calling by Integrating Various Tools”	Prof. Dr. Taufique Sayeed	Masum Biilah	1402610200662
			Moumita Barua	1402610200654
7	Automated Registration System for Alumni Association	Mr. Kingshuk Dhar	Md. Riaz Murshed	1202310200447
			Kazi Mizanul Hoq	1202310200473
8		Asma Joshita Trisha	Md. Aslam Hossain	1202310200450

	A project on Government Market Information System		Dewan Saeeduzzaman Qurashi	1202310200491
9	Data Search Adaptive forwarding Strategies in Named Data Networking	Mr. Sheikh Md Rukunuddin Osmani	Riyadh Uddin	1302510200535
			Piual Das	1302510200554
			K.M. Faisal Epon	1302510200537
10	Accident Tracer	Ms. Senjuthi Bhattcharjee	Irin Parvin	1202310200454
			Mohammad Asifur Rahman	1302410200497
11	"Design and Implementation of gestrue Vocalizer for Deaf & Mute People Interaction"	Mr. Ariful Islam Bhuyan	Azizul Hoque	1202210200388
12	Smart Home Automation	Mr. Ariful Islam Bhuyan	Prothom Acharjee	1402610200629
			Easin Arfat	1402610200661
13	Design & bImplementation of GSM Access Door Lock System with Database	Mr. Ariful Islam Bhuyan	Sudipta Dey	1402610200659
			Md. Jahangir Alam Ripon	1402610200645
14	Smart Irrigation Control System	Faisal Ahmed	Kazi Asifur Rahman	1402610200670
			Amit Chakraborty	1402610200673
15	Automatic Digit & Alphabet Recognition Based Online Toll Collection System	Khaled Mohammad Saifuddin	Md. Sahadat Hossen	1302510200602
			Mahmuda Akter	1402610200646
16	A Report on Traffic Information System Android Application	Syed Md. Minhaz Hossain	Vabna Dey	1302410200525
			Papri Sarker Priya	1202310200437
17	A Report on Classified Auction System & It's Android Applicaton	Mr. Sheikh Md Rukunuddin Osmani	Nasrin Sultana	1202310200448
			Sanjida Hasan Shati	1202310200449
18	"Boat Adventure"-An Android Game	Mohammad Hasan	Prapty Guha	1302510200550
			Mizbatul jannat Refat	1302510200548
19	"Cycle Rider"-A PC Game	Mohammad Hasan	Rana Barua	1302510200538
			Pranta Nandy	1402710200779
20	Social Security Application:An Android	Mohammad Hasan	Ummeh Mahian Tabassum	1302410200493

	Application for an Emergency Response		Tonmoy Chakraborty	1402410200505
21	"Smart Floor Cleaner"	Mr. Kingshuk Dhar	Arup Sharma	1302510200556
			Shuvra Biswas	1302510200568
22	Intellgent Parking Allotment System Based on Automatic Vehicle Detection from Real-Time Video Frame	Khaled Mohammad Saifuddin	Asmatun Nur Ruhi	1402710200786
			Shuvecha Shill	1402710200765
23	Project Report On Merchandising	Faisal Ahmed	Shakila Siddiquea	1302410200523
			Palash Kanti Nath	1302410200492
24	Hungry Ghost-An Android 3D Game	Ms. Senjuthi Bhattcharjee	Mitu Barua	1402710200766
			Irfat Mahmood Turkey	1402710200775
25	Paper Cricket	Mr. Subrata Das	Durjoy Das	1302510200540
			Moumita Chakraborty	1302510200539
26	Online Study-A Study Tool To Support Educational Course	Mrs. Tanni Dhoom	Joy Chowdhury	1202210200384
			H.M. Mohiuddin Islam	1202210200387
27	Gesture Based Computer Control System	Sabrina Tarannum	Kazi Zahedul Hoque	1121120344
			Saju Kumar Nath	1121120353
28	Smart Ai Intruder Alert System	Syed Md. Minhaz Hossain	Mohammad Ashfaq Udddin	1402710200686
			Mohammad Hoque Nasif	1402710200697
29	Vehicle Tracking With Accident Alert and Alcohol Detection System	Syed Md. Minhaz Hossain	MD. Rahat Nur Talikder	1402710200797
			Zia Uddin Chowdhury	1402710200747
30	"Heart Blocks Classification and Prediction Using Association Rules"	Syed Md. Minhaz Hossain	Samar Das	1402610200640
			Sultan MD. Aslam	1402610200658
31	"Online Job Portal"		Md. Akid Hasan	1302510200575

		Syed Md. Minhaz Hossain	MD. Morsheddul Alam	1302510200576
32	‘Microcontroller Based Cloth Folding Machine”	Mr. Khaeque Md. Aashiq Kamal	Tanmoy Barua	1402710200812
			Ananya Palit Mou	1402710200819
33	Doctors Finder	Prof. Dr. Taufique Sayeed	Raktim Acharjee	1202210200372
			Shoeb Bin Abedin	1202210200379
34	“Automated Energy Saving System”	Mrs. Tanni Dhoom	Imtiaz Ahmed Sadi	1402710200789
			Jawad Hasan	1402710200769
35	BloodDF-Blood Donor Finder	Mr. Khaeque Md. Aashiq Kamal	Rubel Chowdhury	1202310200411
			Mohammad Mohiuddin	1202310200423
36	Voice controlled Smart Chess Game	Faisal Ahmed	Joy Barua	1402610200631
			Ashish Kumar Debnath	1402610200648
37	“Traffic Monitoring”	Rumapa Das Gupta	Amit Biswas	1402710200678
			Md. Sayem Uddin	1402710200708
38	Online Library Management System	Mr. Khaeque Md. Aashiq Kamal	Shurav Talukder	58120118
			Samarjit Das	58120119
			Rimon Banik	815120189
39	“Rock on Diabetes”	Ms. Farhana Shirin Chowdhury	Mamtaj Akter	1202310200410
40	“E-Car System”	Mr. Khaeque Md. Aashiq Kamal	Aney Biswas	1302510200616
			Afia Sultana	1302510200608
			Md. Stafizur Rahman	1302510200589
41	Online Bus Ticket Reservation Rahman	Faisal Ahmed	Mohammad Shakawur Rahman	1302510200542
			Anik Das	1302510200536
42	Algorithm Visualizer Mobile Application “ALGOVIZ”	Mrs.Asma Joshita Trisha	Robiul Hasan	1402710200714
			Sharif Ullah Saad	1402710200731
43	Sensor Based Wheelchair For Blind People	Mohammad Hasan	Md. Asraful Alaam Chy	1402710200839

			Pial Chowdhury	1402710200843
44	Law And Advice	Prof. Dr.Taufique Sayeed	Tipu Sultan	1402710200749
			Aritro Das	1402710200752
45	“ICT jobs-ctg: A Jobsite for ICT Sector"	Mr. Khaeque Md. Aashiq Kamal	Trishna Dey	1402710200744
			Chite Debnath	1402710200804
46	A Smart Cane for Visually Impaired Using Raspberry Pi	Prof. Dr. Taufique Sayeed	Binita Das Gupta	1402710200783
			Nishita Das Gupta	1402710200787
47	Augmented Reality Based Travel Assistant	Mrs. Tanni Dhoom	Tahmina Sultana	1402710200820
			Sumita Chowdhury	1402710200811
48	Vat Transfer System	Rumapa Das Gupta	Aminur Rahman	1402710200675
			Omar Sharif Ansary	1402710200702
49	“Premier Iniversity Online Admission System”	Rumapa Das Gupta	Mou Palit	1402710200830
			Tanjina Mim Nur	1402710200829
50	“Analysis of Rice Leaf Disease Classfication Using Color Segmentation”	Syed Md. Minhaz Hossain	Anika Mubash-Shira Kabir	1402710200688
			Anany Barua	1402710200788
51	Digit Writing & Digit Recognition	Syed Md. Minhaz Hossain	Suon Chowdhury	1402710200803
			Akash Kanti Susill	1402710200816
52	“Baby Care”	Rumapa Das Gupta	Reshmy Akter	1402710200721
			Hashna Hena	1402710200700
53	An Interactive Alert System With Android An Android Application To Help Senior Citizen	MD. Neamul Hoque	Monika Biswas	1402710200712
			Ananya Sarker	1402710200715
54	Card Maker: An Android Application	Mr. Kingshuk Dhar	Sabiha Sabnam shaon	1402710200767
			Sahara Kumkum Asma	1402710200801
55	Emotion Detection from Bangla Text Corpus Using Naïve Bayes Classifier	Mr. Kingshuk Dhar	Sara Azmin	1402710200815

56	"Design and Implementation of AMG8833 Based Efficient Power Management System"	Mr. Ariful Islam Bhuyan	Paria Das	1402710200676
			Tasnova Rahman	1402710200722
57	"Design and implementation of EMG based prosthetic hand for handicapped People"	Mr. Ariful Islam Bhuyan	Umme Imoon	1402610200647
			Raffat Ara Nazmin	1402610200671
58	Rice Leaf Diseases Detection Using Convolutional Neural Network	Syed Md. Minhaz Hossain	Md. Monjur Morshed Tanjil	1502810200938
			Mohammad Abser Bin Ali	1402710200825
			Mohammad Zuhadul Islam	1502810200932
59	Scalable Medical System for Managing Patients and Prescription History	Mr. Anik Sen	Ashraful Dowla	1502810200872
			Ajmal Hossain	1502810200874
60	Fighter is Tank	Asma Joshita Trisha	Payel Bhattacharjee	1202310200421
			Md. Suddam Hossen	1202310200412
61	Home Automation	Mr. Ariful Islam Bhuyan	Md. Nurul Kabir Chowdhury	1302510200569
			Sheukh Sydur Rahman	1302510200531
62	Smart Cycling	Ms. Senjuthi Bhattcharjee	Irfan Ali Sakib	1302410200499
			Tahmina Mozammel	1302410200506
63	"Helping Hand-An Android Based Platform For Service Professional"	Mr. Khaeque Md. Aashiq Kamal	Prasenjit Acharjee	1402710200809
			Shahadat Hossain Shefat	1402710200850
64	"An E-Commerce And M-Commerce Platform For Farmers"	Mr. Khaeque Md. Aashiq Kamal	Mowsumy Das	1402710200837
			Nasrin Hossen Tania	1402710200864

Annexure: 6.6
Academic Calendar

PREMIER UNIVERSITY
CHATTOGRAM

ACADEMIC CALENDAR 2023

SEMESTER	PROGRAM	COURSE ENROLLMENT	REGULAR SEMESTER	MID TERM	FINAL EXAM	RESULT
SPRING	BBA	JANUARY, 2023	JANUARY, 2023	MARCH, 2023	MAY, 2023	JUNE, 2023
	MBA 1ST & 2ND (1 YEAR)	JANUARY, 2023	JANUARY, 2023	MARCH, 2023	JUNE, 2023	JUNE, 2023
	MBA (REGULAR & EXECUTIVE)	JANUARY, 2023	JANUARY, 2023	MARCH, 2023	JUNE, 2023	JUNE, 2023
	BACHELOR OF ARCHITECTURE	FEBRUARY, 2023	MARCH, 2023	MAY, 2023	JULY, 2023	AUGUST, 2023
	B.SC. (ENG.) IN CSE	APRIL, 2023	MAY, 2023	JULY, 2023	SEPTEMBER, 2023	OCTOBER, 2023
	M.SC. (ENG.) IN CSE	APRIL, 2023	MAY, 2023	JULY, 2023	SEPTEMBER, 2023	OCTOBER, 2023
	B.SC. (ENG.) IN EEE	FEBRUARY, 2023	MARCH, 2023	APRIL, 2023	JUNE, 2023	AUGUST, 2023
	B.SC. (HONORS) IN MATHEMATICS	MARCH, 2023	MARCH, 2023	MAY, 2023	JULY, 2023	AUGUST, 2023
	M.SC. IN MATHEMATICS (1 YEAR & 2 YEARS)	MARCH, 2023	MARCH, 2023	MAY, 2023	JULY, 2023	AUGUST, 2023
	LL.B. (HONORS)	JANUARY, 2023	JANUARY, 2023	MARCH, 2023	JUNE, 2023	JUNE, 2023
	LL.M.	JANUARY, 2023	JANUARY, 2023	FEBRUARY, 2023	MAY, 2023	JUNE, 2023
	BA (HONORS) IN ENGLISH	MARCH, 2023	MARCH, 2023	APRIL, 2023	JULY, 2023	AUGUST, 2023
	MA IN ENGLISH	MARCH, 2023	MARCH, 2023	APRIL, 2023	JULY, 2023	AUGUST, 2023
	BSS (HONORS) IN ECONOMICS	MARCH, 2023	MARCH, 2023	MAY, 2023	JULY, 2023	AUGUST, 2023
	MSS IN ECONOMICS (1 YEAR/1.5 YEAR)	MARCH, 2023	MARCH, 2023	MAY, 2023	JULY, 2023	AUGUST, 2023
	BSS (HONORS) IN SOCIOLOGY AND SUSTAINABLE DEVELOPMENT	JANUARY, 2023	JANUARY, 2023	FEBRUARY, 2023	MAY, 2023	JUNE, 2023
	MASTER OF PUBLIC HEALTH	MARCH, 2023	MARCH, 2023	MAY, 2023	AUGUST, 2023	SEPTEMBER, 2023
	BA (HONORS) IN FASHION DESIGN AND TECHNOLOGY	MARCH, 2023	MARCH, 2023	MAY, 2023	JULY, 2023	AUGUST, 2023
	POSTGRADUATE DIPLOMA IN LIBRARY AND INFORMATION SCIENCE	MARCH, 2023	MARCH, 2023	MAY, 2023	JULY, 2023	AUGUST, 2023
SUMMER	LL.B. (HONORS)	MARCH, 2023	MARCH, 2023	MAY, 2023	JULY, 2023	AUGUST, 2023
	LL.M.	APRIL, 2023	APRIL, 2023	JUNE, 2023	AUGUST, 2023	SEPTEMBER, 2023
FALL	BBA	JULY, 2023	JULY, 2023	SEPTEMBER, 2023	OCTOBER, 2023	DECEMBER, 2023
	MBA (1 YEAR - ALL BATCHES)	JULY, 2023	JULY, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	DECEMBER, 2023
	MBA (REGULAR & EXECUTIVE)	JULY, 2023	JULY, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	DECEMBER, 2023
	BACHELOR OF ARCHITECTURE	AUGUST, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	JANUARY, 2024	FEBRUARY, 2024
	B.SC. (ENG.) IN CSE	OCTOBER, 2023	OCTOBER, 2023	DECEMBER, 2023	FEBRUARY, 2024	MARCH, 2024
	M.SC. (ENG.) IN CSE	OCTOBER, 2023	OCTOBER, 2023	DECEMBER, 2023	FEBRUARY, 2024	MARCH, 2024
	B.SC. (ENG.) IN EEE	AUGUST, 2023	SEPTEMBER, 2023	OCTOBER, 2023	DECEMBER, 2023	JANUARY, 2024
	B.SC. (HONORS) IN MATHEMATICS	SEPTEMBER, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	JANUARY, 2024	FEBRUARY, 2024
	M.SC. IN MATHEMATICS (1 YEAR & 2 YEARS)	SEPTEMBER, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	JANUARY, 2024	FEBRUARY, 2024
	LL.B. (HONORS)	JULY, 2023	JULY, 2023	SEPTEMBER, 2023	DECEMBER, 2023	DECEMBER, 2023
	LL.M.	JULY, 2023	JULY, 2023	AUGUST, 2023	NOVEMBER, 2023	DECEMBER, 2023
	BA (HONORS) IN ENGLISH	AUGUST, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	JANUARY, 2024	FEBRUARY, 2024
	MA IN ENGLISH	AUGUST, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	JANUARY, 2024	FEBRUARY, 2024
	BSS (HONORS) IN ECONOMICS	SEPTEMBER, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	JANUARY, 2024	FEBRUARY, 2024
	MSS IN ECONOMICS (1 YEAR/1.5 YEAR)	SEPTEMBER, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	JANUARY, 2024	FEBRUARY, 2024
	BSS (HONORS) IN SOCIOLOGY AND SUSTAINABLE DEVELOPMENT	JUNE, 2023	JUNE, 2023	JULY, 2023	OCTOBER, 2023	NOVEMBER, 2023
	MASTER OF PUBLIC HEALTH	SEPTEMBER, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	MARCH, 2024	APRIL, 2024
	BA (HONORS) IN FASHION DESIGN AND TECHNOLOGY	SEPTEMBER, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	JANUARY, 2024	FEBRUARY, 2024
	POSTGRADUATE DIPLOMA IN LIBRARY AND INFORMATION SCIENCE	SEPTEMBER, 2023	SEPTEMBER, 2023	NOVEMBER, 2023	JANUARY, 2024	FEBRUARY, 2024

- Class test, Quiz, Oral test and presentation dates will be announced and conducted by the respective course teachers.
- The dates for beginning of a semester & examinations can not be changed without the approval of administration.

Premier University

1/A, O. R. Nizam Road, Prabartak Circle, Panchlaish, Chattogram, Bangladesh.
Phone (PABX): +88-031-2856352-55, 241355198, 01313 044515-19, Fax: +88-031-657892, e-mail: info@puc.ac.bd, web: www.puc.ac.bd.

2022

Tentative dates of the classes and examinations of the year 2022

Semester	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Spring	BBA	February, 2022	March, 2022	April, 2022	June, 2022	August, 2022
	MBA (1 year)	May, 2022	June, 2022	August, 2022	October, 2022	November, 2022
	MBA (Regular & Executive)	February, 2022	March, 2022	May, 2022	July, 2022	August, 2022
	Bachelor of Architecture	January, 2022	January, 2022	April, 2022	June, 2022	June, 2022
	B.Sc. (Eng.) in CSE	April, 2022	May, 2022	June, 2022	August, 2022	September, 2022
	B.Sc. (Eng.) in EEE	March, 2022	March, 2022	April, 2022	June, 2022	August, 2022
	B.Sc. (Honors) in Mathematics	March, 2022	March, 2022	April, 2022	June, 2022	August, 2022
	M.Sc. in Mathematics (1 year/2 years)	March, 2022	March, 2022	April, 2022	July, 2022	August, 2022
	LL.B. (Honors)	January, 2022	February, 2022	March, 2022	June, 2022	June, 2022
	LL.M.	February, 2022	February, 2022	March, 2022	June, 2022	June, 2022
	BA (Honors) in English	March, 2022	March, 2022	May, 2022	July, 2022	August, 2022
	MA in English	March, 2022	March, 2022	May, 2022	July, 2022	August, 2022
	BSS (Honors) in Economics	March, 2022	March, 2022	May, 2022	July, 2022	August, 2022
	MSS in Economics (1 year/1.5 year)	March, 2022	March, 2022	May, 2022	July, 2022	August, 2022
	B.S.S. (Honors) in Sociology and Sustainable Development	June, 2022	June, 2022	August, 2022	November, 2022	December, 2022
	Master of Public Health	April, 2022	April, 2022	May, 2022	August, 2022	August, 2022

Semester	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Summer	LL.B. (Honors)	April, 2022	May, 2022	June, 2022	September, 2022	September, 2022
	LL.M.	April, 2022	April, 2022	May, 2022	August, 2022	September, 2022

Semester	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Fall	BBA	August, 2022	September, 2022	October, 2022	December, 2022	February, 2023
	MBA (1 year)	November, 2022	December, 2022	January, 2023	April, 2023	May, 2023
	MBA (Regular & Executive)	August, 2022	September, 2022	November, 2022	January, 2023	February, 2023
	Bachelor of Architecture	June, 2022	July, 2022	September, 2022	November, 2022	November, 2022
	B. Sc. (Eng.) in CSE	September, 2022	October, 2022	November, 2022	January, 2023	March, 2023
	B. Sc. (Eng.) in EEE	August, 2022	September, 2022	October, 2022	January, 2023	February, 2023
	B.Sc. (Honors) in Mathematics	September, 2022	September, 2022	October, 2022	December, 2022	February, 2023
	M.Sc. in Mathematics (1 year/2 years)	September, 2022	September, 2022	October, 2022	January, 2023	February, 2023
	LL.B. (Honors)	July, 2022	August, 2022	September, 2022	December, 2022	December, 2022
	LL.M.	August, 2022	August, 2022	October, 2022	December, 2022	January, 2023
	BA (Honors) in English	September, 2022	September, 2022	October, 2022	January, 2023	January, 2023
	MA in English	September, 2022	September, 2022	October, 2022	January, 2023	January, 2023
	BSS (Honors) in Economics	September, 2022	September, 2022	November, 2022	January, 2023	February, 2023
	MSS in Economics (1 year/1.5 year)	September, 2022	September, 2022	November, 2022	January, 2023	February, 2023
	B.S.S. (Honors) in Sociology and Sustainable Development	January, 2022	January, 2022	February, 2022	May, 2022	June, 2022
	Master of Public Health	November, 2021	November, 2021	December, 2021	March, 2022	March, 2022

- *Class test, Quiz, Oral test and presentation dates will be announced and conducted by the respective course teachers.*
- *The dates for beginning of a semester & examinations can not be changed without the approval of administration.*

Premier University

541, O. R. Nizam Road, GEC Circle, Chattogram, Bangladesh.

Stay Home, Stay Safe



Tentative dates of the classes and examinations of the year 2021

Semester/ Trimesters	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Spring	BBA	23 February, 2021	01 March, 2021	26 April, 2021	21 June, 2021	31 July, 2021
	MBA 1st & 2nd (1 year)	23 November, 2020	01 December, 2020	23 February, 2021	13 April, 2021	18 May, 2021
	MBA (Regular & Executive)	23 December, 2020	01 January, 2021	27 March, 2021	04 June, 2021	05 July, 2021
	B. of Architecture	03 January, 2021	12 January, 2021	14 March, 2021	17 May, 2021	21 June, 2021
	B.Sc. (Eng.) in CSE	27 February, 2021	01 March, 2021	10 April, 2021	26 June, 2021	14 August, 2021
	B.Sc. (Eng.) in EEE	27 February, 2021	01 March, 2021	10 April, 2021	26 June, 2021	14 August, 2021
	B.Sc. (Honors) in Mathematics	07 April, 2021	12 April, 2021	27 May, 2021	13 July, 2021	01 September, 2021
	M.Sc. in Mathematics	09 April, 2021	16 April, 2021	29 May, 2021	23 July, 2021	01 September, 2021
	LL.B. (Honors)-All Trimesters	17 January, 2021	19 January, 2021	27 Februray, 2021	28 April, 2021	12 May, 2021
	LL.B. (Honors)-All Semesters	25 January, 2021	27 January, 2021	27 March, 2021	26 May, 2021	15 June, 2021
	LL.M.	15 February, 2021	17 February, 2021	18 April, 2021	07 July, 2021	28 July, 2021
	BA (Honors) in English	07 March, 2021	18 March, 2021	20 April, 2021	11 July, 2021	21 August, 2021
	MA in English/MA (Major) in ELT	07 March, 2021	18 March, 2021	20 April, 2021	11 July, 2021	21 August, 2021
	BSS (Honors) in Economics	25 February, 2021	01 March, 2021	21 April, 2021	15 June, 2021	18 August, 2021
	MSS in Economics	25 February, 2021	01 March, 2021	21 April, 2021	15 June, 2021	18 August, 2021
	B. of Science in Sociology & Sustainable Development (H)	26 November, 2020	01 December, 2020	10 January, 2021	11 April, 2021	02 May, 2021

Trimesters	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Summer	LL.B. (Honors)-All Semesters	15 March, 2021	17 March, 2021	24 April, 2021	17 July, 2021	04 August, 2021
	LL.M.	20 April, 2021	24 April, 2021	02 June, 2021	07 August, 2021	28 August, 2021

Semester/ Trimesters	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Fall	BBA	23 August, 2021	01 September, 2021	03 November, 2021	15 December, 2021	25 January, 2022
	MBA 1st & 2nd (1 year)	15 May, 2021	22 May, 2021	14 August, 2021	09 October, 2021	10 November, 2021
	MBA (Regular & Executive)	12 July, 2021	24 July, 2021	18 September, 2021	13 November, 2021	07 December, 2021
	B. of Architecture	01 July, 2021	12 July, 2021	14 September, 2021	16 November, 2021	21 December, 2021
	B. Sc. (Eng.) in CSE	24 August, 2021	01 September, 2021	16 October, 2021	26 December, 2021	10 February, 2022
	B. Sc. (Eng.) in EEE	24 August, 2021	01 September, 2021	16 October, 2021	26 December, 2021	10 February, 2022
	B.Sc. (Honors) in Mathematics	14 September, 2021	20 September, 2021	28 Octoberr, 2021	22 December, 2021	22 February, 2022
	M.Sc. in Mathematics	17 September, 2021	24 September, 2021	05 November, 2021	31 December, 2021	22 February, 2022
	LL.B. (Honors)-All Trimesters	19 May, 2021	22 May, 2021	26 July, 2021	26 August, 2021	25 September, 2021
	LL.B. (Honors)-All Semesters	03 July, 2021	05 July, 2021	06 September, 2021	01 December, 2021	29 December, 2021
	LL.M.-All Semesters	10 July, 2021	12 July, 2021	11 September, 2021	04 December, 2021	28 December, 2021
	BA (Honors) in English	01 November, 2020	15 November, 2020	13 December, 2020	07 February, 2021	06 March, 2021
	MA in English/MA (Major) in ELT	01 November, 2020	15 November, 2020	13 December, 2020	07 February, 2021	06 March, 2021
	BSS (Honors) in Economics	28 August, 2021	01 September, 2021	15 November, 2021	02 January, 2022	02 February, 2022
	MSS in Economics	28 August, 2021	01 September, 2021	15 November, 2021	02 January, 2022	02 February, 2022
	B. of Science in Sociology & Sustainable Development (H)	13 May, 2021	20 May, 2021	11 July, 2021	14 October, 2021	10 November, 2021

- *Class test, Quiz, Oral test and presentation dates will be announced and conducted by the respective course teachers.*
- *The dates for beginning of a semester & examinations can not be changed without the approval of administration.*

541, O. R. Nizam Road, GEC Circle, Chattogram and 1/A, O. R. Nizam Road, Prabartak Circle, Panchlaish, Chattogram, Bangladesh.
Phone (PABX): +88-031-2856352-55, 241355198, 01313 044515-19, Fax: +88-031-657892, e-mail: info@puc.ac.bd, web: www.puc.ac.bd

Premier University

Tentative dates of the classes and examinations of the year 2020

Semester/ Trimesters	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Spring	BBA	16 February, 2020	23 February, 2020	26 April, 2020	21 June, 2020	10 August, 2020
	MBA - 1 year	01 December, 2019	08 December, 2019	09 February, 2020	10 April, 2020	12 May, 2020
	MBA - 2 year	26 January,,2020	31 January, 2020	10 April, 2020	24 July, 2020	31 August, 2020
	BSS (Honors) in Economics	23 February, 2020	01 March, 2020	20 April, 2020	13 June, 2020	01 August, 2020
	MSS in Economics	23 February, 2020	01 March, 2020	20 April, 2020	13 June, 2020	01 August, 2020
	B. Arch.	01 March, 2020	08 March, 2020	03 May, 2020	15 July, 2020	12 August, 2020
	B. Sc. (Eng.) in CSE	23 February, 2020	01 March, 2020	18 April, 2020	24 June, 2020	12 August, 2020
	B. Sc. (Eng.) in EEE	23 February, 2020	01 March, 2020	18 April, 2020	24 June, 2020	12 August, 2020
	B.Sc. (Honors) in Mathematics	01 March, 2020	03 March, 2020	19 April, 2020	21 June, 2020	16 August, 2020
	M.Sc. in Mathematics	01 March, 2020	06 March, 2020	24 April, 2020	21 July, 2020	25 August, 2020
	LL.B. (Honors)-All Trimesters	12 January, 2020	15 January, 2020	25 Februray, 2020	12 May, 2020	20 June, 2020
	LL.B. (Honors)-All Semesters	12 January, 2020	15 January, 2020	25 Februray, 2020	12 May, 2020	20 June, 2020
	LL.M.	20 January, 2019	26 January, 2020	15 March, 2020	10 June, 2020	07 July, 2020
	BA (Honors) in English	17 February, 2020	01 March, 2020	26 April, 2020	12 July, 2020	18 August, 2020
	MA in English/MA (Major) in ELT	17 February, 2020	01 March, 2020	26 April, 2020	12 July, 2020	18 August, 2020
	BSS (H) in Sociology & Sustainable Development	12 March, 2020	15 March, 2020	14 June, 2020	06 September, 2020	23 September, 2020

Trimesters	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Summer	LL.B. (Honors)-All Semesters	08 March, 2020	08 March, 2020	02 May, 2020	17 August, 2020	01 September, 2020
	LL.M.	15 April, 2020	15 April, 2020	13 June, 2020	01 September, 2020	20 September, 2020

Semester/ Trimesters	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Fall	BBA	23 August, 2020	01 September, 2020	01 November, 2020	17 December, 2020	07 February, 2021
	MBA - 1 year	20 May, 2020	31 May, 2020	07 August, 2020	18 Octoberr, 2020	30 November, 2020
	MBA - 2 year	18 August, 2020	21 August, 2020	23 October, 2020	08 January, 2021	07 Febuary, 2021
	BSS (Honors) in Economics	22 August, 2020	01 September, 2020	11 November, 2020	19 December, 2020	31 January, 2021
	MSS in Economics	22 August, 2020	01 September, 2020	11 November, 2020	19 December, 2020	31 January, 2021
	B. Arch.	08 September, 2020	15 September, 2020	15 November, 2020	15 February, 2021	28 February, 2021
	B. Sc. (Eng.) in CSE	23 August, 2020	01 September, 2020	20 October, 2020	15 December, 2020	09 February, 2021
	B. Sc. (Eng.) in EEE	23 August, 2020	01 September, 2020	20 October, 2020	15 December, 2020	09 February, 2021
	B.Sc. (Honors) in Mathematics	25 August, 2020	01 September, 2020	11 October, 2020	15 December, 2020	18 February, 2021
	M.Sc. in Mathematics	28 August, 2020	04 September, 2020	18 October, 2020	15 January, 2021	18 February, 2021
	LL.B. (Honors)-All Semesters	01 July, 2020	07 July, 2020	05 September, 2020	05 December, 2020	31 December, 2020
	LL.M.	11 July, 2020	12 July, 2020	05 September, 2020	05 December, 2020	31 December, 2020
	BA (Honors) in English	20 August, 2020	01 September, 2020	01 November, 2020	03 January, 2021	10 February, 2021
	MA in English/MA (Major) in ELT	20 August, 2020	01 September, 2020	01 November, 2020	03 January, 2021	10 February, 2021
	BSS (H) in Sociology & Sustainable Development	14 October, 2019	21 October, 2019	22 December, 2019	18 February, 2020	10 March, 2020

- *Class test, Quiz, Oral test and presentation dates will be announced and conducted by the respective course teachers.*
- *The dates for beginning of a semester & examinations can not be changed without the approval of administration.*

1/A, O. R. Nizam Road, Prabartak Circle, Panchlaish, Chattogram, Bangladesh. Phone (PABX): +88-031-656917, 657654, 656612-15, 241355198
01313 044515-19, Fax: +88-031-657892, e-mail: info@puc.ac.bd, www.facebook.com/PremierUniversityChittagong, web: www.puc.ac.bd



Tentative dates of the classes and examinations of the year 2019

Semester/ Trimesters	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Spring	BBA (Mar - Aug, 2019)	18 February, 2019	24 February, 2019	28 April, 2019	17 June, 2019	06 August, 2019
	MBA - 1year	16 November, 2018	02 December, 2018	22 February, 2019	26 April, 2019	13 June, 2019
	MBA - (2 year + 1.5 year)	25 January, 2019	01 February, 2019	19 April, 2019	09 July, 2019	06 August, 2019
	BSS (Honors) in Economics	16 February, 2019	23 February, 2019	20 April, 2019	25 June, 2019	07 August, 2019
	MSS in Economics	16 February, 2019	23 February, 2019	20 April, 2019	25 June, 2019	07 August, 2019
	B. Arch.	03 March, 2019	17 March, 2019	19 May, 2019	17 July, 2019	18 August, 2019
	B. Sc. (Eng.) in CSE	16 February, 2019	23 February, 2019	20 April, 2019	23 June, 2019	05 August, 2019
	B. Sc. (Eng.) in EEE	24 February, 2019	02 March, 2019	13 April, 2019	23 June, 2019	05 August, 2019
	B.Sc. (Honors) in Mathematics	23 February, 2019	02 March, 2019	20 April, 2019	15 June, 2019	04 August, 2019
	LL.B. (Honors)-All Trimesters	01 February, 2019	01 February, 2019	01 March, 2019	20 April, 2019	30 April, 2019
	LL.B. (Honors)-All Semesters	05 January, 2019	05 January, 2019	15 February, 2019	01 June, 2019	25 June, 2019
	LL.M.	11 February, 2019	11 February, 2019	10 March, 2019	30 April, 2019	10 May, 2019
	BA (Honors) in English	24 February, 2019	10 March, 2019	05 May, 2019	14 July, 2019	18 August, 2019
	MA in English/MA (Major) in ELT	24 February, 2019	10 March, 2019	05 May, 2019	14 July, 2019	18 August, 2019

Trimesters	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Summer	LL.B. (Honors)	06 May, 2018	06 May, 2018	05 June, 2018	16 August, 2018	30 August, 2018

Semester/ Trimesters	Program	Course Enrollment	Regular Semester	Mid Term	Final Exam	Result
Fall	BBA (Sep - Feb, 2020)	25 August, 2019	08 September, 2018	11 November, 2019	22 November, 2019	05 February, 2020
	MBA - 1year	14 June, 2019	14 June, 2019	08 September, 2019	12 November, 2019	09 December, 2019
	MBA - (2 year + 1.5 year)	18 August, 2019	20 August, 2019	12 November, 2019	07 January, 2020	30 January, 2020
	BSS (Honors) in Economics	26 August, 2019	01 September, 2019	02 November, 2019	28 December, 2019	02 February, 2020
	MSS in Economics	26 August, 2019	01 September, 2019	02 November, 2019	28 December, 2019	02 February, 2020
	B. Arch.	16 September, 2018	20 September, 2018	17 November, 2018	02 February, 2019	21 February, 2019
	B. Sc. (Eng.) in CSE	04 August, 2019	18 August, 2019	13 October, 2019	08 December, 2019	19 January, 2019
	B. Sc. (Eng.) in EEE	18 August, 2019	01 September, 2019	19 October, 2019	18 December, 2019	05 February, 2020
	B.Sc. (Honors) in Mathematics	26 August, 2019	01 September, 2019	20 October, 2019	22 December, 2019	15 February, 2020
	LL.B. (Honors)-All Trimesters	05 September, 2019	01 September, 2019	20 Octtober, 2019	22 December, 2019	15 February, 2020
	LL.B. (Honors)-All Semesters	05 September, 2019	05 September, 2019	20 Octtober, 2019	01 February, 2019	28 February, 2020
	LL.M.	15 September, 2019	05 September, 2019	20 Octtober, 2019	01 February, 2019	28 February, 2020
	BA (Honors) in English	01 September, 2019	10 September, 2019	10 November, 2019	12 January, 2020	13 February, 2020
	MA in English/MA (Major) in ELT	01 September, 2019	10 September, 2019	10 November, 2019	12 January, 2020	13 February, 2020

- *Class test, Quiz, Oral test and presentation dates will be announced and conducted by the respective course teachers.*
- *The dates for beginning of a semester & examinations can not be changed without the approval of administration.*

1/A, O. R. Nizam Road, Prabartak Circle, Panchlaish, Chattogram. Phone (PABX): +88 031 656917, 657654, 656612-14, Cell: +880 1313 044515-19
Fax: +88 031 657892, e-mail: info@puc.ac.bd, web: www.puc.ac.bd, www.facebook.com/PremierUniversityChittagong